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# A Greedy PDE Router for Blending Neural Operators and Classical Methods

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1 **Revision note.** All text, tables and figures added in this revision are typeset in red (new figures  
2 carry a red frame); everything in black is unchanged from the submitted version. The iteration-based  
3 experiments of the submitted version on  $31^2$  and  $63^2$  grids with the LSTM router, and their appendix  
4 analyses, have been removed: they are superseded by the wall-clock study of Sections E and 7, which  
5 covers the same solvers and equations (plus anisotropic and variable-coefficient diffusion) on  $128^2$ –  
6  $512^2$  grids and also reports the iteration-based metrics. The Dirichlet experiment of the submitted  
7 version has been withdrawn (Section D); two black sentences of the abstract (the greedy-guarantee  
8 sentence and the AUC sentence) were edited and are now marked in red.

9 **Status of the confirmatory re-run (draft for review).** In response to five rounds of review, every  
10 wall-clock result is being regenerated under a revised, frozen protocol (Section E.1): compiled  
11 classical kernels, a corrected forcing sampler, per-operation costs measured in the form in which  
12 they are charged, the deployed rule equal to Algorithm 1 on macro-actions, HINTS with its two-  
13 dimensional period  $\tau = 15$ , residual-decay controls executed live, classical baselines timed in the  
14 same session as the policies with validated Krylov replays, a drift guard with re-timing, two-sided  
15 tests with bootstrap intervals and explicit censoring classes, a fresh test seed and three warmed-up  
16 timed replays. At the time of this draft the following are *final*: the theory corrections (Sections 4  
17 and 5 and their proofs); the wording of every claim is macro-driven and will be recomputed from the  
18 complete tables. The  $128^2$  pairwise cells of Poisson and convection–diffusion were produced by an  
19 earlier revision of the driver (before the live decay rule, the Krylov validation, the stored decision  
20 sequences and the in-cell re-timing existed), and every one of their instances is re-timed with the final  
21 driver once the other  $128^2$  cells are complete; those of anisotropic diffusion were produced by the  
22 final driver except for two details of the drift guard (it slept rather than ran the reference operation  
23 while waiting, and it compared an instance only with the reference at the time of its check), and  
24 the re-timing pass re-times any of their instances that were slow against the cell’s final reference.  
25 Until the re-timing, the macros that depend on those fields (decision-sequence agreement, Krylov  
26 validity, re-timing counts) print as pending. Every  $128^2$  pairwise cell and ensemble of the four  
27 equations is final (re-timing included), and every number computed from them is. Every Poisson and  
28 convection–diffusion cell at  $128^2$ ,  $256^2$  and  $512^2$  is final as well, with their ensembles. Everything  
29 marked **[pending]** in the text or shown as a **[results pending]** table is still running, in the order of  
30 `run_all.sh`: the development-set schedule selection, the  $128^2$  seed trials and assumption, screening  
31 and theorem checks, the rest of the decision-granularity ablation (Section E.6), the anisotropic and  
32 variable-coefficient cells at  $256^2$  with their ensembles, the checks at the larger grids, overheads and  
33 the discretisation study. Summary ranges quoted in the abstract and in Section 7 are computed from  
34 the cells available at build time. Where a first confirmatory run (before the protocol revision) had  
35 produced a number that is quoted in the text, it is labelled as such.

## Abstract

36 When solving PDEs, classical numerical solvers are often computationally ex-  
37 pensive, while machine learning methods can suffer from spectral bias, failing  
38 to capture high-frequency components. Designing an optimal hybrid iterative  
39 solver—where, at each iteration, a solver is selected from an ensemble of solvers to

leverage their complementary strengths—poses a challenging combinatorial problem. While greedy selection is desirable for its **guarantee of a constant fraction of the optimal error reduction** under Lipschitz and shared-eigenbasis assumptions, it requires knowledge of the true error at each step, which is unavailable in practice. We address this by proposing an approximate greedy router that efficiently mimics a greedy approach to solver selection. **Empirical results on the Poisson, convection–diffusion and anisotropic diffusion equations compare our method with single-solver baselines and with existing hybrid approaches such as HINTS on the area-under-the-curve (AUC) of the error trajectory: over the  $128^2$  cells the mean-AUC ratios of HINTS and of the solver alone to our method are  $3\text{--}174709\times$  and  $3\text{--}2666251\times$  (one-sided paired  $t$ -tests on  $\log$  AUC,  $p \leq 10^{-10}$ ). With a cost-aware instantiation of the greedy rule, a lightweight router, and a band-limited scale-equivariant corrector, we also measure wall-clock time to a target accuracy (the time charged to each method in a replay of its decisions, Section E.1) on  $128^2$  to  $512^2$  grids with compiled classical kernels, four equations (Poisson, convection–diffusion, anisotropic and variable-coefficient diffusion) and classical solvers from Jacobi to geometric multigrid, against HINTS, the best fixed schedule and the best residual-decay rule chosen per setting on the test set, and multigrid, preconditioned Krylov and direct solves timed in the same session. At truncation-level accuracy the learned router executes the same decision sequence as the best static schedule found by a test-set search on 0–100% of the instances, and its paired time ratios are  $0.94\times\text{--}0.97\times$  against that schedule and  $0.95\times\text{--}5.11\times$  against HINTS with its published period; below truncation level the ratios are  $0.92\times\text{--}3.07\times$  against HINTS,  $0.97\times\text{--}1373\times$  against the one-shot schedule and  $0.83\times\text{--}1.61\times$  against the best fixed schedule (the router is faster in 6 of 24 pairings), and the ratio of multigrid alone to the router is  $0.03\times\text{--}20.5\times$  (all at  $128^2$ ; a ratio above one favours the router). Routing over ensembles of classical solvers gains nothing at  $128^2$  (work-unit ratios  $0.74\times\text{--}1.00\times$  against the best single-member router at  $10^{-8}$ ) but pays at  $256^2$ , where the two-member Jacobi ensemble is  $1.23\text{--}1.24\times$  faster than the best single-member router on the Poisson and convection–diffusion equations on every test instance (**[pending]**–**[pending]** over all equations), and the router’s losses to the best schedule at  $10^{-8}$  are confined to the 10 pairings whose best schedule calls the corrector more often than one macro-action allows (elsewhere it is faster in 6 of 14); the larger grids and the finer decision granularity are **[pending]**.**

## 1 Introduction

Natural phenomena and engineered systems are often governed by ordinary and partial differential equations (PDEs). Solving these equations enables a variety of tasks - predicting the evolution of the system through simulation (e.g., forecasting weather using the Navier-Stokes equations) [Kalnay, 2003, Bauer et al., 2015, Staniforth and Côté, 1991], addressing control problems (e.g., optimizing heat shield design for spacecrafts using heat transfer equations) [Anderson, 1989, Tröltzsch, 2010], and tackling inverse problems based on external measurements (e.g., reconstructing brain activity from EEG data using electrophysiological models) [Baillet et al., 2001, Grech et al., 2008].

Traditionally, PDEs are solved using finite difference methods [Smith, 1985, LeVeque, 2007], which discretize the spatial and/or temporal domain and approximate the solution by solving a system of equations at the discrete grid points. Similarly, finite element methods [Hughes, 2003, Bathe, 2006, Brenner, 2008] construct a system of equations based on local basis functions, and then rely on numerical linear algebra techniques—such as the Jacobi and Gauss-Seidel method [Saad, 2003]—to compute the solution. These iterative methods, however, **must be re-run from scratch for every new** initial condition, boundary condition, or forcing function, as even minor changes to any of these parameters require re-solving the entire system of equations from scratch.

These challenges have motivated the development of neural operators [Kovachki et al., 2023]—a class of machine learning models that aim to learn the solution operator directly, enabling fast inference across a range of parameters. Neural operators lift the classical linear layer in neural networks to an operator—typically a kernel integral operator—between function spaces. Despite strong approximation guarantees [Chen and Chen, 1995], neural operators are prone to spectral bias [Liu and Cai, 2021, Liu

et al., 2024, You et al., 2024, Khodakarami et al., 2025, Xu et al., 2025], similar to traditional neural networks [Rahaman et al., 2019, Xu et al., 2019, Luo et al., 2019]. As a result, they favor learning low-frequency components of the solution function and often struggle to capture high-frequency features that iterative solvers excel at.

Recognizing this complementarity, Zhang et al. [2024] proposed HINTS, a hybrid solver that interleaves a neural operator pass every  $\tau^{\text{th}}$  iteration of a classical solver, with  $\tau$  fixed a priori. While HINTS reports significant empirical gains in convergence and accuracy over the standalone neural operator and Jacobi solver, this fixed schedule can be detrimental: a poorly timed neural correction may increase the error and undo recent progress.

Although state-dependent solver schedules are desirable, this naturally suggest a reinforcement learning formulation for learning routing policies. However, in our setting, the structure of iterative PDE solvers allows for a simpler approach: each solver induces a known update, and its effect can be evaluated through error-based signals. Leveraging this, we adopt a greedy rule that selects, at each iteration, the solver with the largest immediate error reduction. Such a rule can be near-optimal when the final error satisfies (weak) supermodularity—so that applying a beneficial solver earlier is at least as valuable as applying it later. To support this approach, we make the following contributions:

- We present a general hybrid PDE solver in which a routing rule chooses, at each iteration, from a set of classical solvers and neural operators. With oracle access to the true error at every step, we show that a greedy routing rule achieves **a constant fraction of the optimal error reduction** for linear PDEs, provided the updates are **non-expansive (Lipschitz with constant  $\leq 1$ , at least one of those in the optimal sequence strictly contractive)** and zero-preserving—conditions under which weak supermodularity holds—and **share a unitary eigenbasis, which gives the postfix monotonicity the guarantee also needs; for other ensembles the slack  $\mu$  is measured, over short horizons only.**
- Given that the true error is not observed at test time, a convex surrogate loss is introduced, which, when minimized, enables the learned model to imitate the greedy router without access to true error information. This alignment is guaranteed through Bayes consistency **of the surrogate for the full-information router (Section 5).**
- We evaluate the approximate greedy router against HINTS and individual solvers on the Poisson, convection-diffusion, anisotropic and variable-coefficient diffusion equations, **in iterations and in wall-clock time on  $128^2$  to  $512^2$  grids with compiled classical kernels and with the classical baselines timed in the same session. At truncation-level accuracy the router executes the same decision sequence as the best static schedule on 0–100% of the instances without a search, and its paired time ratios are  $0.94\times$ – $0.97\times$  against that schedule and  $0.95\times$ – $5.11\times$  against HINTS with its published period; below truncation level the ratios are  $0.92\times$ – $3.07\times$  against HINTS,  $0.97\times$ – $1373\times$  against the one-shot schedule and  $0.83\times$ – $1.61\times$  against the best fixed schedule chosen on the test set (faster in 6 of 24 pairings at  $128^2$ ), while the ratio of multigrid alone to the router is  $0.03\times$ – $20.5\times$  there. The construction extends to ensembles of classical solvers (Section E.7), which at  $128^2$  never beat the best single-member router (0 of 20 cells; work-unit ratios  $0.74\times$ – $1.00\times$  at  $10^{-8}$ ) and at  $256^2$  do ([pending] of [pending] cells, [pending]–[pending]; Poisson and convection–diffusion:  $1.23$ – $1.26\times$  on every instance).**

## 2 Background

Consider the following linear PDE:

$$\mathcal{L}_x^a(u) = f, x \in D; \quad \mathcal{B}_x(u) = b, x \in \partial D \quad (1)$$

where  $\mathcal{L}_x^a$  is a differential operator with respect to the spatial variable  $x \in D$  parameterized by the coefficient function  $a$ ,  $f$  is a forcing function,  $u$  is the PDE solution,  $\mathcal{B}_x$  is the boundary operator, and  $b$  is the boundary term. Such PDEs can be solved numerically using various discretization strategies, such as finite differences [Smith, 1985, LeVeque, 2007], finite element, and spectral methods [Boyd, 2001, Canuto et al., 2006]. Here, we focus on finite differences, which replace derivatives with difference quotients (e.g.,  $\partial_x u(x) \approx (u(x+h) - u(x))/h$ ) on a uniform grid.

Formally, let  $h$  be the grid size and  $G_h(D)$  be a uniform grid with spacing  $h$  over  $D$ . For a function  $g : D \rightarrow \mathbb{R}$ , we denote its restriction to  $G_h(D)$  by  $g_h \in \mathbb{R}^N$ , where  $N = |G_h(D)|$ . If  $\mathcal{L}_h^a \in \mathbb{R}^{N \times N}$

147 is the discretized operator, the PDE reduces to a linear system of equations

$$\mathcal{L}_h^a u_h = f_h, \quad (2)$$

148 with the boundary conditions incorporated in  $\mathcal{L}_h^a$ . Unlike much of the neural operator literature, which  
 149 retains function space formulations for discretization invariance, we follow the conventions of classical  
 150 numerical analysis and represent functions and operators as vectors and matrices. Discretization  
 151 invariance is not essential here, since we focus on fixed-grid problems.

## 152 2.1 Iterative PDE Solvers

153 Direct methods like Gaussian elimination and the Thomas algorithm can be computationally expensive  
 154 in high-dimensional domains when solving systems like Equation (2). In contrast, iterative methods  
 155 like Jacobi and Gauss-Seidel offer computational speedups by iteratively updating the solution,  
 156 gradually converging to the true solution for the PDE. The general iterative update is

$$u^{(t+1)} = u^{(t)} + C \left( f_h - \mathcal{L}_h^a u^{(t)} \right), \quad (3)$$

157 where  $u^{(t)}$  is the  $t^{\text{th}}$  iterate of the solution and  $C$  is a preconditioning matrix. Jacobi uses  $C = D^{-1}$ ,  
 158 where  $D$  is the diagonal of  $\mathcal{L}_h^a$ , and Gauss-Seidel uses  $C = (D + L)^{-1}$ , where  $L$  denotes the strict  
 159 lower triangular part. However, these methods tend to damp low frequency parts of the error slowly.

## 160 2.2 Neural Operators

161 Suppose we observe samples  $\{(a_i, f_i, u_i)\}_{i=1}^N$  such that  $(a_i, f_i) \sim \mathcal{P}$  are i.i.d. samples, and  $u_i =$   
 162  $\mathcal{G}^*(a_i, f_i)$  be generated by a deterministic solution operator  $\mathcal{G}^*$ . A neural operator (NO)  $\mathcal{G}_\theta$  seeks to  
 163 approximate  $\mathcal{G}^*$  by minimizing the expected squared  $L^2(D)$  error:

$$\mathcal{R}_{\text{NO}}(\mathcal{G}_\theta) = \mathbb{E}_{(a,f) \sim \mathcal{P}} \left[ \int_D \|\mathcal{G}^*(a, f)(x) - \mathcal{G}_\theta(a, f)(x)\|_2^2 dx \right]$$

164 Neural operators use discretization-invariant layers. Prominent instantiations include Fourier Neural  
 165 Operators (FNO), which use spectral transforms [Li et al., 2020a], Graph Neural Operators, which  
 166 perform graph-based aggregation over sampled points [Li et al., 2020b], and DeepONets, which  
 167 employ a trunk-branch decomposition to map input functions to output functions [Lu et al., 2021].

## 168 2.3 Greedy Optimization

169 Consider maximizing  $g(S)$  over  $S \subseteq \Omega$  with  $|S| \leq T$  where  $g : 2^\Omega \rightarrow \mathbb{R}$  is a set function defined  
 170 on subsets of a set  $\Omega$ . An exhaustive search is infeasible, so an efficient alternative is the greedy  
 171 algorithm. It builds the solution subset iteratively by adding  $\omega^*$  that yields the largest marginal  
 172 gain, i.e.  $\omega^* = \arg \max_{\omega \in \Omega \setminus S} g(S \cup \omega)$ . When  $g$  is non-negative, monotone (adding elements  
 173 never decreases the value), and submodular (diminishing returns), the greedy algorithm achieves a  
 174  $(1 - e^{-1})$  approximation to the optimal solution [Nemhauser et al., 1978]. Formally, a function  $g$   
 175 is submodular if  $g(A \cup \{\omega\}) - g(A) \geq g(B \cup \{\omega\}) - g(B)$  for all  $A \subseteq B \subseteq \Omega$  and  $\omega \in \Omega \setminus B$ ;  
 176 intuitively, delaying an addition cannot increase its benefit. For set function minimization problems,  
 177 supermodularity, where the inequality flips, plays an analogous role, and the greedy rule achieves  
 178 constant-factor approximation guarantees [Liberty and Sviridenko, 2017].

179 The suboptimality of the greedy algorithm has been extensively studied for both set-function mini-  
 180 mization [Bounia and Koriche, 2023] and maximization [Das and Kempe, 2018, Feige et al., 2011,  
 181 Bian et al., 2017, Harshaw et al., 2019] when standard assumptions—non-negativity, monotonicity, and  
 182 sub/supermodularity—are weakened. In contrast, results on greedy sequence maximization [Streeter  
 183 and Golovin, 2008, Alaei et al., 2021, Zhang et al., 2015, Bernardini et al., 2020, Van Over et al.,  
 184 2024, Tschitschek et al., 2017]—where the ordering of the elements affects the function—have led  
 185 to sequential analogues of submodularity and monotonicity. We leverage these tools to analyze the  
 186 suboptimality of the greedy solution to minimizing the final error.

## 187 3 General framework for Hybrid Solvers

188 To solve Equation (2), consider the following hybrid iterative update:

$$u_h^{(t+1)} = u_h^{(t)} + C_{S_t} \left( f_h - \mathcal{L}_h^a u_h^{(t)} \right) \quad (4)$$

where  $\mathcal{C} = \{C_j\}_{j=1}^K$  is a set of preconditioning functions and  $S_t \in [K] = \{1, \dots, K\}$  indexes the function chosen at step  $t$ . Here, we use “preconditioning function” broadly to include classical linear updates ( $C_j(x) = C_j x$  where  $C_j$  is a matrix), and learned models, such as neural operators. This update generalizes the classical iterative update in Equation (3), by allowing  $C_j$  to be non-linear and enabling the preconditioning function to be adaptively chosen at every step.  $\mathcal{C}$  can also accommodate parameterized solver families (e.g., different Jacobi relaxation weights), enabling adaptive selection of solver parameters. As the number of solvers  $K$  grows, the chance of selecting a more effective update increases. Furthermore, HINTS [Zhang et al., 2024] is a special case of this hybrid solver with  $K = 2$ , where  $C_1$  is a neural operator and  $C_2$  is a classical preconditioner. Its routing rule is given by  $S_t = \mathbf{1}_{(t+1) \bmod \tau > 0} + 1$ , which selects  $C_1$  every  $\tau$  steps (first at iteration  $\tau$ , after  $\tau - 1$  classical steps, as in Algorithm S3 of Zhang et al. [2024]) and  $C_2$  otherwise.

Let  $e_h^{(t)} = u_h - u_h^{(t)}$  denote the error at step  $t$ . Then,

$$e_h^{(t+1)} = u_h - u_h^{(t)} - C_{S_t} \left( \mathcal{L}_h^a u_h - \mathcal{L}_h^a u_h^{(t)} \right) = (I - C_{S_t} \circ \mathcal{L}_h^a) (e_h^{(t)})$$

where  $I$  is the identity map and  $I - C_j \circ \mathcal{L}_h^a$  is the error propagation function for solver  $j$ . Here, “ $\circ$ ” denotes function composition (matrix multiplication for linear updates). The objective is to select a sequence  $S = (S_1, \dots, S_T)$  that minimizes the error norm after  $T$  steps or  $\|e_h^{(T)}\|_2^2$ :

$$\min_{S \in [K]^T, |S| \leq T} h(S) := \left\| (I - C_{S_{|S|}} \circ \mathcal{L}_h^a) \circ \dots \circ (I - C_{S_1} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \quad (5)$$

with compositions applied from right to left, so that the  $S_1$  update acts on the initial error.

## 4 Greedy Algorithm

Equation (5) defines a combinatorial optimization problem with a search space exponential in  $T$ , making exact optimization intractable for large  $T$  or  $K$ . As a starting point, we consider an “omniscient” greedy algorithm that assumes access to the true initial error  $e_h^{(0)}$ . This is unrealistic since knowledge of  $e_h^{(0)}$  could directly recover the solution via  $u_h = u_h^{(0)} + e_h^{(0)}$ , but it provides a useful benchmark. In Section 5, we relax this assumption using a practical learning strategy that is Bayes consistent with this omniscient rule: for the full-information router, in the infinite-sample limit, it drives the normalised routing risk to its Bayes value, i.e. it recovers the greedy decision in the risk sense on the state distribution it is trained on, which is a per-decision statement (Section 5 states what the feature-based router inherits); the sequential guarantee below is stated for the rule with access to the true error.

As discussed in Section 2.3, when supermodularity and monotonicity hold, the greedy rule—such as the one described in Algorithm 1—enjoys constant-factor approximation guarantees. However, classical results focus on *set* functions, with only recent extensions made to sequences. Building on this line, we introduce a sequence-based notion of weak supermodularity.

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### Algorithm 1 Greedy Algorithm for a Hybrid PDE solver

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**Require:**  $\{C_j\}_{j=1}^K, T, \mathcal{L}_h^a, e_h^{(0)}$   
 $S^0 \leftarrow \emptyset$   
**for**  $t < T$  **do**  
 $S^{t+1} \leftarrow S^t \oplus \arg \min_{j \in [K]} \|(I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)})\|_2^2$   
 $e_h^{(t+1)} \leftarrow (I - C_{S^{t+1}} \circ \mathcal{L}_h^a)(e_h^{(t)})$   
**end for**  
**return**  $S^T$

---

Let  $\Omega^*$  denote the space of sequences with elements in  $\Omega$ . For  $S, S' \in \Omega^*$ , we denote their concatenation as  $S \oplus S'$ .  $S$  is a prefix of  $S'$  or  $S \preceq S'$  if  $S' = S \oplus L$  for some  $L \in \Omega^*$ . A function  $g : \Omega^* \rightarrow \mathbb{R}$  is prefix monotonically non-increasing if  $g(S \oplus S') \leq g(S)$  for all  $S, S' \in \Omega^*$ , and postfix monotonically non-increasing if  $g(S' \oplus S) \leq g(S)$  for all  $S, S' \in \Omega^*$ . A prefix non-increasing function  $g$  is sequence supermodular if, for all  $S', S \in \Omega^* : S \preceq S'$ , it holds that

$$g(S) - g(S \oplus \omega) \geq g(S') - g(S' \oplus \omega), \quad \forall \omega \in \Omega \quad (6)$$

225 However,  $h(S)$  (defined in Equation (5)) may not satisfy this property in general. Therefore, we  
 226 introduce weak sequence supermodularity. A prefix non-increasing function  $g$  is weakly supermodular  
 227 with respect to  $S' \in \Omega^*$  if there exists a single constant  $\alpha(S') \geq 1$  such that, for every  $S \in \Omega^*$ ,

$$g(S) - g(S \oplus S') \leq \alpha(S') \sum_{i \in [|S'|]} g(S) - g(S \oplus S'_i) \quad (7)$$

228 The parameter  $\alpha(S')$  or the supermodularity ratio quantifies deviation from exact sequence super-  
 229 modularity. Expanding the  $g(S) - g(S \oplus S')$  as a telescoping sum  $\sum_{i=1}^{|S'|} g(S \oplus (S'_1, \dots, S'_{i-1})) -$   
 230  $g(S \oplus (S'_1, \dots, S'_i))$  shows that the marginal decrease from appending  $S'_i$  after its predecessors is  
 231 controlled by the effect of appending  $S'_i$  directly to  $S$ . Thus, postponing the inclusion of  $S'_i$  cannot  
 232 yield a significantly larger benefit than adding it earlier. The supermodularity ratio  $\alpha(S)$  quantifies  
 233 the extent to which future gains from delays may exceed immediate gains.

234 Postfix monotonicity enters the analysis only through the greedy prefixes, so we also allow a slack:  
 235  $g$  is  $\mu$ -postfix monotone with respect to  $O$  along a sequence  $S^T$  if  $g(S^t \oplus O) \leq \mu g(O)$  for every  
 236 prefix  $S^t, t < T$ ; a postfix non-increasing function satisfies this with  $\mu = 1$ , and  $\mu$  can be computed  
 237 exactly for short horizons (Section E.10). Having introduced these notions, we now characterize the  
 238 suboptimality of greedy solutions of weakly supermodular and postfix monotonic sequence functions  
 239 in Theorem 4.1.

240 **Theorem 4.1.** Let  $g : \Omega^* \rightarrow \mathbb{R}$  be a non-negative, prefix non-increasing function that is weakly  
 241 supermodular with respect to the optimal solution  $O = \arg \min_{S \in \Omega^T} g(S)$  with a supermodularity  
 242 ratio of  $\alpha(O)$ , and let the greedy solution of length  $T$  be  $S^T$ . Suppose  $g$  is  $\mu$ -postfix monotone with  
 243 respect to  $O$  along  $S^T$  ( $\mu = 1$  if  $g$  is postfix non-increasing). If  $\phi_T(\alpha) = (1 - \frac{1}{\alpha T})^T$ , then

$$g(S^T) \leq \mu (1 - \phi_T(\alpha(O))) g(O) + \phi_T(\alpha(O)) g(\emptyset), \quad \text{i.e.,} \quad g(\emptyset) - g(S^T) \geq (1 - \phi_T(\alpha(O))) (g(\emptyset) - \mu g(O)).$$

244 The proof of Theorem 4.1 appears in Section A.2. As  $T \rightarrow \infty$ , the factor  $1 - \phi_T(\alpha(O))$  decreases  
 245 to  $1 - e^{-1/\alpha(O)}$ , so, for a fixed  $\alpha$ , the worst-case performance of the greedy rule saturates rather  
 246 than degrades with horizon. Since  $\alpha(O)$  and  $\mu$  are defined through the optimal sequence of length  
 247  $T$ , a horizon-uniform statement needs uniform bounds  $\alpha(O_T) \leq A$  and  $\mu_T \leq M$ , under which the  
 248 guaranteed reduction is at least  $(1 - e^{-1/A}) [g(\emptyset) - M g(O_T)]$  whenever that bracket is nonnegative,  
 249 and only the trivial nonnegative reduction of prefix monotonicity otherwise. The bound concerns  
 250 the reduction of the error over  $T$  steps relative to the best sequence of  $T$  steps: with  $\alpha = \mu = 1$  it  
 251 guarantees a  $(1 - e^{-1})$  fraction of the optimal reduction, uniformly over ensembles, but it is not a  
 252 convergence rate—a single convergent solver applied  $T$  times already reduces the error by  $\rho^{2T}$ , which  
 253 for large  $T$  is far below the  $e^{-1} g(\emptyset)$  term. Section E.10 reports, for short horizons, how far the greedy  
 254 sequence is from the exhaustive optimum in the settings of Section E. Additionally, larger  $\alpha(O)$ ,  
 255 which indicates higher reward for delayed inclusions, loosens the suboptimality bound. Theorem 4.1  
 256 requires that the sequence objective  $h$  be weakly supermodular with respect to the optimal solution  
 257 and postfix monotone—properties established in Proposition 4.2. We defer the proof to Section A.3.

258 **Proposition 4.2.** Suppose that for all  $j \in [K]$ , the error propagation function  $I - C_j \circ \mathcal{L}_h^a$  is  
 259  $\rho_j$ -Lipschitz continuous with  $\rho_j \leq 1$ , that  $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$ , and that  $\sum_{i=1}^T (1 - \rho_{O_i}^2) > 0$ ,  
 260 i.e., at least one action of the optimal sequence  $O$  is a strict contraction. Then, the function  $h$  is  
 261 weakly supermodular with respect to the optimal solution  $O$ , with

$$\alpha(O) = \max \left\{ \frac{1}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}$$

262 Furthermore, if the error propagation maps are linear and simultaneously diagonalizable by a  
 263 unitary matrix with eigenvalues of modulus at most one (the setting of Theorem 4.3),  $h$  is also postfix  
 264 non-increasing; invertibility alone does not suffice (Section A.3).

265 The conditions of Proposition 4.2 are natural. The corrector of Section 7 leaves the modes outside its  
 266 band untouched and has  $\rho = 1$  exactly; the proposition permits this provided the optimal sequence  
 267 contains at least one action that is a strict contraction in the norm of  $h$  (the Euclidean norm) on  
 268 a subspace that contains the initial error and is invariant under every action. For the Jacobi-type  
 269 smoothers the mean-free, Nyquist-free subspace is such a subspace and they are strict contractions  
 270 on it; for Gauss–Seidel, SOR and multigrid neither the invariance of that subspace nor contraction

in the Euclidean norm is established in our settings (SOR(1.5) expands on it, Section E.10), so for them Proposition 4.2 in the norm of  $h$  is not established. For the exact common-unitary case Proposition 4.3 applies without any strict contraction. For classical solvers, the Lipschitz constant is  $\|I_N - C_j \mathcal{L}_h^a\|$ , which is typically less than 1 for well-posed linear elliptic PDEs (i.e., the update is damping errors). For neural networks, Lipschitz continuity can be enforced via weight regularization [Gouk et al., 2021] and a sufficiently trained model should approximate  $(\mathcal{L}_h^a)^{-1}$  well enough to make the Lipschitz constant small. The requirement  $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$  is both natural and desirable: it precludes spurious updates when the residual  $f_h - \mathcal{L}_h^a u_h^{(t)}$  is 0. This holds by design for classical schemes, and it can be enforced for any learned model by excluding bias terms.

The form of  $\alpha(O)$  in Proposition 4.2 highlights that the suboptimality factor in Theorem 4.1 is governed by the collective contraction factors of the solvers chosen by the optimal solution: as  $\sum_{i=1}^T \rho_{O_i}^2 \rightarrow T$ ,  $\alpha(O)$  grows, resulting in a weaker bound. Postfix monotonicity ( $\mu = 1$ ) holds when the error propagation maps share a unitary eigenbasis—constant-coefficient periodic operators with Jacobi-type smoothers and band-limited spectral correctors—but it can fail for non-commuting contractions even when all of them are invertible, and a Gauss–Seidel sweep or a learned corrector need not be invertible at all. For the non-commuting solvers of our experiments no a priori bound on  $\mu$  is available; Section E.10 computes  $\mu$ ,  $\alpha(O)$  and the bound of Theorem 4.1 by exhaustive enumeration for short horizons ( $T \leq 4$  macro-actions) on every pairing and on four representative ensembles, including ones with Gauss–Seidel, SOR and multigrid, and reports the largest  $\mu$  (1.104), the largest candidate  $\alpha(O)$  over greedy prefixes (1.003) and the number of instance–ensemble pairs on which the on-path premises of the proof were verified numerically (183/296); at the horizons of tens to hundreds of macro-actions at which the oracle actually runs, neither  $\alpha$  in the Euclidean norm nor  $\mu$  is established for the non-normal sweeps, and no guarantee is claimed for them.

Theorem 4.1 thus indicates that the approximation guarantee is strongest when the sequence is nearly supermodular ( $\alpha(O) \approx 1$ ). Generally, if all error propagation maps share an eigenbasis, supermodularity is established. This occurs, for example, for linear, constant-coefficient PDEs with periodic boundary conditions when the solver ensemble consists of (damped) Jacobi iterations and band-limited spectral correctors such as a single-layer linear Fourier neural operator, all of which are diagonal in the Fourier basis; lexicographic Gauss–Seidel and SOR are not, because their sweep order breaks translation invariance. The proof of Theorem 4.3 is deferred to Section A.5.

**Proposition 4.3.** *Let  $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$  for all  $j \in [K]$  and  $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$  with a common unitary matrix  $P$  (orthogonal in the real symmetric case; the Jacobi matrix of the convection–diffusion operator is a real circulant with complex eigenvalues, hence unitarily but not orthogonally diagonalisable). Then,  $h$  is supermodular.*

**Cost-aware greedy routing.** Algorithm 1 treats every update as equally expensive, whereas in practice a neural-operator evaluation and a classical sweep have very different costs, and the quantity of interest is the error reached per unit of wall-clock time. We make the greedy rule cost-aware by changing its action set, not its comparison: let  $c_j$  denote the (measured) cost of one application of  $C_j$ , take the cost  $u$  of one neural-operator call as the unit, and replace each preconditioner by the macro-action  $C_j^{(m_j)}$  that applies  $C_j$  for  $m_j = \max\{1, \text{round}(u/c_j)\}$  consecutive iterations, so that every action not dearer than the unit costs  $u$  up to rounding ( $m_j c_j / u$  within  $1/(2m_j - 1)$  of 1). For a linear update,  $m$  consecutive sweeps are again a stationary iteration,  $(I - C_j \circ \mathcal{L}_h^a)^m = I - C_j^{(m)} \mathcal{L}_h^a$  with  $C_j^{(m)} = \sum_{i=0}^{m-1} (I - C_j \mathcal{L}_h^a)^i C_j$ , and for a nonlinear  $C_j$  the macro-action is the  $m$ -fold composition of its error propagation map. Hence  $\{C_j^{(m_j)}\}_{j=1}^K$  is again a set of preconditioning functions in the sense of Section 4: if  $I - C_j \circ \mathcal{L}_h^a$  is  $\rho_j$ -Lipschitz and zero-preserving, then so is its  $m_j$ -fold composition (with constant  $\rho_j^{m_j}$ ), a shared eigenbasis is inherited by powers, and Theorem 4.1, Propositions 4.2 and 4.3, and the consistency result of Section 5 apply verbatim to an ensemble of macro-actions, all of which are equalised to the unit cost. The deployed rule is exactly Algorithm 1 on these macro-actions—the plain comparison of the errors after one macro-action—and a horizon of  $T$  macro-actions corresponds to a wall-clock budget of  $\approx Tu$ ; it selects, at every decision epoch, the action with the largest error reduction per unit cost up to the rounding of the  $m_j$ . The one exception is an action dearer than the unit ( $c_j > u$ , hence  $m_j = 1$ ), whose error reduction is compared per unit of cost,  $(\|e_j\|/\|e^{(t)}\|)^{u/c_j}$ : this is a geometric-rate heuristic rather than an equal-cost step, and the analysis does not cover it. Which actions are dearer than the unit is a fact of the measured cost

tables (Tables 7, 13, 19 and 25): on the four  $128^2$  grids none is, so those cells run Algorithm 1 on macro-actions, which is also the rule that the exhaustive check of Section E.10 verifies; on the larger grids only a multigrid cycle can be dearer than the unit (**pending**), and in the granularity ablation of Section E.6 the corrector itself is. Even with all exponents equal to one, the wall-clock reading of a horizon of  $T$  macro-actions is approximate: the rounding of the  $m_j$  and the router's decision cost are not part of the equal-count analysis. (An earlier version applied the exponent  $u/(m_j c_j)$  to every action, including the rounding error of the cheap ones; that rule is not Algorithm 1, and it was changed so that the theory covers what is deployed.) Section E uses this cost-aware instantiation throughout.

## 5 Approximate Greedy Router

The results in Section 4 indicate that the error reduction from the greedy solution closely matches that of the optimal sequence, but this is predicated on having access to the initial error,  $e_h^{(0)}$ . Inaccurate error estimates can result in poor solver decisions, causing errors to amplify. To remedy this, we learn a router  $r$  that selects solvers myopically, as in Algorithm 1, without access to true errors.

We adopt the following learning setup. Let  $\mathcal{A}, \mathcal{F}, \mathcal{U} \subseteq \mathbb{R}^N$  denote the spaces of coefficient, forcing, and solution functions on the grid  $G_h(D)$ . We assume an application-specific data distribution  $\mathcal{P}_{\mathcal{A} \times \mathcal{F}}$  over  $\mathcal{A} \times \mathcal{F}$  that reflects test time conditions. During training,  $(a_h, f_h)$  is drawn from  $\mathcal{P}_{\mathcal{A} \times \mathcal{F}}$ , and a high-accuracy reference solution  $u_h$  is computed, providing the true per-step error. The router  $r$  is learned offline on this data; at test time, it operates without access to true errors.

At iteration  $t$ , router  $r$  selects a solver using the coefficient  $a_h \in \mathcal{A}$ , forcing  $f_h \in \mathcal{F}$ , and the current iterate  $u_h^{(t)} \in \mathcal{U}$ . If  $r(a_h, f_h, u_h^{(t)}) = j$ , the next iterate is computed with solver  $j$ , resulting in an error of  $\|(I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)})\|_2^2$ . Learning such a router requires minimizing the following loss:

$$l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) = \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)})=j} \quad (8)$$

with risk  $\mathcal{R}_{\text{route}}(r) = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} [l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h)]$ . For analysis, we fix the iterate generation via teacher-forcing [Williams and Zipser, 1989, Lamb et al., 2016]: during training, the iterates fed to the router are produced by an oracle greedy rollout, not by the router's own past choices. Formally, with  $u_h^{(0)} = 0_N$ , for  $t \geq 1$ ,

$$j_t^* \in \operatorname{argmin}_{j \in [K]} \left\| (I - C_j \circ \mathcal{L}_h^a) (e_h^{(t-1)}) \right\|_2^2, \quad u_h^{(t)} = u_h^{(t-1)} + C_{j_t^*} (f_h - \mathcal{L}_h^a u_h^{(t-1)})$$

Thus,  $\{u_h^{(t)}\}$  is a deterministic function of  $(a_h, f_h)$ . At each  $t$ ,  $l_{\text{route}}$  is evaluated on  $r(a_h, f_h, u_h^{(t)})$  which takes in the teacher-forced iterate. Finally, the greedy choice  $j_{t+1}^*$  is used to advance  $u_h^{(t+1)}$ . Thus the input distribution seen by  $r$  depends on  $(a_h, f_h)$ , not on the router's predictions. However, full teacher forcing induces a distributional mismatch at test time (exposure bias). In experiments, we mitigate this with scheduled sampling [Bengio et al., 2015]; see Section C for details.

Minimizing  $\mathcal{R}_{\text{route}}(r)$  yields the following Bayes-Optimal Router:

$$r^*(a_h, f_h, u_h^{(t)}) \in \operatorname{argmin}_{j \in [K]} \left\| (I - C_j \circ \mathcal{L}_h^a) (e_h^{(t)}) \right\|_2^2 \quad (9)$$

which aligns with the update in Algorithm 1. Hence,  $l_{\text{route}}$  is consistent with learning the greedy rule.

### 5.1 Surrogate Loss

Since Equation (8) is discontinuous and non-convex, direct minimization is intractable in practice. We instead introduce a surrogate loss that (1) is convex **in the scores**; (2) upper bounds the original loss; and (3) is Bayes consistent, ensuring the Bayes optimal decision of the original loss is preserved upon minimization. Formally, a surrogate  $\phi$  is considered to be Bayes consistent with respect to the loss  $l$  if

$$\lim_{n \rightarrow \infty} \mathcal{R}_\phi(f_n) - \mathcal{R}_\phi^* = 0 \implies \lim_{n \rightarrow \infty} \mathcal{R}_l(f_n) - \mathcal{R}_l^* = 0$$

where  $\mathcal{R}_l(f) = \mathbb{E}[l(f(X), Y)]$  and  $\mathcal{R}_l^* = \inf_f \mathcal{R}_l(f)$ . In other words, in the limit of infinite data, if the risk of a sequence of learned hypotheses  $\{f_n\}$  converges to the optimal risk under  $\phi$ , it also converges to the optimal risk with respect to the original loss  $l$ .

To define a surrogate loss for the routing problem, consider a set of scoring functions  $\mathbf{g} = \{g_j\}_{j=1}^K$  with  $g_j : \mathcal{A} \times \mathcal{F} \times \mathcal{U} \rightarrow \mathbb{R}$  and we define the router as  $r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$ . For example,  $\mathbf{g}$  can be a neural network with  $K$  outputs. Then, minimizing the following surrogate loss yields the same decision as minimizing Equation (8):

$$\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h) = - \sum_{j=1}^K \sum_{k=1}^K \tilde{c}_k(a_h, u_h^{(t)}, u_h) \mathbf{1}_{k \neq j} \log \left( \frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{m=1}^K \exp(g_m(a_h, f_h, u_h^{(t)}))} \right) \quad (10)$$

where  $\tilde{c}_j(a_h, u_h^{(t)}, u_h) = \|(I - C_j \circ \mathcal{L}_h^a)(u_h - u_h^{(t)})\|_2^2$ . The  $\Psi$ -risk is denoted by  $\mathcal{R}_\Psi(\mathbf{g}) = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}}[\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)]$ . The convexity of  $\Psi$  with respect to the score vector  $\mathbf{g}$  follows from the convexity of the negative log-softmax function in its inputs (the surrogate is convex in the scores, not in the parameters of a network that produces them). Moreover,  $\Psi$  upper bounds  $l_{\text{route}}$  up to a constant factor, and we refer the reader to Section B.2 for the proof. Finally, Theorem 5.1 shows that the per-state normalised surrogate  $\hat{\Psi}$  is Bayes consistent with respect to the normalised routing loss  $\hat{l}_{\text{route}}$ , which has the same Bayes decision as  $l_{\text{route}}$  at every state; the proof can be found in Section B.3.

**Theorem 5.1.** *Let  $K \geq 2$  and let the per-step costs be normalised state by state,  $\hat{c}_j = \tilde{c}_j / \sum_{k=1}^K \tilde{c}_k$ , at every state at which  $\sum_k \tilde{c}_k > 0$ , i.e., at which at least one solver does not annihilate the error (states at which every solver annihilates it carry no routing loss and are discarded), and let  $\hat{l}_{\text{route}}$  and  $\hat{\Psi}$  denote the routing loss and the surrogate with  $\tilde{c}_j$  replaced by  $\hat{c}_j$ ; the normalisation does not change the Bayes-optimal decision equation 9 at any state. Then, for any collection of solvers  $\{C_j\}_{j=1}^K$  and linear discrete operator  $\mathcal{L}_h^a$ ,  $\hat{\Psi}$  is a Bayes consistent surrogate for  $\hat{l}_{\text{route}}$ .*

Theorem 5.1 is a cost-sensitive analogue of the classical Bayes-consistency of multiclass cross-entropy for 0-1 loss: in the infinite-sample limit, minimizing cross-entropy recovers the true conditional class probabilities, so the induced decision is Bayes optimal. Our result extends this to cross-entropy with instance-dependent weights  $\sum_{k \neq j}^K \hat{c}_k(a_h, u_h^{(t)}, u_h)$ . The normalisation is what the router is trained with (Section E.1) and what makes the calibration constants meaningful: an earlier version assumed uniform absolute bounds  $E_{\min} \leq \max_j \tilde{c}_j \leq \bar{E}$ , which cannot hold along a convergent solve, where every  $\tilde{c}_j \rightarrow 0$ ; with the normalised weights the only hypothesis is that at least one solver leaves a positive fraction of the error at every visited state, which Section E.10 checks. Under this condition, minimizing  $\hat{\Psi}$  recovers the Bayes-optimal router of Equation (9), i.e., the greedy solution of Equation (5). The proof bounds the conditional routing excess by the square root of the conditional surrogate excess through Pinsker’s inequality, a direct calibration argument in the spirit of Mao et al. [2023], and takes expectations; it concerns unrestricted measurable scores, not a fixed network class or optimiser. This is a per-decision statement on the state distribution the router is trained on: it says that minimising  $\hat{\Psi}$  drives the normalised routing risk to its Bayes value, i.e. recovers the greedy decision in the risk sense, not that the induced sequence, its time to tolerance, or its distance to the optimal sequence are controlled; the latter would require a sequential argument about the router’s own state distribution, which we do not provide and mitigate empirically with DAgger (Section E). The deployed router, moreover, does not see  $(a_h, f_h, u_h^{(t)})$  but a few scalar features of the residual history (Section E.1); with respect to those features the Bayes-optimal router minimises the conditional expectation of the normalised costs given the features, which can differ from the full-information greedy decision when states with different error spectra share the same features, and the normalisation preserves the decision state by state but not necessarily after that averaging. Theorem 5.1 therefore concerns the full-information router, and the price of the feature compression is measured rather than proved, by the agreement between the deployed router and the oracle (Section E).

## 6 Related Works

**Hybrid PDE Solvers:** Early data-driven solvers sought convergence guarantees by predicting parameters—e.g., preconditioning matrices, multi-grid smoothers or restriction matrices—within itera-

tive schemes [Taghibakhshi et al., 2021, Caldana et al., 2024, Kopaničáková and Karniadakis, 2025, Huang et al., 2022, Katrutsa et al., 2020]. These works, however, do not leverage the neural surrogates’ ability to generalize across varying coefficients and forcings highlighted in Section 2.2 and thus, offer only modest speedups over classical solvers. HINTS [Zhang et al., 2024] introduced hybrid solvers that interleave a classical method with a pre-trained DeepONet on a fixed schedule (e.g., 24 Jacobi steps, then one DeepONet correction), achieving faster convergence than the standalone numerical solver. Subsequent works extend this idea to new geometries [Kahana et al., 2023] and analyze error mode damping, replacing DeepONet with alternatives such as MIONet [Jin et al., 2022] [Hu and Jin, 2025] or FNO [Li et al., 2020a, Cui et al., 2022]. However, these hybrid solvers are limited in two ways: they, firstly, only combine the trained surrogate with a *single* numerical solver, and secondly rely on a fixed, heuristic schedule. Our proposed method addresses these shortcomings, resulting in significant empirical improvements (Section 7).

**Model Routing:** Routing [Shnitzer et al., 2023, Hu et al., 2024, Ding et al., 2024, Huang et al., 2025] selects, for each input, a model from a fixed set to optimize a task metric under cost or latency constraints. Simple heuristics—e.g., thresholding a cheap model’s uncertainty estimates [Chuang et al., 2024, 2025]—often poorly balance the cost-accuracy tradeoff, motivating many systems to instead learn a router that maps inputs to the best model [Hari and Thomson, 2023, Mohammadshahi et al., 2024, Šakota et al., 2024]. We similarly learn a router, but over numerical and neural solvers for PDEs at the iteration level. Our work is the first to apply routing to the problem of learning hybrid PDE solvers, in contrast to prior approaches that use fixed schedules. Additionally, by exploiting the algebraic structure of PDE solvers, we derive theoretical guarantees for our routing strategy (Section 4), unlike typical model-routing settings.

**Mixture of Experts:** Classical mixture-of-experts (MoE) [Jacobs et al., 1991, Jordan and Jacobs, 1994] uses a gating network to combine the outputs of multiple experts, trained jointly with the gate. In modern LLMs, MoE instead performs sparse, token-level routing to a subset of in-layer experts, enabling large capacity without proportional compute, typically with load-balancing mechanisms [Shazeer et al., 2017, Lepikhin et al., 2020, Fedus et al., 2022, Jiang et al., 2024]. Our method can be regarded as gating of a different kind: “experts” (solvers) are not jointly trained with the router, and a single router is reused across iterations, unlike MoE which commonly employs layer-specific gates.

## 7 Experiments

We evaluate the approximate greedy router on four PDEs posed on the unit domain  $D = [0, 1]^2$  with periodic boundary conditions:

**Poisson:**  $-\Delta u = f$ , **ConvDiff:**  $-\Delta u + 20 \partial_{x_1} u + 20 \partial_{x_2} u = f$ , **AnisoDiff:**  $-\epsilon \partial_{x_1}^2 u - \partial_{x_2}^2 u = f$ ,  $\epsilon = 10^{-2}$ , **VarCoef**

where  $f$  is centred to satisfy the compatibility condition  $\int_D f(\mathbf{x}) d\mathbf{x} = 0$  and is drawn from a zero-mean hierarchical Gaussian random field with covariance operator  $\alpha(-\Delta + \beta I)^{-\gamma}$  and sample-dependent  $(\alpha, \beta, \gamma)$  (Section C). Anisotropic diffusion is the standard problem on which point relaxation fails to smooth: an error mode oscillatory in  $x_1$  and smooth in  $x_2$  is damped by only  $(1 - \epsilon)/(1 + \epsilon) \approx 0.98$  per Jacobi sweep. Variable-coefficient diffusion uses a fixed smooth coefficient field  $a(\mathbf{x}) = \exp(\kappa z(\mathbf{x}))$  whose contrast  $\max a / \min a$  is at most 10 by construction (9.99 measured): its discrete operator is not diagonalised by the FFT, so no direct spectral solve exists, the corrector’s band-limited inverse has no closed form and is fitted, and the fitted corrector is only approximately accurate (Section E.5). The domain is discretised on  $128^2$ ,  $256^2$  and  $512^2$  grids; Dirichlet boundary conditions are not covered by this study (Section D). Because iteration counts favour methods whose iterations are expensive, our primary metric is the *wall-clock time to a target relative error* charged to each method in a replay of its decisions (Section E.1)  $\varepsilon = \|e_h^{(t)}\|/\|u_h\|$  on 64, 32 and 16 test instances per grid, with  $\varepsilon = h^2$ —a grid-dependent tolerance chosen to lie below the discretisation error of the test instances (the smallest ratio of the estimated discretisation error to  $h^2$  over the confirmatory instances is **[pending]**, Section E.1)—highlighted; paired comparisons of times use two-sided Wilcoxon signed-rank tests with a Holm correction and bootstrap intervals, the AUC comparisons one-sided paired  $t$ -tests on log AUC. We also report the iteration-based metric of the original submission, the area under the error curve,  $\text{AUC} = \sum_{t=1}^T \|e_h^{(t)}\|/\|u_h\|$  over  $T = 300$  iterations. Section E gives the full protocol and collects every result per equation.

**Setup.** The classical solvers are Jacobi, damped Jacobi ( $\omega = 0.67$ ), Gauss–Seidel (GS), symmetric GS (SymGS), SOR ( $\omega = 1.5$ ), line Gauss–Seidel (anisotropic diffusion only: every grid line along the strongly coupled axis solved exactly, the classical remedy for anisotropy) and geometric multigrid (V(2,2)-cycles with GS smoothing; line-GS smoothing for anisotropic diffusion), all implemented as compiled single-threaded C stencil kernels (a GS sweep costs  $61 \mu\text{s}$  and a V-cycle  $326 \mu\text{s}$  at  $128^2$ , measured in the form in which the benchmark charges them), so the classical baselines are not slowed by the implementation. The neural operator is a DeepONet corrector that acts on the residual through a  $64 \times 64$  sensor grid with a Fourier trunk basis and is applied scale-equivariantly: its deployed form is a linear, Fourier-structured map (a least-squares fit of the band-limited inverse of the discrete operator, which it reproduces to  $\approx 10^{-6}$  on the constant-coefficient equations), so one call solves the smooth band of the spectrum at the cost of about 32 Jacobi or 11 GS sweeps at  $128^2$  (and 6 Jacobi sweeps at  $512^2$ , since its cost is nearly grid-independent) and never injects error outside its band. The router is made *cost-aware* through the cost-equalised macro-actions of Section 4 and selects from  $6+K$  scalar features (log residual, its recent decay rate, the previous action and call counts) with a two-layer MLP whose decision costs  $7\text{--}21 \mu\text{s}$  in situ, independently of  $N$ ; it is trained with the surrogate of Section 5.1 to imitate the cost-aware greedy oracle on oracle rollouts followed by DAgger rounds. We compare against the classical solver alone; HINTS with the period  $\tau=15$  that Zhang et al. [2024] use in all of their two-dimensional experiments ( $\tau=25$ , their one-dimensional period, is reported in Section E); the strongest fixed schedule chosen *a posteriori* on the test set for each cell (the schedule with the smallest paired time ratio to the router) from HINTS with  $\tau \in \{2, 5, 10, 15, 25, 50\}$ , phase-shifted HINTS (first call at iteration 0) and the static one-shot schedule (corrector once at iteration 0, then the solver); the best of three residual-decay rules—deterministic controls at the router’s granularity that call the corrector again whenever the residual’s per-iteration contraction over the last macro-action was worse (closer to one) than a threshold, also chosen on the test set; the cost-aware greedy oracle (Algorithm 1 with the true error; not deployable); and the classical methods a practitioner would use on these problems without a corrector—multigrid alone (line-smoothed for anisotropic diffusion), multigrid-preconditioned CG or BiCGSTAB, and the FFT or sparse direct solve—timed inside the same replay loop as the policies. The forcings are sampled from the hierarchical GRF with the Hermitian symmetry of a real field and without Nyquist modes (Section E.1). All configurations were fixed on a development test set, and every number reported is a single confirmatory evaluation on a fresh test set with three timed replays in random order and a drift guard per instance (Section E.9); paired comparisons use two-sided Wilcoxon tests with a Holm correction and bootstrap intervals.

**Paired solvers.** Table 1 reports, for every pairing  $\{\text{NO}, \text{solver}\}$ , equation and grid, the median wall-clock time to  $\varepsilon = h^2$  of HINTS, of the best fixed schedule, and of the learned router, together with the classical methods without corrector. Every ratio below is a paired median time ratio (baseline over router; above one favours the router; the one exception, the router-to-oracle ratio, is a ratio of medians) computed over the  $128^2$  pairwise cells unless a grid is named, and “faster in  $n$  of  $m$ ” counts pairings whose ratio exceeds one irrespective of significance (the tests are in Section E). Two regimes appear. *At truncation level* the router makes exactly one corrector call on 80–100% of the instances (Table 6), executes the same decision sequence to  $h^2$  as the cost-aware oracle on 50–100% of the instances and as the strongest static schedule ( $\tau=10$  (first call at 0) (8),  $\tau=25$  (first call at 0) (6), one-shot (4),  $\tau=50$  (first call at 0) (3),  $\tau=5$  (2),  $\tau=5$  (first call at 0) (1) at  $h^2$ , pairings in parentheses) on 0–100% (zero where that schedule opens with sweeps rather than a call), and its ratios against the strongest fixed schedule of the test-set search are  $0.94 \times \text{--} 0.97 \times$  at  $128^2$  ([pending]–[pending] at  $256^2$ ,  $0.94 \times \text{--} 1.01 \times$  at  $512^2$ ), a difference consistent with its decision overhead (Section E.8), and  $0.95 \times \text{--} 1.43 \times$  against the schedule selected on the development instances instead, the deployable choice (Section E.1; faster in 2 of 24 pairings, the two selections coinciding in 2); against HINTS with its published period  $\tau=15$ , whose first call comes  $\tau - 1$  sweeps too late, they are  $0.95 \times \text{--} 5.11 \times$  ([pending]–[pending] at  $256^2$ ,  $1.10 \times \text{--} 1.86 \times$  at  $512^2$ ;  $0.95 \times \text{--} 7.08 \times$  against  $\tau=25$ ); against the best residual-decay rule, which also calls the corrector first,  $0.95 \times \text{--} 0.97 \times$ ; and against the classical solver alone  $0.95 \times \text{--} 7083 \times$  (censored runs entered at the cap; the low end is the Multigrid pairing). *Below truncation level* the schedule matters and the picture is pairing-dependent: down to  $\varepsilon = 10^{-8}$  the ratios are  $0.92 \times \text{--} 3.07 \times$  against HINTS ( $\tau=15$ ; the router is faster in 22 of 24 pairings) and  $0.97 \times \text{--} 1373 \times$  against the one-shot schedule (faster in 21), because the router calls the corrector again when the smooth error has re-emerged (Figure 1), but  $0.83 \times \text{--} 1.61 \times$  against the strongest fixed schedule (faster in 6;  $0.83 \times \text{--} 1.62 \times$  against the development-selected schedule, faster in 8) and  $0.98 \times \text{--} 2.07 \times$  against the best residual-decay rule (faster in 21). The best fixed schedules at  $10^{-8}$  are  $\tau=15$  (first call at 0) (7),  $\tau=10$  (first call at 0) (7),  $\tau=5$  (first call at 0) (4), one-shot (3),  $\tau=50$

(first call at 0) (1),  $\tau=25$  (first call at 0) (1),  $\tau=10$  (1) (number of pairings in parentheses): a periodic schedule with a short period calls the corrector more often than a cost-equalised macro-action of 11 Gauss–Seidel or 32 Jacobi sweeps allows the router to, and the losses are confined to that case: in the 10 pairings whose best schedule has a period shorter than the macro-action the router is faster in 0 (ratios  $0.83\times$ – $0.98\times$ ), in the other 14 it is faster in 6 ( $0.95\times$ – $1.61\times$ , median  $0.99\times$ ); Section E.6 tests whether a finer unit removes the losses **[pending]**. The ratio of the router’s median time to the cost-aware oracle’s is  $1.01\times$ – $1.09\times$ , and its decision sequence to  $10^{-8}$  is identical to the oracle’s on 30–94% of the instances; the stability of its decisions over five independent training seeds is reported in Section E **[pending]**. Against the classical methods without corrector, the ratio of multigrid alone to the router is  $0.95\times$ – $11.2\times$  at  $h^2$  and  $0.03\times$ – $20.5\times$  at  $10^{-8}$ , that of multigrid-preconditioned CG or BiCGSTAB  $0.95\times$ – $3.00\times$  and  $0.03\times$ – $3.46\times$ , and routing over {NO, multigrid} has ratios  $0.94\times$ – $2.30\times$  and  $1.11\times$ – $1.63\times$  against multigrid alone; on the constant-coefficient equations the router takes  $3.08\times$ – $3.25\times$  times as long as the FFT direct solve, and on variable-coefficient diffusion, which has no such solve, it is compared with the sparse direct solve (Section E.5). For the iteration-based metric of the original submission, at  $128^2$  the mean AUC over  $T = 300$  iterations of HINTS and of the solver alone are  $3\text{--}174709\times$  and  $3\text{--}2666251\times$  the router’s (one-sided paired  $t$ -tests on log AUC,  $p \leq 10^{-10}$ ; Tables 8, 14, 20 and 26).

**Solver ensembles.** The same construction routes over ensembles  $\text{NO} \cup \mathcal{W}$  of several classical solvers. Since the greedy rule is a per-decision comparison, a larger ensemble can only help where the error passes through regimes that favour *different* members; an oracle-level screening of every subset of up to three members (Section E.7) showed, in the development runs, that this never happens at  $\varepsilon = h^2$ —one corrector call and one macro-action of the cheapest smoother already reach truncation level—and happens at  $10^{-8}$  only through a nonstationary Jacobi schedule (undamped Jacobi on the band-edge modes, damped Jacobi on the near-Nyquist modes). We evaluate the learned router over the nested sets {Jacobi}, {Jacobi (0.67)}, {Jacobi, Jacobi (0.67)} and {Jacobi, Jacobi (0.67), GS} and the four larger sets of Table 30 on every equation, with the single-member routers evaluated in the same session on the same instances. At  $128^2$  no ensemble is significantly faster than the best single-member router (0 of 20 cells) and the ensemble oracle is  $0.95\times$ – $1.02\times$  of the best single-member oracle in work units, which we attribute to the decision granularity: a single damped-Jacobi macro-action of 32 sweeps contracts the band-edge modes by  $0.665^{32} \approx 2 \times 10^{-6}$ , which suggests that the band the corrector leaves at  $128^2$  is removed within one or two macro-actions, leaving a nonstationary schedule little to exploit. The ensemble routers’ work-unit ratios against the best single-member router are  $1.00\times$ – $1.00\times$  at  $h^2$  and  $0.74\times$ – $1.00\times$  at  $10^{-8}$ , with 9 of 20 cells more than 5% slower (a greedy rule is not monotone in its action set, and its learned imitation less so); on the  $256^2$  grids, where the band left to the smoothers is twice as wide and needs several macro-actions, the gain appears: on the Poisson and convection–diffusion equations the ensembles {Jacobi, Jacobi (0.67)} and {Jacobi, Jacobi (0.67), GS} are  $1.23$ – $1.26\times$  faster than the best single-member router at  $10^{-8}$  on every one of the 32 instances, in work units and in live time alike, and their oracles  $1.26$ – $1.28\times$  faster than the best single-member oracle (Table 30; over all equations at  $256^2$ : **[pending]** of **[pending]** cells significantly faster, ratios **[pending]**–**[pending]**). Whether a finer granularity restores the gain at  $128^2$  (Section E.6) is **[pending]**. Finally, Section E.10 checks numerically which assumptions of Sections 4 and 5 hold in which setting: exactly for the Jacobi-type ensembles with the exact band-limited inverse on the constant-coefficient equations, and only as finite diagnostics for the fitted corrector, the non-normal sweeps and multigrid, for which no guarantee is claimed.

Figure 4 shows the individual operation sequences of the learned router on every test instance, next to HINTS on the same instances. Every result per equation and grid, the ensemble study, the accounting of overheads and training cost, and the verification of the theoretical assumptions are collected in Section E; Section D reports the Dirichlet experiment of the original submission.

## 8 Discussion

We have introduced an adaptive method for selecting solvers that efficiently minimizes the final error of a PDE in iterative methods, opening many directions for future work. Our current DeepONet is trained without accounting for its downstream role in correction term prediction; jointly training the ML solver and router could yield larger gains. Another avenue is to use reinforcement learning to learn a cost-aware routing strategy that optimizes error under compute budgets; Sections 4 and E show that a myopic cost-aware variant of the greedy rule—Algorithm 1 on cost-equalised macro-

Table 1: Median wall-clock time to truncation-level relative error  $\varepsilon = h^2$  for every solver pairing  $\{\text{NO, solver}\}$ , equation and grid ( $64, 32$  and  $16$  test instances at  $128^2, 256^2$  and  $512^2$ ): HINTS with its published two-dimensional period  $\tau=15$ , the strongest fixed schedule chosen per cell on the test set itself (smallest paired ratio to the router) from HINTS ( $\tau \in \{2, 5, 10, 15, 25, 50\}$ ), phase-shifted HINTS ( $\tau \in \{5, 10, 15, 25, 50\}$ , first call at 0) and the one-shot schedule (subscript), and the learned router (paired median speedup over HINTS-15/over the best fixed schedule in parentheses; bold: both ratios at least one). The paired ratio of the classical solver alone to the router is  $0.95 \times -7083 \times$  at  $128^2$  (above one favours the router; censored runs entered at the cap; the low end is the Multigrid pairing; superscripts count censored and invalid runs as in Section E.1; intervals and tests are in the appendix speed tables, this table being descriptive; Section E). The last rows of each block give geometric multigrid alone (point-smoothed; line-smoothed as well for anisotropic diffusion), the best multigrid-preconditioned Krylov method and the direct solve, without corrector, timed in the same session on the same instances. Anisotropic diffusion is not run at  $512^2$ : its corrector keeps full resolution along  $x_1$ , and the resulting  $65,536 \times 65,536$  branch matrix does not fit in memory (Section E). † $n$ :  $n$  instances did not reach  $h^2$  within the iteration cap.

| Equation  | Pairing                        | 128 × 128   |                                 |                              | 256 × 256 |                             |                          | 512 × 512 |                             |                                 |
|-----------|--------------------------------|-------------|---------------------------------|------------------------------|-----------|-----------------------------|--------------------------|-----------|-----------------------------|---------------------------------|
|           |                                | HINTS-15    | best schedule                   | router (ours)                | HINTS-15  | best schedule               | router (ours)            | HINTS-15  | best schedule               | router (ours)                   |
| Poisson   | Jacobi                         | 977 $\mu$ s | 708 $\mu$ s $\tau=25, t_0=0$    | 723 $\mu$ s (1.40 × /0.96 ×) | 1.97 ms   | 1.55 ms <sub>one-shot</sub> | 1.57 ms (1.75 × /0.99 ×) | 29.6 ms   | 18.4 ms <sub>one-shot</sub> | 20.6 ms (1.86 × /1.00 ×)        |
|           | Jacobi (0.67)                  | 1.01 ms     | 734 $\mu$ s $\tau=25, t_0=0$    | 763 $\mu$ s (1.32 × /0.95 ×) | 1.92 ms   | 1.71 ms $\tau=50, t_0=0$    | 1.76 ms (1.38 × /0.97 ×) | 42.3 ms   | 25.1 ms <sub>one-shot</sub> | 25.5 ms (1.67 × /0.98 ×)        |
|           | GS                             | 1.58 ms     | 712 $\mu$ s $\tau=10, t_0=0$    | 758 $\mu$ s (2.09 × /0.96 ×) | 4.99 ms   | 2.28 ms $\tau=15, t_0=0$    | 2.32 ms (2.24 × /0.95 ×) | 42.5 ms   | 39.0 ms $\tau=50, t_0=0$    | 39.4 ms (1.16 × /0.99 ×)        |
|           | SymGS                          | 1.99 ms     | 744 $\mu$ s $\tau=5, t_0=0$     | 793 $\mu$ s (2.57 × /0.95 ×) | 6.81 ms   | 2.00 ms $\tau=10, t_0=0$    | 2.02 ms (3.45 × /0.97 ×) | 36.4 ms   | 33.0 ms $\tau=25, t_0=0$    | 32.1 ms (1.13 × /0.99 ×)        |
|           | SOR (1.5)                      | 3.06 ms     | 733 $\mu$ s <sub>one-shot</sub> | 777 $\mu$ s (3.65 × /0.96 ×) | 8.60 ms   | 2.08 ms $\tau=15, t_0=0$    | 2.13 ms (4.11 × /0.96 ×) | 36.1 ms   | 19.5 ms <sub>one-shot</sub> | <b>19.8 ms</b> (1.71 × /1.01 ×) |
|           | Multigrid                      | 1.01 ms     | 1.01 ms $\tau=5$                | 880 $\mu$ s (0.97 × /0.97 ×) | 3.71 ms   | 2.20 ms $\tau=5, t_0=0$     | 2.29 ms (1.62 × /0.98 ×) | 21.1 ms   | 13.3 ms $\tau=10, t_0=0$    | 13.1 ms (1.63 × /0.99 ×)        |
|           | Multigrid alone (no corrector) |             |                                 | 1.01 ms                      |           |                             | 4.07 ms                  |           |                             | 21.0 ms                         |
|           | PCG (MG) (no corrector)        |             |                                 | 982 $\mu$ s                  |           |                             | 3.82 ms                  |           |                             | 16.3 ms                         |
|           | FFT direct solve (exact)       |             |                                 | 235 $\mu$ s                  |           |                             | 1.06 ms                  |           |                             | 5.20 ms                         |
|           |                                |             |                                 |                              |           |                             |                          |           |                             |                                 |
| ConvDiff  | Jacobi                         | 986 $\mu$ s | 712 $\mu$ s <sub>one-shot</sub> | 746 $\mu$ s (1.39 × /0.95 ×) | 1.99 ms   | 1.84 ms <sub>one-shot</sub> | 1.90 ms (1.73 × /0.98 ×) | 39.9 ms   | 28.0 ms <sub>one-shot</sub> | 25.4 ms (1.67 × /1.00 ×)        |
|           | Jacobi (0.67)                  | 1.03 ms     | 768 $\mu$ s $\tau=25, t_0=0$    | 813 $\mu$ s (1.26 × /0.96 ×) | 2.12 ms   | 2.13 ms <sub>one-shot</sub> | 2.16 ms (1.43 × /0.98 ×) | 62.6 ms   | 37.5 ms <sub>one-shot</sub> | 44.9 ms (1.77 × /0.94 ×)        |
|           | GS                             | 1.88 ms     | 810 $\mu$ s $\tau=10, t_0=0$    | 853 $\mu$ s (2.30 × /0.95 ×) | 7.89 ms   | 2.81 ms <sub>one-shot</sub> | 2.83 ms (2.54 × /0.98 ×) | 58.1 ms   | 54.6 ms $\tau=50, t_0=0$    | <b>52.7 ms</b> (1.18 × /1.01 ×) |
|           | SymGS                          | 2.15 ms     | 807 $\mu$ s $\tau=25, t_0=0$    | 864 $\mu$ s (2.51 × /0.94 ×) | 7.49 ms   | 2.65 ms $\tau=10, t_0=0$    | 2.72 ms (2.76 × /0.97 ×) | 46.9 ms   | 44.1 ms $\tau=25, t_0=0$    | 42.8 ms (1.10 × /0.99 ×)        |
|           | SOR (1.5)                      | 3.14 ms     | 876 $\mu$ s $\tau=5, t_0=0$     | 906 $\mu$ s (3.46 × /0.96 ×) | 8.81 ms   | 2.46 ms $\tau=10, t_0=0$    | 2.59 ms (3.35 × /0.98 ×) | 34.0 ms   | 23.0 ms $\tau=25, t_0=0$    | 23.3 ms (1.43 × /0.97 ×)        |
|           | Multigrid                      | 1.05 ms     | 1.05 ms $\tau=5$                | 1.09 ms (0.95 × /0.95 ×)     | 4.26 ms   | 2.34 ms <sub>one-shot</sub> | 2.37 ms (1.54 × /0.97 ×) | 20.9 ms   | 13.6 ms <sub>one-shot</sub> | 13.7 ms (1.57 × /0.98 ×)        |
|           | Multigrid alone (no corrector) |             |                                 | 1.02 ms                      |           |                             | 4.53 ms                  |           |                             | 21.3 ms                         |
|           | BiCGSTAB (MG) (no corrector)   |             |                                 | 1.46 ms                      |           |                             | 5.46 ms                  |           |                             | 22.3 ms                         |
|           | FFT direct solve (exact)       |             |                                 | 235 $\mu$ s                  |           |                             | 1.05 ms                  |           |                             | 5.19 ms                         |
|           |                                |             |                                 |                              |           |                             |                          |           |                             |                                 |
| AnisoDiff | Jacobi                         | 1.04 ms     | 729 $\mu$ s $\tau=50, t_0=0$    | 764 $\mu$ s (1.37 × /0.95 ×) | 9.74 ms   | 8.82 ms <sub>one-shot</sub> | 8.82 ms (1.11 × /1.00 ×) | —         | —                           | —                               |
|           | Jacobi (0.67)                  | 970 $\mu$ s | 681 $\mu$ s $\tau=25, t_0=0$    | 700 $\mu$ s (1.38 × /0.95 ×) | 9.69 ms   | 8.74 ms $\tau=10, t_0=0$    | 8.80 ms (1.10 × /0.99 ×) | —         | —                           | —                               |
|           | GS                             | 1.54 ms     | 672 $\mu$ s $\tau=10, t_0=0$    | 710 $\mu$ s (2.18 × /0.95 ×) | 12.5 ms   | 8.74 ms $\tau=10, t_0=0$    | 8.76 ms (1.42 × /0.99 ×) | —         | —                           | —                               |
|           | SymGS                          | 1.98 ms     | 677 $\mu$ s $\tau=50, t_0=0$    | 714 $\mu$ s (2.72 × /0.95 ×) | 14.6 ms   | 8.73 ms $\tau=10, t_0=0$    | 8.75 ms (1.67 × /1.00 ×) | —         | —                           | —                               |
|           | SOR (1.5)                      | 1.53 ms     | 665 $\mu$ s $\tau=10, t_0=0$    | 690 $\mu$ s (2.20 × /0.95 ×) | 12.5 ms   | 8.74 ms $\tau=5, t_0=0$     | 8.81 ms (1.42 × /0.99 ×) | —         | —                           | —                               |
|           | Line GS                        | 2.00 ms     | 656 $\mu$ s $\tau=10, t_0=0$    | 686 $\mu$ s (2.81 × /0.95 ×) | 13.8 ms   | 8.73 ms $\tau=15, t_0=0$    | 8.76 ms (1.57 × /1.00 ×) | —         | —                           | —                               |
|           | Multigrid alone (no corrector) |             |                                 | 8.19 ms                      |           |                             | 48.8 ms                  | —         | —                           | —                               |
|           | PCG (MG) (no corrector)        |             |                                 | 2.08 ms                      |           |                             | 11.3 ms                  | —         | —                           | —                               |
|           | FFT direct solve (exact)       |             |                                 | 245 $\mu$ s                  |           |                             | 1.07 ms                  | —         | —                           | —                               |
|           |                                |             |                                 |                              |           |                             |                          |           |                             |                                 |
| VarCoeff  | Jacobi                         | 1.24 ms     | 737 $\mu$ s $\tau=25, t_0=0$    | 784 $\mu$ s (1.65 × /0.96 ×) | —         | —                           | —                        | —         | —                           | —                               |
|           | Jacobi (0.67)                  | 1.20 ms     | 705 $\mu$ s $\tau=50, t_0=0$    | 752 $\mu$ s (1.59 × /0.95 ×) | —         | —                           | —                        | —         | —                           | —                               |
|           | GS                             | 2.58 ms     | 770 $\mu$ s $\tau=10, t_0=0$    | 792 $\mu$ s (3.19 × /0.96 ×) | —         | —                           | —                        | —         | —                           | —                               |
|           | SymGS                          | 3.59 ms     | 780 $\mu$ s <sub>one-shot</sub> | 831 $\mu$ s (4.32 × /0.96 ×) | —         | —                           | —                        | —         | —                           | —                               |
|           | SOR (1.5)                      | 4.90 ms     | 813 $\mu$ s $\tau=10, t_0=0$    | 848 $\mu$ s (5.11 × /0.95 ×) | —         | —                           | —                        | —         | —                           | —                               |
|           | Multigrid                      | 2.53 ms     | 998 $\mu$ s <sub>one-shot</sub> | 1.06 ms (2.42 × /0.96 ×)     | —         | —                           | —                        | —         | —                           | —                               |
|           | Multigrid alone (no corrector) |             |                                 | 2.66 ms                      | —         | —                           | —                        | —         | —                           | —                               |
|           | PCG (MG) (no corrector)        |             |                                 | 2.01 ms                      | —         | —                           | —                        | —         | —                           | —                               |
|           | Sparse LU direct solve         |             |                                 | 932 $\mu$ s                  | —         | —                           | —                        | —         | —                           | —                               |
|           |                                |             |                                 |                              |           |                             |                          |           |                             |                                 |

576 actions—executes the decision sequence of the best static schedule at truncation level on 0–100%  
577 of the instances and gives pairing-dependent results below it (Section 7), and RL could extend it  
578 to non-myopic budgeted routing. RL would additionally enable extending the ensemble routing to  
579 continuous parameterization of solvers. Finally, framing routing as an online learning problem could  
580 enable adaptation to unknown deployment conditions.

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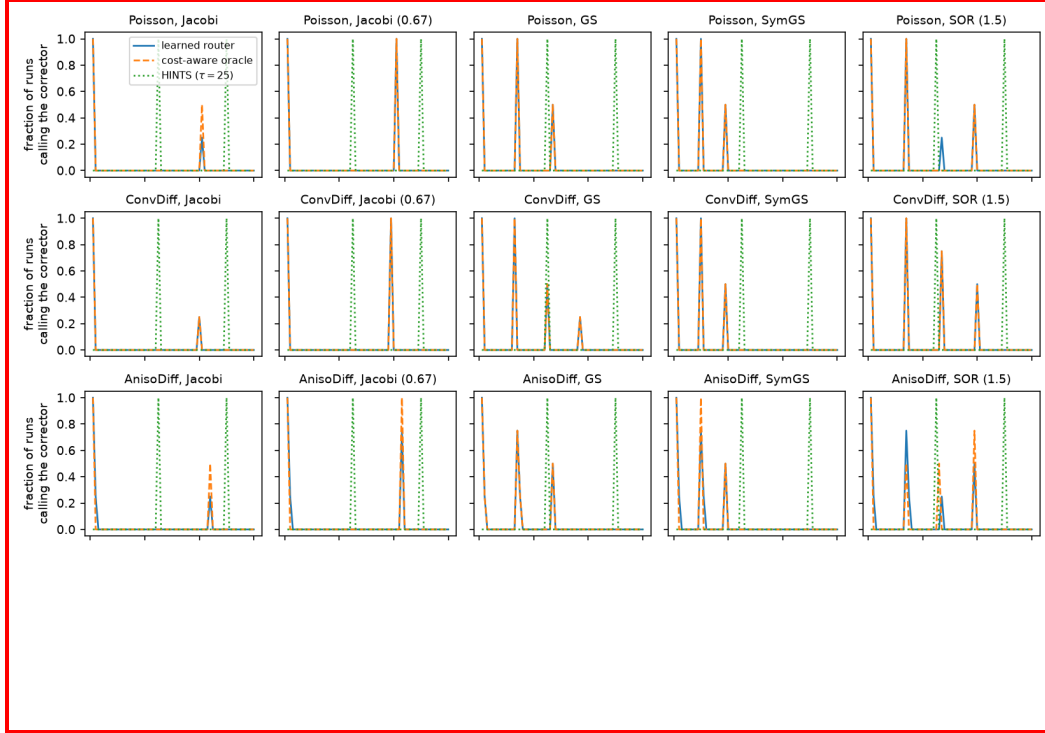


Figure 1: Corrector usage on the  $128 \times 128$  grid: fraction of test runs that execute a corrector call at each iteration, for the learned router, the cost-aware oracle and HINTS ( $\tau=15$ ), for every solver pairing (columns) and all four equations (rows); the same figure for the  $256^2$  and  $512^2$  grids is in Section E.

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## 768 A Proofs for Section 4

### 769 A.1 Proof of Proposition A.1

770 **Proposition A.1.** Any prefix monotonically non-increasing sequence supermodular function  $g$  is  
 771 weakly supermodular with respect to all sequences  $S \in \Omega^*$  with  $\alpha(S) = 1$

772 *Proof.* This proof is adapted from Liberty and Sviridenko [2017].

773 If  $g$  is sequence supermodular then,

$$\begin{aligned}
 & g(S) - g(S \oplus S') \\
 &= \sum_{i=1}^{|S'|} g(S \oplus (S'_1, \dots, S'_{i-1})) - g(S \oplus (S'_1, \dots, S'_i)) \\
 &\stackrel{(a)}{\leq} \sum_{i=1}^{|S'|} g(S) - g(S \oplus S'_i) \\
 &\leq |S'| \max_{i \in [|S'|]} g(S) - g(S \oplus S'_i)
 \end{aligned}$$

774 (a) by supermodularity

775 □

### 776 A.2 Proof of Theorem 4.1

777 **Theorem 4.1.** Let  $g : \Omega^* \rightarrow \mathbb{R}$  be a weakly supermodular function with respect to the optimal  
 778 solution  $O = \arg \min_{S \in \Omega^T} g(S)$  with a supermodularity ratio of  $\alpha(O)$  and  $\mu$ -postfix monotonicity  
 779 with respect to  $O$  along the greedy solution  $S^T$  of length  $T$ . If  $\phi_T(\alpha) = (1 - \frac{1}{\alpha T})^T$ , then

$$g(S^T) \leq \mu (1 - \phi_T(\alpha(O))) g(O) + \phi_T(\alpha(O)) g(\emptyset)$$

780 *Proof.* This proof strategy is inspired by Streeter and Golovin [2008] and Liberty and Sviridenko  
 781 [2017]

$$\begin{aligned}
g(S^t) - \mu g(O) &\stackrel{(a)}{\leq} g(S^t) - g(S^t \oplus O) \\
&= \sum_{i=1}^{|O|} g(S^t \oplus (o_1, \dots, o_{i-1})) - g(S^t \oplus (o_1, \dots, o_i)) \\
&\stackrel{(b)}{\leq} \alpha(O) \sum_{i=1}^{|O|} g(S^t) - g(S^t \oplus o_i) \\
&\leq \alpha(O)|O| \max_{i \in [|O|]} g(S^t) - g(S^t \oplus o_i) \\
&\leq \alpha(O)Tg(S^t) - \alpha(O)T \min_{\omega \in \Omega} g(S^t \oplus \omega) \\
&= \alpha(O)Tg(S^t) - \alpha(O)Tg(S^{t+1})
\end{aligned}$$

782 (a) by  $\mu$ -postfix monotonicity along  $S^T$ , i.e.,  $g(S^t \oplus O) \leq \mu g(O)$ ; (b) by weak supermodularity with  
783 respect to  $O$  (applied to the prefix  $S^t$ ).

784 After rearranging the inequality, we get:

$$\begin{aligned}
g(S^{t+1}) &\leq \frac{1}{\alpha(O)T} (\mu g(O) + (\alpha(O)T - 1)g(S^t)) \\
&= \frac{\mu}{\alpha(O)T} g(O) + \left(1 - \frac{1}{\alpha(O)T}\right) g(S^t)
\end{aligned}$$

785 When recursively applying this inequality, we get:

$$\begin{aligned}
g(S^T) &\leq \frac{\mu}{\alpha(O)T} g(O) \sum_{i=0}^{T-1} \left(1 - \frac{1}{\alpha(O)T}\right)^i + \left(1 - \frac{1}{\alpha(O)T}\right)^T g(\emptyset) \\
&= \mu \left(1 - \left(1 - \frac{1}{\alpha(O)T}\right)^T\right) g(O) + \left(1 - \frac{1}{\alpha(O)T}\right)^T g(\emptyset)
\end{aligned}$$

786

□

### 787 A.3 Proof of Proposition 4.2

788 **Proposition 4.2.** Suppose that for all  $j \in [K]$ , the error propagation function  $I - C_j \circ \mathcal{L}_h^a$  is  
789  $\rho_j$ -Lipschitz continuous with  $\rho_j \leq 1$ , that  $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$ , and that  $\sum_{i=1}^T (1 - \rho_{O_i}^2) > 0$ ,  
790 i.e., at least one action of the optimal sequence  $O$  is a strict contraction. Then, the function  $h$  is  
791 weakly supermodular with respect to the optimal solution  $O$ , with

$$\alpha(O) = \max \left\{ \frac{1}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}$$

792 Furthermore, if the error propagation maps are linear and simultaneously diagonalizable by a  
793 unitary matrix with eigenvalues of modulus at most one (the setting of Theorem 4.3),  $h$  is also postfix  
794 non-increasing; invertibility alone does not suffice (Section A.3).

795 *Proof.* For brevity, we use the notation  $(g_1 \circ \dots \circ g_T)(x) = \circ_{t=1}^T g_t(x)$ , where composition is  
796 applied from right to left so that  $g_T$  acts first. In this proof, we use a few properties of Lipschitz  
797 continuous functions:

- 798 • **Property 1:** If  $g$  is  $\rho$ -Lipschitz continuous and  $g(0) = 0$ , then  $\|g(x)\|_2 = \|g(x) - 0\|_2 =$   
799  $\|g(x) - g(0)\|_2 \leq \rho \|x - 0\|_2 = \rho \|x\|_2$

- 800 • **Property 2:** If  $g_1$  and  $g_2$  are Lipschitz continuous functions with Lipschitz constants of  $\rho_1$   
801 and  $\rho_2$  respectively, the Lipschitz constant of  $g_1 + g_2$  and  $g_1 - g_2$  is **at most**  $\rho_1 + \rho_2$ .
- 802 • **Property 3:** If  $g_1$  and  $g_2$  are Lipschitz continuous functions with Lipschitz constants of  $\rho_1$   
803 and  $\rho_2$  respectively, the Lipschitz constant of  $g_1 \circ g_2$  is **at most**  $\rho_1 \rho_2$ .

804 In order to prove weakly- $\alpha$ -supermodularity, we must first prove prefix monotonicity.

805 **Prefix monotonicity:** Let  $S \preceq S'$  where  $S' = S \oplus N$ .

$$\begin{aligned}
h(S') &= \left\| \circ_{t=|S'|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \\
&= \left\| \circ_{t=|S'|}^{|S|+1} (I - C_{S'_t} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \\
&\stackrel{(a)}{\leq} \left( \prod_{t=|S|+1}^{|S'|} \rho_{S'_t}^2 \right) \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \\
&\leq h(S)
\end{aligned}$$

806 (a) by Property 1

807 **Weak supermodularity:** We will upper bound  $\alpha(O)$  by providing an upper bound for  $h(S) - h(S \oplus$   
808  $O)$  and a lower bound for  $\sum_{i=1}^T h(S) - h(S \oplus O_i)$ . **Since  $h \geq 0$ ,  $h(S) - h(S \oplus O) \leq h(S) =$**   
809  **$\left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2$ . (An earlier version of this step used  $\|x\|^2 - \|Gx\|^2 \leq \|x - Gx\|^2$ ,**  
810 **which is false, e.g. for  $G = \frac{1}{2}I$ ; the direct bound is sharper by a factor 4.)**

811 To lower bound  $\sum_{i=1}^{|O|} h(S) - h(S \oplus O_i)$

$$\begin{aligned}
\sum_{i=1}^{|O|} h(S) - h(S \oplus O_i) &= \sum_{i=1}^{|O|} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 - \left\| (I - C_{O_i} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \\
&\geq \sum_{i=1}^{|O|} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 - \rho_{O_i}^2 \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \\
&= \left\| \circ_{t=|S|}^1 (I_N - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \left( T - \sum_{i=1}^T \rho_{O_i}^2 \right)
\end{aligned}$$

812 **If  $h(S) = 0$  then, by prefix monotonicity, every term above vanishes and the weak-supermodularity**  
813 **inequality holds for any  $\alpha \geq 1$ ; assume  $h(S) > 0$ . The lower bound just obtained is positive because**  
814  **$T - \sum_i \rho_{O_i}^2 > 0$  by hypothesis (this is the only place where a strict contraction is needed;  $\rho_j \leq 1$**   
815 **suffices everywhere else). Finally,**

$$\begin{aligned}
\frac{h(S) - h(S \oplus O)}{\sum_{i=1}^T (h(S) - h(S \oplus O_i))} &\leq \max \left\{ \frac{\left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2}{\left\| \circ_{t=|S|}^1 (I_N - C_{S_t} \circ \mathcal{L}_h^a) \left( e_h^{(0)} \right) \right\|_2^2 \left( T - \sum_{i=1}^T \rho_{O_i}^2 \right)}, 1 \right\} \\
&= \max \left\{ \frac{1}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}
\end{aligned}$$

816 **Postfix monotonicity.** Let  $S' = S \oplus N$  and suppose that  $I - C_j \mathcal{L}_h^a = P \Lambda_j P^*$  for all  $j$  with a  
817 common unitary  $P$  and  $|\lambda_{ji}| \leq 1$  (the products below are of  $|\lambda_{ji}|^2$  and  $|z_i|^2$ ; we write them without  
818 moduli for the real case). By Theorem A.2,

$$h(S') = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(N)} \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \leq \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(N)} = h(N),$$

819 since every factor  $\lambda_{ji}^{2m_j(S)}$  lies in  $[0, 1]$ . The shared eigenbasis cannot be replaced by invertibility of  
 820 the individual maps: for

$$A = \begin{bmatrix} 0.0674240 & -0.1864331 \\ -0.1473150 & -0.8707252 \end{bmatrix}, \quad B = \begin{bmatrix} 0.7575738 & 0.4816284 \\ -0.1369844 & 0.3257283 \end{bmatrix}, \quad x = \begin{bmatrix} 0.2812107 \\ -0.5538228 \end{bmatrix},$$

821 both matrices are invertible with spectral norm 0.9, yet  $\|BAx\|_2 = 0.330 > 0.225 = \|Bx\|_2$ , i.e.,  
 822  $h((A) \oplus (B)) > h((B))$ . (An earlier version of this proof bounded  $\|G^{-1}\|$  by the reciprocal of the  
 823 Lipschitz constant of  $G$ , which is not valid.)

824

□

#### 825 A.4 Proof of Theorem A.2

826 **Lemma A.2.** Let  $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$  where  $P$  is a unitary matrix and  $\Lambda_j =$   
 827  $\text{diag}(\lambda_{j1}, \dots, \lambda_{jN})$ . If  $P^{-1}e_h^{(0)} = z$ , then the following equality holds (written for a real or-  
 828 thogonal  $P$  and real eigenvalues; in general  $z_i^2$  and  $\lambda_{ji}^{2m}$  are to be read as  $|z_i|^2$  and  $|\lambda_{ji}|^{2m}$  here and  
 829 in the proofs of Propositions 4.2 and 4.3, which use only  $\|Px\| = \|x\|$  and  $|\lambda_{ji}| \leq 1$ ):

$$h(S) = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \quad (11)$$

830 where  $m_j(S) = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j}$

*Proof.*

$$\begin{aligned} h(S) &= \left\| \prod_{t=|S|}^1 (I_N - C_{S_t} \mathcal{L}_h^a) e_h^{(0)} \right\|^2 \\ &= \left\| \prod_{t=|S|}^1 (P \Lambda_{S_t} P^{-1}) e_h^{(0)} \right\|^2 \\ &= \left\| P \prod_{t=|S|}^1 \Lambda_{S_t} P^{-1} e_h^{(0)} \right\|^2 \\ &= \left\| \prod_{t=|S|}^1 \Lambda_{S_t} P^{-1} e_h^{(0)} \right\|^2 \\ &= \sum_{i=1}^N z_i^2 \prod_{t=|S|}^1 \lambda_{S_t i}^2 \\ &= \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \end{aligned}$$

831

□

#### 832 A.5 Proof of Theorem 4.3

833 **Proposition 4.3.** Let  $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$  for all  $j \in [K]$  and  $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$  with a common  
 834 unitary matrix  $P$ . Then,  $h$  is supermodular.

835 *Proof.* Let  $S \preceq S'$  where  $S' = S \oplus B$ . By Theorem A.2,

$$h(S) = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)}$$

836 where  $m_j(S) = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j}$  be the number of times a sequence  $S$  calls the solver  $j$ . Recall that  $h$  is  
837 considered sequence supermodular if  $\forall S', S \in \Omega^*$  such that  $S \preceq S'$ , it holds that

$$h(S) - h(S \oplus \omega) \geq h(S') - h(S' \oplus \omega)$$

838

$$\begin{aligned} h(S) - h(S \oplus \omega) &= \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} - \sum_{i=1}^N z_i^2 \lambda_{\omega i}^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \\ &= \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \end{aligned}$$

839 Similarly,

$$\begin{aligned} h(S') - h(S' \oplus \omega) &= \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S')} \\ &\stackrel{(a)}{=} \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2(m_j(S) + m_j(B))} \\ &\stackrel{(b)}{\leq} \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \\ &= h(S) - h(S \oplus \omega) \end{aligned}$$

840 (a) Since  $m_j(S') = \sum_{t=1}^{|S'|} \mathbf{1}_{S'_t=j} = \sum_{t=1}^{|S|} \mathbf{1}_{S'_t=j} + \sum_{t=|S|+1}^{|S'|} \mathbf{1}_{S'_t=j} = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j} +$   
841  $\sum_{t=1}^{|B|} \mathbf{1}_{B_t=j} = m_j(S) + m_j(B)$ , (b) since  $|\lambda_{ji}| \leq 1$  (as  $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$  and  $P$  is unitary)  
842 and  $m_j(B) \geq 0$  for all  $j \in [K]$

843 □

## 844 B Proofs for Section 5

### 845 B.1 Proof of Theorem B.1

846 **Lemma B.1.** For any set of preconditioning functions  $\mathcal{C}$ , any discrete operator  $\mathcal{L}_h^a$ , any router  $r$ , any  
847  $a_h, f_h, u_h^{(t)}, u_h \in \mathcal{A} \times \mathcal{F} \times \mathcal{U} \times \mathcal{U}$ , the following equality holds true:

$$\begin{aligned} l_{route} \left( r, a_h, f_h, u_h^{(t)}, u_h \right) &= \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\ &\quad - (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \end{aligned}$$

848 *Proof.* Note that  $\sum_{j=1}^K \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} = K - 1$

$$\begin{aligned}
l_{\text{route}} \left( r, a_h, f_h, u_h^{(t)}, u_h \right) &= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) = j} \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad + (K - 1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 - (K - 1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad + \sum_{j=1}^K \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \\
&\quad - (K - 1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2
\end{aligned}$$

849

$$\begin{aligned}
&= \sum_{j=1}^K \left( \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 - \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \right) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K - 2) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \\
&= \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K - 2) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2
\end{aligned}$$

850

□

## 851 **B.2 Proof of Theorem B.2**

852 **Proposition B.2.** *Let  $K \geq 2$ . For any router  $r$  defined by  $r(a, f, u^{(t)}) = \arg\max_{j \in [K]} g_j(a, f, u^{(t)})$ ,*

853 *any  $a_h \in \mathcal{A}$ ,  $f_h \in \mathcal{F}$ , and  $u_h^{(t)}, u_h \in \mathcal{U}$ , the routing loss  $l_{\text{route}}$  satisfies:*

$$\log(2) l_{\text{route}} \left( r, a_h, f_h, u_h^{(t)}, u_h \right) \leq \Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)$$

854 *Proof.* By Theorem B.1, we know that

$$\begin{aligned}
\log(2)l_{\text{route}}\left(r, a_h, f_h, u_h^{(t)}, u_h\right) &= \log(2) \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - \log(2) (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \\
&\leq \log(2) \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\stackrel{(a)}{\leq} - \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left( u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \log \left( \frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \right) \\
&= \Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)
\end{aligned}$$

855 (a) if  $r(a_h, f_h, u_h^{(t)}) \neq j$ ,  $\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \leq 0.5$  (with equality only at a tie for the largest  
856 score), which implies that  $-\log \left( \frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \right) \geq \log(2) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j}$

857  $\square$

### 858 B.3 Proof of Theorem 5.1

859 **Theorem 5.1.** *Let  $K \geq 2$  and let the per-step costs be normalised state by state,  $\hat{c}_j = \tilde{c}_j / \sum_{k=1}^K \tilde{c}_k$ ,  
860 at every state at which  $\sum_k \tilde{c}_k > 0$ , i.e., at which at least one solver does not annihilate the error  
861 (states at which every solver annihilates it carry no routing loss and are discarded), and let  $\hat{l}_{\text{route}}$   
862 and  $\hat{\Psi}$  denote the routing loss and the surrogate with  $\tilde{c}_j$  replaced by  $\hat{c}_j$ ; the normalisation does  
863 not change the Bayes-optimal decision equation 9 at any state. Then, for any collection of solvers  
864  $\{C_j\}_{j=1}^K$  and linear discrete operator  $\mathcal{L}_h^a$ ,  $\hat{\Psi}$  is a Bayes consistent surrogate for  $\hat{l}_{\text{route}}$ .*

865 **Proof.** Fix a state  $x = (a_h, f_h, u_h^{(t)})$  at which  $\sum_k \tilde{c}_k > 0$  and write  $\hat{c}_j = \tilde{c}_j / \sum_k \tilde{c}_k$ , so that  
866  $\hat{c}_j \in [0, 1]$  and  $\sum_j \hat{c}_j = 1$ ; let  $K \geq 2$ . Set  $\hat{B}_j = \sum_{k \neq j} \hat{c}_k = 1 - \hat{c}_j$  and  $q_j = \hat{B}_j / (K-1)$ ; then  $q$  is  
867 a probability vector on  $[K]$ , and  $\arg \max_j q_j = \arg \min_j \hat{c}_j$  is the Bayes decision equation 9. Let  
868  $p = \text{softmax}(\mathbf{g}(x))$ .

869 **Surrogate excess.** By the definition equation 10 of the surrogate with the weights  $\hat{c}$ ,  $\hat{\Psi}(\mathbf{g}, x) =$   
870  $-\sum_j \hat{B}_j \log p_j = (K-1) \text{CE}(q, p) = (K-1) [H(q) + \text{KL}(q \| p)]$ , with  $H$  the entropy and KL  
871 the Kullback–Leibler divergence. Since the scores are unrestricted measurable functions,  $\hat{\Psi}^*(x) =$   
872  $\inf_{\mathbf{g}} \hat{\Psi}(\mathbf{g}, x) = (K-1) H(q)$  (attained at  $p = q$  when  $q > 0$ , approached in the limit otherwise), so  
873 the conditional surrogate excess is  $\hat{\Psi}(\mathbf{g}, x) - \hat{\Psi}^*(x) = (K-1) \text{KL}(q \| p)$ .

874 **Routing excess.** With  $r(x) = \arg \max_j g_j(x) = \arg \max_j p_j$  and  $j^* \in \arg \max_j q_j$ ,  $\hat{l}_{\text{route}}(r, x) =$   
875  $\hat{c}_{r(x)} = 1 - (K-1) q_{r(x)}$  and  $\hat{l}_{\text{route}}^*(x) = 1 - (K-1) q_{j^*}$ , so the conditional routing excess is  
876  $(K-1) (q_{j^*} - q_{r(x)})$ . Since  $p_{r(x)} \geq p_{j^*}$ ,

$$q_{j^*} - q_{r(x)} \leq (q_{j^*} - p_{j^*}) + (p_{r(x)} - q_{r(x)}) \leq \|q - p\|_1 \leq \sqrt{2 \text{KL}(q \| p)}$$

877 by Pinsker's inequality, hence

$$\hat{l}_{\text{route}}(r, x) - \hat{l}_{\text{route}}^*(x) \leq (K-1) \sqrt{2 \text{KL}(q \| p)} = \sqrt{2(K-1) [\hat{\Psi}(\mathbf{g}, x) - \hat{\Psi}^*(x)]}.$$

878 **Risks.** Taking expectations over the state distribution and using Jensen's inequality for the concave  
879 square root,

$$\mathcal{R}_{\hat{l}_{\text{route}}}(r_n) - \mathcal{R}_{\hat{l}_{\text{route}}^*}^* \leq \sqrt{2(K-1) (\mathcal{R}_{\hat{\Psi}}(\mathbf{g}_n) - \mathcal{R}_{\hat{\Psi}}^*)},$$

where  $\mathcal{R}_{\hat{l}_{\text{route}}}^* = \mathbb{E}[\hat{l}_{\text{route}}^*]$  and  $\mathcal{R}_{\hat{\Psi}}^* = \mathbb{E}[\hat{\Psi}^*]$  because the infima are over all measurable functions. Hence  $\mathcal{R}_{\hat{\Psi}}(\mathbf{g}_n) - \mathcal{R}_{\hat{\Psi}}^* \rightarrow 0$  implies  $\mathcal{R}_{\hat{l}_{\text{route}}}(\mathbf{r}_n) - \mathcal{R}_{\hat{l}_{\text{route}}}^* \rightarrow 0$ , which is Bayes consistency. (An earlier version of this proof bounded the excess through the calibration function  $\Gamma$  of Mao et al. [2023], whose inverse is defined only on  $[0, \log 2]$  while the surrogate excess is unbounded, and assumed uniform absolute bounds on the unnormalised costs, which cannot hold along a convergent solve; the argument above needs neither.)  $\square$

## C Training Details

The periodic experiments of Section 7 use the corrector and the lightweight router described in Section E.1; the data distribution below is shared by all experiments, and the DeepONet and LSTM router described afterwards are those of the original submission’s pipeline, whose Dirichlet experiment has been withdrawn (Section D); they are kept for the record.

Data for both DeepONet and the routers is sampled from a zero-mean Hierarchical Gaussian Random Field on the periodic domain with covariance operator  $\alpha(-\Delta + \beta I)^{-\gamma}$ , where  $\alpha, \beta, \gamma$  are sampled as:

$$\begin{aligned}\alpha &\sim \text{Log-Uniform}(0.01, 100) \\ \beta &\sim \text{Log-Uniform}(0.1, 1000) \\ \gamma &\sim \text{Uniform}(\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0\})\end{aligned}$$

Given  $(\alpha, \beta, \gamma)$ , samples are generated in the Fourier space. For each non-zero frequency mode  $k$ , we draw an independent complex coefficient from a Gaussian distribution with mean 0 and variance  $\alpha(4\pi^2\|k\|_2^2 + \beta)^{-\gamma}$ . We enforce a Hermitian symmetry to obtain a real-valued field, set the zero-frequency (DC) mode to 0 to obtain a zero mean field, and apply the inverse Discrete Fourier Transform to obtain the field in physical space.

For each sample, we compute reference solutions with a least squares solver and treat them as ground truth.

This data is used to trained our DeepONet models and LSTM routers. All the models were implemented using PyTorch and all the models were trained on one Nvidia A40 GPU.

Table 2 contains all hyperparameter details for the DeepONet. DeepONet took 30 minutes to train. We then use the model with the best validation loss.

Table 2: Hyperparameter settings for DeepONet

| Hyperparameter                     | Value |
|------------------------------------|-------|
| Learning rate                      | 1e-3  |
| Branch Dimension                   | 128   |
| Hidden dimension for branch net    | 256   |
| No. of hidden layers in branch net | 4     |
| Hidden dimension for trunk net     | 256   |
| No. of hidden layers in trunk net  | 4     |
| Gradient Clipping Norm             | 1.0   |
| Weight Decay                       | 0.005 |
| Batch size                         | 256   |
| Training samples                   | 10000 |
| Validation samples                 | 2000  |
| Epochs                             | 1000  |

The routers are LSTM models. Along with the inputs specified in Section 5, namely  $a_h, f_h, u_h^{(t)}$ , we also supply the routers with the iteration index  $t$ , current residual  $f_h - \mathcal{L}_h^a u_h^{(t)}$ , and the routing decisions from the previous iteration. While the iteration index and residual are deterministic functions of  $(a_h, f_h, u_h^{(t)})$  and align with the theory presented in Section 5, incorporating previous routing decisions is a mild deviation from the idealized setting. Nevertheless, this modification is motivated by the same considerations that justify the use of scheduled sampling: under deployment,

911 routing decisions are predicted autoregressively and influence future inputs, whereas training under  
 912 teacher forcing assumes independence from past predictions. Including previous routing decisions  
 913 bridges the gap induced by exposure bias by providing the model with trajectory-level information.

914 All the routers are trained with scheduled sampling. We use a warm-up of  $e_w$  epochs with teacher-  
 915 forcing probability  $p_{tf}(e) = ss_{start}$ . After the warm-up, the  $p_{tf}$  decays geometrically by a factor of  
 916  $\gamma_{tf} < 1$  per epoch and is floored by  $ss_{end}$ :

$$p_{tf}(e) = \begin{cases} ss_{start} & e \leq e_w \\ \max(ss_{start} \gamma_{tf}^{e-e_w}, ss_{end}) & e > e_w \end{cases}$$

917 At each time step, with probability  $p_{tf}(e)$ , we feed the teacher-forced greedy iterate; otherwise, we  
 918 feed the router’s own predicted iterate.

919 Since LSTMs on long rollouts can suffer from exploding/vanishing gradients, we use truncated  
 920 backpropagation through time (TBPTT) [Mozier, 2013, Robinson and Fallside, 1987, Werbos, 1988]:  
 921 the forward pass unrolls the entire trajectory, but gradients are propagated only through the most  
 922 recent  $w_{bptt}(e)$  steps at epoch  $e$ . Hidden states are passed forward between segments, while earlier  
 923 segments are treated as stop-gradient.

924 We employ a curriculum learning approach analogous to scheduled sampling. Let  $T_{max}$  be the  
 925 horizon (300 iterations). With a warm-up of  $e_w$  epochs,

$$w_{bptt}(e) = \begin{cases} w_{start} & e \leq e_w \\ \min \left( T_{max}, w_{start} \gamma_{bptt}^{\lfloor \frac{e-e_w}{f_{bptt}} \rfloor} \right) & e > e_w \end{cases} \quad (12)$$

926 so the window grows geometrically by a factor of  $\gamma_{bptt} > 1$  every  $f_{bptt}$  epochs and is capped at the  
 927 full trajectory length.

928 Table 3 contains all hyperparameter details for the LSTM routers. The routers for the Paired Solver  
 929 experiment took a maximum of 4 hours and 30 minutes to train while the routers for the Solver  
 930 Ensemble experiment took a maximum of 6 hours to train. We then use the model with the best  
 931 validation loss for testing. The architecture and dataset size is larger for the Solver ensemble  
 932 experiments to encourage the model to learn some of the nuanced differences between the classes.

Table 3: Hyperparameter settings for routers in Paired solver and Solver Ensemble Experiments

| Hyperparameter         | Paired Solver | Solver Ensemble |
|------------------------|---------------|-----------------|
| Learning rate          | 1e-3          | 1e-3            |
| Hidden dimension       | 256           | 512             |
| No. of hidden layers   | 4             | 3               |
| Gradient Clipping Norm | 1.0           | 1.0             |
| Weight Decay           | 0.005         | 0.005           |
| Batch size             | 32            | 32              |
| Training samples       | 256           | 1024            |
| Validation samples     | 32            | 128             |
| Epochs                 | 200           | 200             |
| $ss_{start}$           | 1.0           | 1.0             |
| $\gamma_{tf}$          | 0.95          | 0.95            |
| $ss_{end}$             | 0.0           | 0.0             |
| $w_{start}$            | 50            | 50              |
| $\gamma_{bptt}$        | 1.25          | 1.25            |
| $e_w$                  | 10            | 10              |
| $f_{bptt}$             | 1             | 1               |

## 933 D Additional Experimental Results

934 The original submission’s Dirichlet experiment ( $31 \times 31$  grid, LSTM router, MLP-branch DeepONet),  
 935 which this appendix previously kept, has been withdrawn. A review of its evaluation code found

that HINTS’ predictions had been mean-centred before comparison with the (uncentred) Dirichlet reference solution while the other methods’ predictions had not, which penalises HINTS by an amount we cannot reconstruct without the original per-instance predictions; in addition, its symmetric-GS rows used the mis-scaled SSOR reference noted in Section E.1. The evaluation code has been corrected (`results.py`), but the experiment has not been rerun, so the wall-clock study of Section E, which is restricted to periodic boundary conditions, is the paper’s only experimental evidence. Extending the corrector to Dirichlet boundaries requires a sine-transform trunk in place of the Fourier trunk and is left to future work.

## E Cost-Aware Routing and Wall-Clock Evaluation

This appendix collects the complete wall-clock study that Section 7 summarises: the *wall-clock time to a target accuracy* (charged replay time, Section E.1) of the learned router against HINTS, the classical solver alone and the classical methods a practitioner would reach for on these problems (geometric multigrid and preconditioned Krylov methods), for all five classical solvers and three PDEs, on  $128^2$ ,  $256^2$  and  $512^2$  grids, using the cost-aware instantiation of the greedy rule introduced at the end of Section 4. Besides the Poisson and convection–diffusion equations of Section 7 we add *variable-coefficient diffusion* (Section E.5), on which no Fourier diagonalisation exists and the corrector must be learned from data, and *anisotropic diffusion*,  $-\epsilon \partial_{x_1}^2 u - \partial_{x_2}^2 u = f$  with  $\epsilon = 10^{-2}$  (diffusion tensor  $\text{diag}(\epsilon, 1)$ ), on the same periodic domain with the same forcing distribution. It is the standard model problem on which point-relaxation methods fail to smooth: an error component oscillatory in  $x_1$  and smooth in  $x_2$  is damped by only  $(1 - \epsilon)/(1 + \epsilon) \approx 0.98$  per Jacobi sweep, and geometric multigrid with point smoothing loses its grid-independent convergence (line relaxation or semi-coarsening would be needed). Section E.1 describes the protocol; Sections E.2 to E.5 collect all results per equation (every grid, every tolerance, every baseline, with significance tests and seed trials); Section E.7 studies ensembles of one, two and three classical solvers; Section E.8 accounts for overheads and training cost; and Section E.10 verifies numerically that the assumptions of Theorem 4.1, Propositions 4.2–4.3 and Theorem 5.1 hold in these settings.

### E.1 Protocol

**Fast, matched implementations.** A wall-clock comparison is meaningful only if the classical baseline is not handicapped by implementation. All classical operations are therefore implemented as compiled, single-threaded C kernels on the periodic 5-point stencil (`stencil.c`, built with `cc -O3`): the residual together with its norm (the stopping test, in one pass), fused (damped) Jacobi sweeps, in-place lexicographic Gauss–Seidel, SOR and symmetric SOR sweeps (forward and backward), a lexicographic line Gauss–Seidel sweep in which every grid line along the strongly coupled axis is solved exactly by a cyclic Thomas algorithm (the line relaxation that anisotropic problems call for), and the full-weighting and bilinear transfers of geometric multigrid, whose V(2,2)-cycles use Gauss–Seidel smoothing—or line Gauss–Seidel smoothing for the anisotropic equation—with rediscritised coarse operators and a direct solve on the  $32 \times 32$  coarsest grid (measured on one  $128^2$  instance, first cycle and asymptotic rate: 0.03 and 0.08 on Poisson with point smoothing, 0.005 and 0.04 on the anisotropic equation with line smoothing, against 0.25 and 0.86 with point smoothing there). At  $128^2$  a residual evaluation with its norm costs  $14 \mu\text{s}$ , a Gauss–Seidel sweep  $61 \mu\text{s}$ , a V-cycle  $326 \mu\text{s}$  and one corrector call  $672 \mu\text{s}$ , all measured in the exact per-iteration form in which the benchmark charges them (Table 7); an earlier version of this study used numpy/scipy sweeps that were  $4\text{--}5\times$  slower and correspondingly overstated the router’s margins (Section E.9). The compiled kernels agree with the numpy/scipy implementations they replace to  $2 \times 10^{-14}$  (relative), the line sweep agrees with a sparse block-triangular solve to  $10^{-13}$ , and their operators and one-step iterates match the dense reference implementations of the original submission to the reference’s own precision (relative error below  $10^{-7}$ , the reference assembling its matrices in single precision;  $10^{-12}$  where the reference is exact); the comparison is an assertion in `validate_fast_pde.py`. In the course of this check we found that the original dense symmetric-GS reference mis-implemented the SSOR scaling (the factor  $\omega/(2 - \omega)$  was added to  $D^{-1}$  instead of multiplying it); it has been corrected, and the wall-clock study uses the standard SSOR throughout. Reference solutions are computed by FFT diagonalization of the same discrete operator, i.e., they are exact solutions of the discrete system, so true errors are available for evaluation at no charged cost; the FFT solve is also reported as a baseline in every table (it is a direct method that exists because these test problems have constant coefficients and periodic

boundaries; at  $128^2$  the router takes  $3.08 \times - 3.25 \times$  times as long as it, and the variable-coefficient problem of Section E.5 has no such solve). All timings are single-threaded CPU measurements (Apple M4 Pro, numpy/scipy/PyTorch, one thread). The benchmark guards against machine drift, which on this machine is caused by macOS background services (media analysis, Spotlight indexing and window-server rendering, which slow the memory-bound corrector call by up to  $3 \times$  for minutes at a time) and by the scheduler, which moves an idle process to the efficiency cores: before the timed replays of every test instance it re-times a reference operation (one corrector call) and waits for up to 2 min, re-checking every 5 s and running the reference operation in between (a sleeping process stays on the efficiency cores, where the reference remains slow for the whole wait), while that time is more than 15% slower than the session’s reference (the 10th percentile of the reference measurements taken so far, seeded after 10 s of running the reference operation, which is robust to measurement noise and tightens only as genuinely faster states are observed); an instance whose reference is still too slow after the wait, or more than 15% slower than the cell’s final reference (the 10th percentile over all its instances, which catches a session that started in a slow state) or than the median final reference of the other cells of the same equation and grid (which catches a session that was slow throughout), is timed anyway and re-timed at the end of the cell, after waiting up to 5 min more (the Poisson and convection–diffusion  $128^2$  cells were produced before this in-cell pass existed, and every one of their instances is re-timed with the final driver in a separate pass once the other  $128^2$  cells are complete). The guard is a per-instance diagnostic; it does not certify that no drift occurred within a replay. Over the 2208 instance timings of the study the guard waited 2517 times, 794 instances were re-timed, 324 remained above the tolerance (largest accepted deviation 14.9%), and the per-instance ratios and load averages are stored with the results (an earlier confirmatory run had been contaminated by a 3.5 h stall of the machine inside one cell, which the harness could not detect). Two accounting checks are reported in Section E.8: the charged time of a router solve is 99–100% of the outer elapsed time of its replay, and its live time to  $h^2$  is 0.62–1.17 times its work-unit time.

**Corrector.** The neural corrector is a DeepONet applied to the residual  $r = f_h - \mathcal{L}_h^a u_h^{(t)}$  and evaluated on a  $64 \times 64$  sensor grid, i.e., on the coarser grid obtained by band-limited (FFT) restriction to the modes  $|k_1|, |k_2| \leq 31$  at every grid size (for the anisotropic equation the sensor grid is  $N \times N/4$ , i.e., full resolution along  $x_1$ , so that the band contains every mode the point smoothers cannot damp; the sensor grid is the only PDE-specific hyperparameter of the pipeline; the variable-coefficient equation uses the isotropic  $64 \times 64$  grid); its output is prolonged back to the fine grid by exact trigonometric interpolation, so the correction is band-limited by construction and never injects error into the high-frequency band that the classical smoother owns (Figure 2). The trunk net is the fixed orthonormal Fourier basis of that band ( $p = 3969$  basis functions, as in POD-DeepONet [Lu et al., 2022]), and the branch net is a linear layer on the 4096 sensor values; it is fit by least squares on 32,000–64,000 exact (residual, error) pairs of the discrete operator drawn from a mixture of spectral shapes (forcing-like, residual-like, white and randomly tilted spectra), and reaches a validation relative error of  $\approx 2 \times 10^{-6}$  on every mode of the band. The corrector is applied scale-equivariantly,  $C_{\text{NO}}(r) = \|Rr\| \mathcal{G}_\theta(Rr/\|Rr\|)$ , which preserves  $C_{\text{NO}}(0) = 0$  (Proposition 4.2). Its deployed form is therefore a linear, Fourier-structured map—a least-squares approximation of the exact band-limited inverse  $P(\mathcal{L}_h)_{\text{band}}^{-1} R$  of the discrete operator. Replacing it by that exact inverse changes nothing (development run): on the GS pairing at  $128^2$  the cost-aware oracle’s median work-unit time to  $10^{-8}$  was 4.472 ms with either corrector and its decision sequence identical on 13 of 16 instances. The gains reported below are thus those of an accurate band-limited spectral correction applied at the right moments; the learned corrector stands in for a spectral inverse that, on these constant-coefficient periodic problems, could also be computed directly. Its cost is dominated by a  $4096 \times 4096$  matrix–vector product and is therefore almost independent of  $N$  ( $672 \mu\text{s}$  at  $128^2$ , 2.62 ms at  $512^2$ , Table 31), i.e., about 32 Jacobi or 11 GS sweeps at  $128^2$  and 6 Jacobi sweeps at  $512^2$  (Tables 7, 13 and 19). The same corrector is used by HINTS and by all routing policies.

**Cost-equalised macro-actions.** Per-iteration costs  $c_j$  are measured once per equation and grid on the benchmark machine, for all operations of the study in one interleaved session, in the exact form in which the timed replay charges them—the update given the current residual, then the residual of the new iterate and its norm, i.e. the stopping test—as the median over eleven interleaved blocks of the block median of 100 repetitions (the minimum and the spread across blocks are stored with the costs); every pairing and ensemble of that grid therefore shares one corrector cost and one calibration. The unit of cost  $u$  is one corrector call; every operation is applied  $m_j = \max\{1, \text{round}(u/c_j)\}$  times

per decision, and these macro-actions are compared by their plain error, i.e. the deployed rule is Algorithm 1 on macro-actions; only an operation dearer than the unit ( $c_j > u$ , hence  $m_j = 1$ ) is compared per unit of cost,  $(\|e_j\|/\|e^{(t)}\|)^{u/c_j}$ , and the surrogate weights  $\tilde{c}_j$  use the same exponent. Which actions are dearer than the unit is read off the cost tables: on the four  $128^2$  grids none is, so those cells run exactly Algorithm 1 on macro-actions (the rule the exhaustive check of Section E.10 verifies); on the larger grids only a multigrid cycle can be dearer than the unit (**pending**), and in the granularity ablation of Section E.6 the corrector itself is. Tables 7, 13 and 19 list  $c_j$ ,  $u$  and  $m_j$  for every cell; the equalisation is exact up to rounding ( $m_j c_j / u$  lies within  $1/(2m_j - 1)$  of 1). Ensembles reuse the cached pairwise costs of their members, so every member has the same macro-action size in the ensemble as in its pairwise run. The cost-aware oracle runs Algorithm 1 on these macro-actions with the true error; the *greedy oracle* rows run the cost-agnostic Algorithm 1 on single operations, as in the main text. (An earlier version of this study applied the exponent  $u/(m_j c_j)$  to every action, including the rounding error of cheap ones, which is not Algorithm 1; the deployed rule was changed so that the theory covers it.)

**Lightweight router.** The learned router selects macro-actions from  $6 + K$  scalar features, all  $O(1)$  given the residual norm that any stopping test computes: the log relative residual, its per-iteration change over the last macro-action and an exponential moving average thereof, the log epoch index, a one-hot encoding of the previous macro-action, and the log counts of corrector calls so far and of iterations since the last call. It is a two-hidden-layer MLP (64 units) evaluated in numpy, costing  $\approx 5 \mu\text{s}$  per decision in a warm loop and  $7\text{--}21 \mu\text{s}$  in situ, where every decision after a macro-action finds cold caches (measured inside the timed replays, Section E.8), independently of  $N$ . It is trained with the surrogate of Section 5.1, with the per-state weights  $\tilde{c}_k$  normalised by  $\sum_k \tilde{c}_k$  (which does not change the Bayes decision), on states visited by oracle rollouts with  $\varepsilon$ -exploration (probability 0.1), followed by two to three DAgger rounds [Ross et al., 2011] in which the current router’s own rollouts are relabelled by the oracle—the batch analogue of the scheduled sampling used in Section C. Training uses 32–128 instances per round (fewer on the larger grids) drawn from the same GRF distribution as the test set but with a different seed; the router of every pairing is trained in seconds to a few minutes. Ensemble routers, which must additionally learn to choose among classical members, use a larger imitation budget (256 rollouts at  $128^2$  and 128 at  $256^2$ , six DAgger rounds, hidden width 128, 300 epochs): with the pairwise recipe the router over  $\{\text{Jacobi}, \text{Jacobi} (0.67)\}$  imitated its oracle imperfectly at tight tolerances (convection–diffusion,  $128^2$ ,  $\varepsilon = 10^{-8}$ : 3.34 against the oracle’s 2.67 work-unit ms), which the larger budget removes (2.70). At  $512^2$  the routers are trained from 32 oracle rollouts (stopping error  $10^{-8}$ ) whose length covers the test horizon: the cap of 8000 operations is 1,333 decisions in the Jacobi pairings (6 sweeps per corrector call) and up to 8000 in the others, so the rollouts run for at most 1400 decisions in the Jacobi pairings and 4000 elsewhere. A first training with rollouts of at most 400 decisions, the recipe of the development phase, produced a Poisson Jacobi router that kept calling the corrector on the 6 of 16 test instances whose error stagnates beyond the horizon it had seen ( $\approx 5,900$  calls in 8000 operations against 2 for its oracle) and therefore missed  $10^{-6}$  within the cap; that run is archived and the horizon rule above replaced it (the development-phase exception for the convection–diffusion Jacobi pairing, 64 rollouts of 800 decisions, is subsumed).

**Baselines and metric.** HINTS applies the corrector every  $\tau$ -th iteration, first at iteration  $\tau$  (Section 3). Zhang et al. [2024] use the proportion  $1/15$  (one DeepONet call per 14 relaxation sweeps) in all of their two- and three-dimensional experiments and  $1/25$  only in the one-dimensional Poisson experiment; since all our problems are two-dimensional, the reference schedule is  $\tau = 15$ , and  $\tau = 25$  is reported as well (an earlier version of this study used it as the reference; the corresponding margins are  $0.95\times\text{--}7.08\times$  at  $h^2$  and  $1.12\times\text{--}4.08\times$  at  $10^{-8}$  on the  $128^2$  pairwise cells). We further report the *best fixed schedule*, selected *per cell and per tolerance* on the test instances themselves as the schedule with the smallest paired median time ratio to the router—the strongest competitor, an optimistic baseline (the schedule with the smallest median time can be a weak paired competitor when medians are close)—and, as the deployable counterpart, the schedule selected on the *development* instances (seed 72, the same number of instances, by median work units over the family; `select_schedule.py`), the choice a practitioner can make without seeing the test instances, from the family of HINTS with  $\tau \in \{2, 5, 10, 15, 25, 50\}$ , phase-shifted HINTS with the first call at iteration 0 and  $\tau \in \{5, 10, 15, 25, 50\}$ , and the *one-shot schedule*—the corrector once at iteration 0, then the classical solver only—which is the static rule closest to what the router learns at truncation level and therefore the natural ablation for state-dependent routing ( $\tau = 1$ , the corrector at every

iteration, never reduces the error outside the corrector’s band and is omitted); and the best of three *residual-decay rules*, deterministic controls that decide at the router’s granularity from the same residual norm that the stopping test computes: the corrector at iteration 0, then the solver’s macro-action, and the corrector again whenever the per-iteration contraction of the residual over the last macro-action was worse than  $\theta \in \{0.9, 0.95, 0.98\}$  (the threshold is selected per cell and tolerance on the test set, like the best fixed schedule); like the router, the rule is executed live in the timed replay, its decision statistic and threshold test are charged, and its decisions are verified against the untimed trace. The classical methods without corrector run on the same stencil, with the same accounting, *inside the same replay loop as the policies*—same instance, same warm-up, same random order, same three replays: multigrid V(2,2) alone; conjugate gradients, CG preconditioned by symmetric GS and by a symmetric V-cycle (forward and backward GS, both compiled) for Poisson; BiCGSTAB, multigrid-preconditioned BiCGSTAB and restarted GMRES(20) for convection–diffusion (GMRES is accounted at the granularity of its restart cycles and is not run at  $512^2$ ); point-smoothed and line-smoothed multigrid, CG and its two preconditioned variants for anisotropic diffusion; and the sparse direct solve with a cached factorisation together with the multigrid and CG variants for variable-coefficient diffusion. The Krylov methods are SciPy implementations driven by the compiled matrix–vector product and preconditioners: their per-iteration true errors come from an untimed pass with a callback, and the timed pass runs exactly the same number of iterations with a timestamp-only callback (a few hundred nanoseconds per iteration) and verifies that it made exactly that many iterations and that its final iterate reaches the error the untimed pass recorded to within 10% (largest ratio of the two final errors over the study: 1.00; runs failing either check are excluded from the comparisons and counted: 0). An earlier version of this study timed the classical baselines in separate sessions; the same computation differed by 4–9% between sessions on identical instances (paired,  $p < 10^{-4}$ ), so every cross-session  $p$ -value was uninformative, which is why all baselines are now timed in-session. All configuration choices of the study were made on a development test set (GRF seed 72, with the earlier numpy kernels, the earlier forcing sampler and  $\tau = 25$  as reference; archived as `results_dev/`); the results reported here are a single confirmatory evaluation of the frozen configurations on a fresh test set (seed 73) with three timed replays per instance, and Section E.9 places the two side by side. The test forcings are drawn from the hierarchical GRF of Section 7 as real white noise filtered in Fourier space, which realises exactly the stated covariance with the Hermitian symmetry of a real field (the development runs used a sampler that drew independent complex coefficients on the half-spectrum, which halved the variance of the modes on the  $k_2 = 0$  line); the Nyquist modes are excluded, so every forcing is a trigonometric polynomial with  $|k| < N/2$  in both directions, and the instances are drawn sequentially so that the first  $n$  instances of any script are the benchmark’s. For each test instance (64 at  $128^2$ ; 32 at  $256^2$ , except 16 for anisotropic diffusion; 16 at  $512^2$ ) we record, after every iteration, the true relative error  $\|e_h^{(t)}\|/\|u_h\|$  (untimed, benchmark-only) and the cumulative charged wall-clock time of a replay that executes only the chosen operations (residual and its norm, update, and—for the learned router—every feature and decision computation; oracle decisions are not charged). The reported time to a tolerance is the charged time at the first iteration whose true error is below it, identically for every method: it is a benchmark of the work needed to reach an error level, not of a deployed stopping rule (which would stop on the residual norm that every method computes anyway). We report the median first time at which the error drops below a tolerance  $\varepsilon$ ; paired speedups—the median over instances of the per-instance time ratio, which is the estimand behind every speedup and test in the speed tables and in Table 1 (the time tables show medians of raw times for scale, which can order differently from the paired median on small samples, and their bold marks compare those raw medians within a pairing)—with bootstrap 95% intervals (2000 resamples over instances) and two-sided Wilcoxon signed-rank tests on the paired log time ratios (the direction is read from the median ratio; “slower” marks a median below one); the study performs several hundred such tests, so the speedup tables mark as significant only comparisons that survive a Holm correction at level 0.05 over all tests of the table. A run that does not reach  $\varepsilon$  within the iteration cap (60,000 at  $128^2$ , 40,000 at  $256^2$ , 8,000 at  $512^2$ ; 5,000 units for the classical baselines, 3,000 at  $512^2$ ) is marked † with the number of censored instances; a median time over runs of which any is censored is the median of the mixed vector of reached times and times-to-cap, a lower bound, and is marked  $\geq$  (invalid runs are dropped and counted with  $\times$ ). In the paired comparisons the four censoring classes are decided from the censoring indicators before any substitution: a censored baseline run enters at its time-to-cap (marked § with its count), so that the median ratio is a lower bound on the true ratio whenever such caps are present and neither of the two following classes occurs (then marked  $\geq$ ); a censored router run enters as a ratio of 0 and as the most extreme negative rank in the test, i.e. a failure penalty counted against the router (marked † with its

1162 count); a pair in which both are censored carries no information about the true ratio and is excluded  
 1163 from the median and the test (marked  $\ddagger$  with its count); and a pair with an invalid run is dropped  
 1164 (marked  $\times$ ). The estimand is therefore the paired median of capped-time ratios with a failure penalty  
 1165 for the router, not a ratio of ordinary completion times, and the  $p$ -values and intervals refer to that  
 1166 convention; a cell with fewer than four informative pairs prints no ratio. Every affected cell carries its  
 1167 class counts. Timed replays are repeated three times per instance in a random order over policies  
 1168 and baselines, each replay preceded by an untimed warm-up (one corrector call and one sweep, or  
 1169 the baseline’s own operation) so that every method starts from the same cache state—the corrector’s  
 1170 matrix exceeds the CPU caches, and without the warm-up a policy’s first call was up to 0.1 ms slower  
 1171 or faster depending on which policy had run before it; the median over replays is used. The measured  
 1172 per-operation costs set the macro-action sizes and define *work units* (measured per-iteration costs  
 1173 times executed operations, plus the measured decision cost per decision), a timer-free measure of  
 1174 the decision sequence used wherever runs from different sessions must be compared; all reported  
 1175 live times are measured directly. We highlight  $\varepsilon = h^2$  as a nominal grid-dependent tolerance and  
 1176 checked on the confirmatory instances themselves ([**pending**] instances over all equations and grids)  
 1177 whether it lies below their discretisation error: comparing the exact discrete solution with the exact  
 1178 discrete solution on a four times finer grid for the same forcing (`discretization_error.py`; for  
 1179 variable coefficients the coefficient is one grid-independent function), the relative discretisation error,  
 1180 estimated as the difference between the two discrete solutions, has median [**pending**]-[**pending**]  $h^2$   
 1181 over the equations and grids, and its smallest ratio to  $h^2$  over all instances is [**pending**]; wherever that  
 1182 ratio exceeds one,  $h^2$  is a tolerance below the (estimated) discretisation error, and driving the algebraic  
 1183 error further is unlikely to improve the computed solution materially (the two error components can  
 1184 partly cancel, so “never” would be too strong); we also report  $10^{-3}$  and  $10^{-8}$ , and the iteration-based  
 1185 metrics of the main text ( $\text{AUC} = \sum_{t=1}^T \|e_h^{(t)}\|/\|u_h\|$  over  $T = 300$  iterations, compared by paired  
 1186  $t$ -tests on  $\log \text{AUC}$ ) on the  $128^2$  grid. Every router was additionally retrained from scratch with five  
 1187 independent seeds (different training instances, exploration and initialisation; the corrector and the  
 1188 test set are fixed) and re-run on the test set; because those trials ran in separate sessions, whose live  
 1189 timings are not comparable with the main run’s, they are compared in work units.

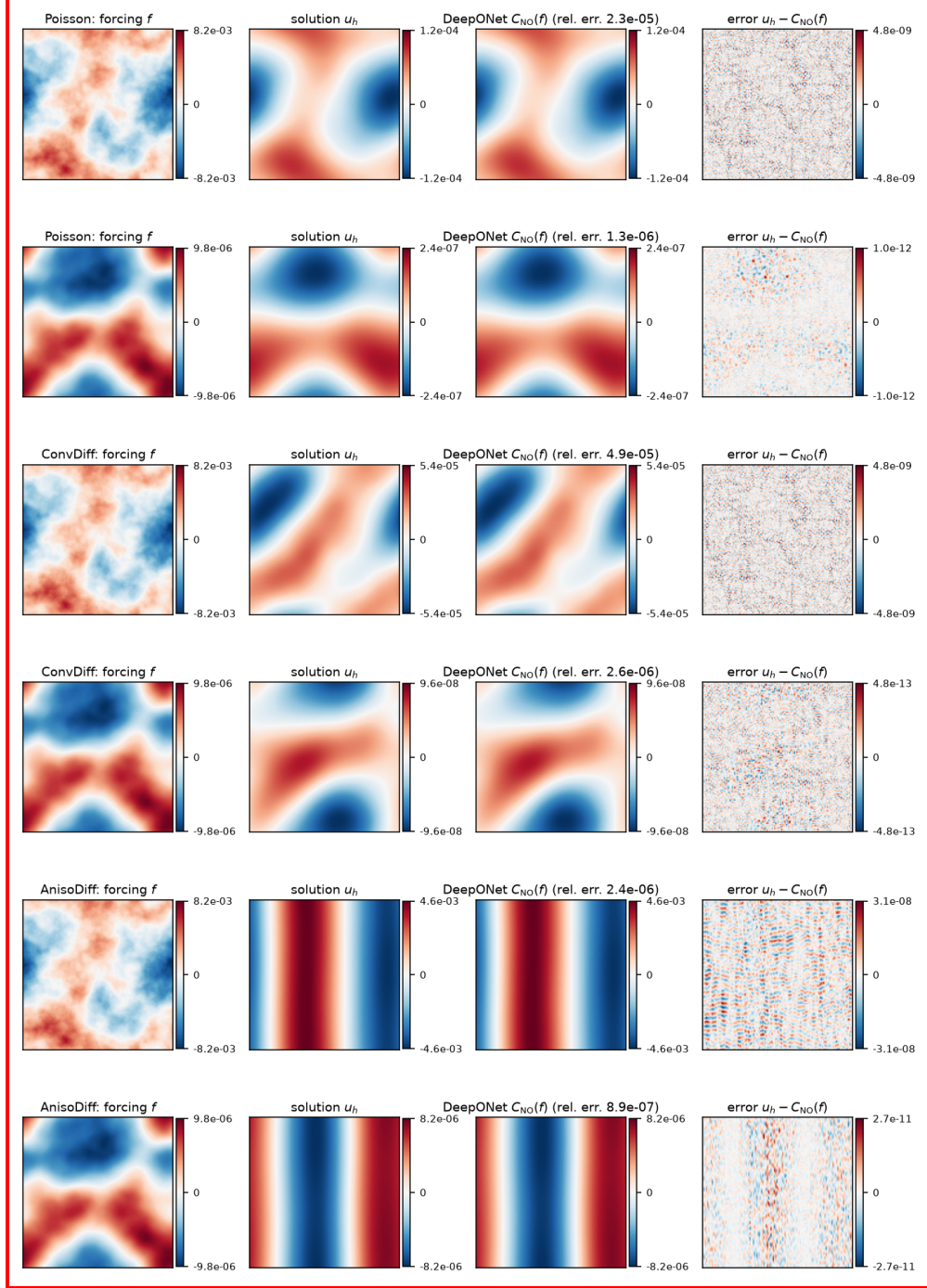


Figure 2: One-shot corrector predictions on the  $128 \times 128$  grid (two test instances per equation): forcing, exact discrete solution, the DeepONet correction applied to the initial residual ( $u_h^{(0)} = 0$ ), and the remaining error, which is confined to the modes outside the corrector's band (the relative error of the prediction is  $10^{-6}$ – $10^{-5}$  on the isotropic equations and  $10^{-6}$  on anisotropic diffusion, whose solution is dominated by modes that are smooth in  $x_2$ ).

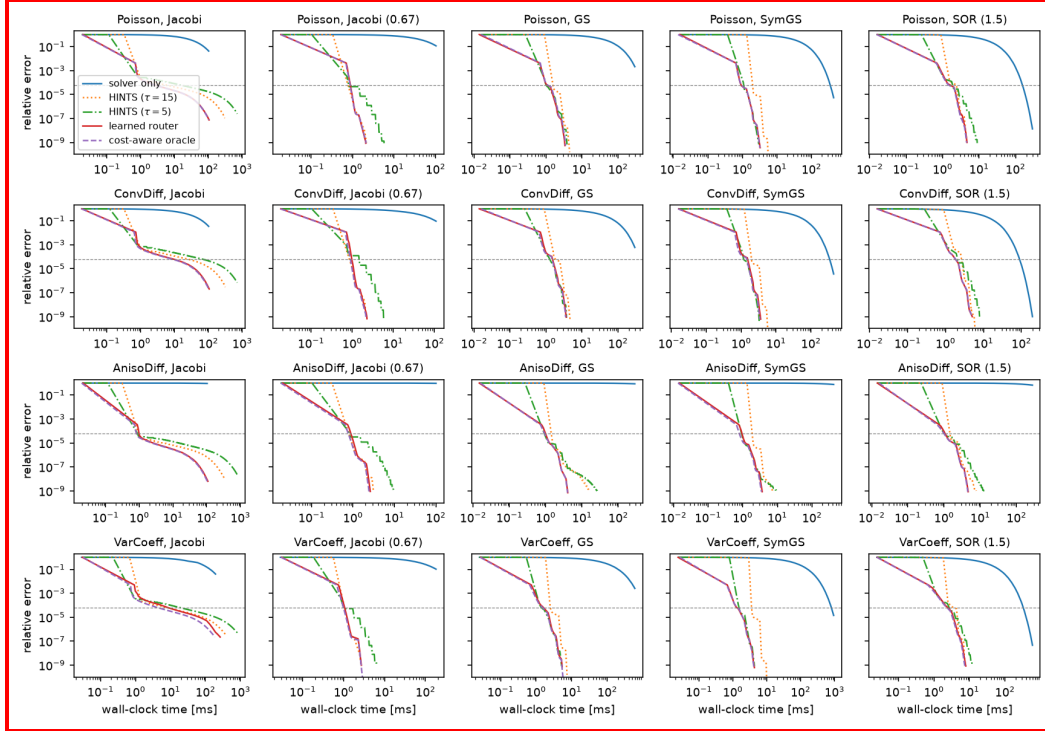


Figure 3: Relative error versus charged wall-clock time on the  $128 \times 128$  grid for one representative test instance of every pairing (columns) and equation (rows); the dashed line is  $\varepsilon = h^2$ . The learned router and the cost-aware oracle are indistinguishable.

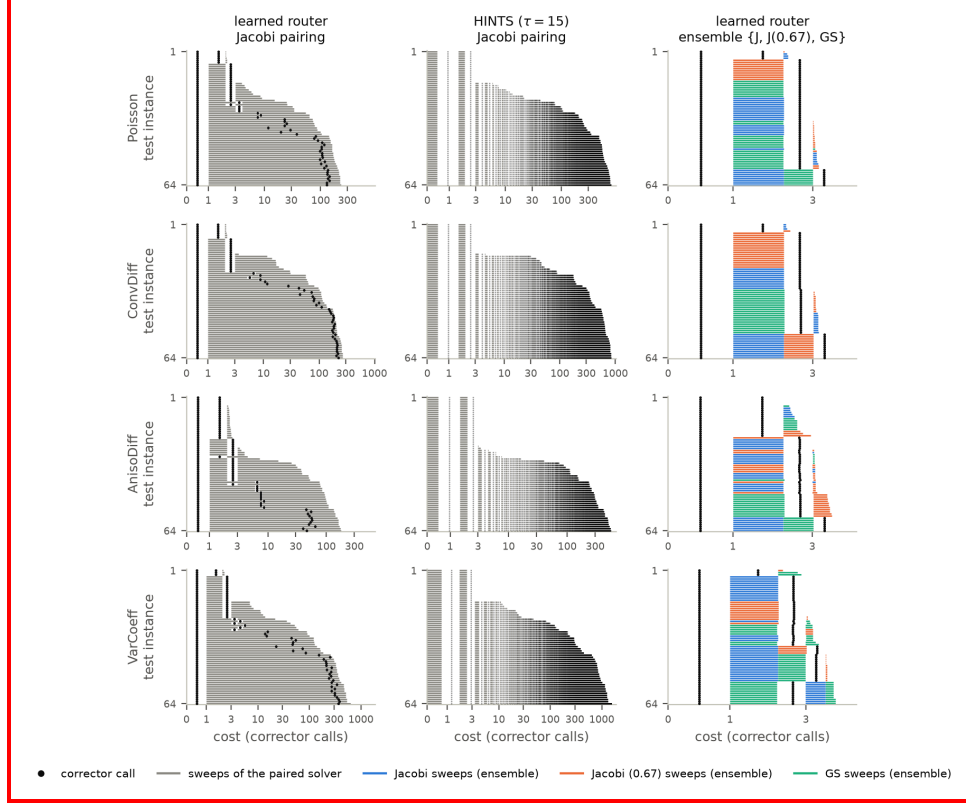


Figure 4: Decisions across test instances on the  $128 \times 128$  grid: for every equation (rows), the operations of the learned router in the Jacobi pairing (left), of HINTS ( $\tau=15$ ) in the same pairing on the same instances in the same order (middle), and of the learned router in the ensemble  $\{\text{Jacobi}, \text{Jacobi}(0.67), \text{GS}\}$  (right), until the relative error first falls below  $10^{-8}$ , as executed in the timed replays. Each row is one test instance; dots mark corrector calls and lines the solver sweeps between them (coloured by solver in the ensemble); the horizontal axis is the cumulative cost in units of one corrector call, linear below one call and logarithmic above. HINTS calls the corrector at the same iterations on every instance by construction; the learned router’s calls, and its choice of solver in the ensemble, are read off per instance. The per-equation versions for every pairing are Figures 6, 8, 10 and 12.

## 1190 E.2 Poisson

1191 Table 4 collects the median time to  $10^{-3}$ ,  $h^2$  and  $10^{-8}$  for every pairing, every policy and the classical  
 1192 baselines on all three grids; Table 5 the paired speedups of the learned router with their  $p$ -values;  
 1193 Figure 5 and table 6 where the iterations go; Table 7 the measured costs and macro-action sizes;  
 1194 Table 8 the iteration-based metrics; and Table 9 the seed trials.

1195 **Two regimes.** At  $\varepsilon = h^2$  the router makes one corrector call on 80–100% of the instances (over all  
 1196  $128^2$  pairwise cells) and then smooths (Table 6): the router executes the same decision sequence to  
 1197  $h^2$  as the best static schedule on 0–100% of the instances and as the cost-aware oracle on 50–100%  
 1198 (over all  $128^2$  pairwise cells, every equation; Section 7) and the router’s paired ratio against the best  
 1199 fixed schedule is close to one (Table 5; bold marks the comparisons that remain significant after  
 1200 the Holm correction), while HINTS with its published period, whose first call comes after  $\tau - 1$   
 1201 wasted sweeps, is slower by the factors in the table; the ratio of the solver alone to the router ranges  
 1202 over  $0.95 \times$ – $7083 \times$  over all  $128^2$  pairwise cells (every equation), the low end being the Multigrid  
 1203 pairing (multigrid alone reaches  $h^2$  in a few cycles:  $0.94 \times$ – $2.30 \times$  against the  $\{\text{NO}, \text{multigrid}\}$   
 1204 router). Below  $h^2$  the remaining error is the band the corrector cannot see, and the schedules separate:  
 1205 the router calls the corrector again only when the smooth error has re-emerged (two or three calls  
 1206 down to  $10^{-8}$ ; Figure 5), the one-shot schedule never calls it again and pays for hundreds of extra

sweeps, and a periodic schedule keeps paying for calls whether or not they help, so at  $10^{-8}$  the paired ratios against HINTS and the one-shot schedule are  $0.92\times$ – $3.07\times$  and  $0.97\times$ – $1373\times$  over all  $128^2$  pairwise cells (every equation), whereas against the best fixed schedule and the best residual-decay rule—both selected on the test set—the outcome depends on the pairing (Table 5; the counts over all pairings are in Section 7, and Section E.6 tests whether the macro-action granularity explains the losses). The cost-agnostic greedy oracle (Algorithm 1 on single operations) makes the same decisions as its cost-aware counterpart on the path to  $h^2$  (they differ later, in when the corrector is called again); the ratio of the learned router’s median time to the cost-aware oracle’s is  $1.01\times$ – $1.09\times$  and the two execute the same decision sequence to  $h^2$  on 50–100% of the instances (over all  $128^2$  pairwise cells), and the five retrained routers are compared with the main router in Table 9 **[pending]**.

**Classical baselines and larger grids.** Against the classical methods without corrector (bottom block of Table 4; timed in the same session as the router of every pairing), multigrid alone has the smallest ratio to the router among the classical methods in the current cells: at  $128^2$  the paired ratio of multigrid alone to the router (above one favours the router; over all  $128^2$  pairwise cells) is  $0.95\times$ – $11.2\times$  at  $\varepsilon = h^2$  and  $0.03\times$ – $20.5\times$  at  $10^{-8}$ , because one corrector call removes the modes  $|k| \leq 31$  at the price of one matrix–vector product whereas a V(2,2)-cycle contracts the whole spectrum by  $\approx 0.03$  on the first cycle and  $\approx 0.08$  asymptotically; routing over {NO, multigrid} calls the corrector once and lets the cycles remove the rest ( $0.94\times$ – $2.30\times$  of multigrid alone at  $h^2$  and  $1.11\times$ – $1.63\times$  at  $10^{-8}$ ), and the ratio of multigrid-preconditioned CG to the router is  $0.95\times$ – $3.00\times$  at  $h^2$  and  $0.03\times$ – $3.46\times$  at  $10^{-8}$ . The FFT direct solve, exact for this constant-coefficient periodic problem, is the reference the router takes  $3.08\times$ – $3.25\times$  times as long as at  $128^2$  and 1.5–2.3 times as long as at  $256^2$  (Table 5). At  $256^2$  the macro-actions are shorter (15 Jacobi, 4 Gauss–Seidel, 2 SymGS sweeps per corrector call, Table 7), so every schedule period that beat the router at  $128^2$  fits inside one macro-action, and the picture below truncation level changes: at  $10^{-8}$  the router is significantly faster than the strongest fixed schedule of the test-set search in the Jacobi ( $1.28\times$ , every instance) and damped-Jacobi ( $1.06\times$ ) pairings and within 1% of it in the other four, with the cost-aware oracle at  $1.00$ – $1.30\times$ ; at  $h^2$  it stays 1–5% behind the strongest schedule and is  $1.4$ – $4.1\times$  faster than HINTS ( $\tau=15$ ) and  $1.6$ – $2.8\times$  faster than multigrid alone in every pairing (Table 5). At  $512^2$  (macro-actions of 6 Jacobi, 2 Gauss–Seidel and 1 SymGS sweeps; 16 instances, cap 8000 operations) the router is again within 1% of the strongest fixed schedule at  $10^{-8}$  in four pairings and significantly faster in the damped-Jacobi ( $1.24\times$ ) and Jacobi ( $1.19\times$ , on the 9 instances that any policy reaches within the cap) pairings, within 2% of it at  $h^2$  in every pairing, and  $1.04$ – $1.86\times$  faster than HINTS ( $\tau=15$ ) throughout; its only losses are at  $10^{-6}$  in the Jacobi ( $0.97\times$ ) and damped-Jacobi ( $0.93\times$ ) pairings, where it calls the corrector a few more times than its oracle (Table 5). The classical comparison, however, turns: the corrector’s fixed band  $|k| \leq 31$  leaves the smoother all modes  $k \geq 32$ , whose slowest member contracts by only 0.98 per Jacobi sweep (against 0.71 at  $N = 128$ ), so every policy needs the same long tail of sweeps after the first call and the corrector—whose cost is nearly grid-independent—is worth fewer of them; multigrid alone is therefore the best classical method at  $512^2$ ,  $1.5$ – $4\times$  faster than the router in the Gauss–Seidel, SymGS and SOR pairings at  $10^{-8}$ , while routing over {NO, multigrid} stays  $1.3$ – $1.6\times$  faster than multigrid alone and the FFT solve  $2.6$ – $7.7\times$  faster than any router at  $h^2$ : the limit of a corrector whose band does not grow with the grid.

Table 4: Poisson: median wall-clock time to relative error  $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$  on the  $128^2$ ,  $256^2$  and  $512^2$  grids (64, 32 and 16 test instances) for every pairing  $\{\text{NO}, \text{solver}\}$  and policy, and for the classical baselines without corrector. “Best fixed schedule” is the best of HINTS ( $\tau \in \{2, 5, 10, 15, 25, 50\}$ ), phase-shifted HINTS ( $\tau \in \{5, 10, 15, 25, 50\}$ ) and the one-shot schedule, and “Best residual-decay rule” the best threshold  $\theta \in \{0.9, 0.95, 0.98\}$ , both chosen per cell and tolerance on the test set itself (the chosen schedule is the subscript). The greedy oracle is the cost-agnostic Algorithm 1, the cost-aware oracle is Algorithm 1 on cost-equalised macro-actions; both use the true error and are not deployable. Bold marks the router where its median is the smallest among the deployable methods of its pairing (solver alone, every fixed schedule, the one-shot and residual-decay rules); the oracles are in italics;  $\dagger n$ :  $n$  instances did not reach the tolerance within the iteration cap ( $>$ : lower bound).

| Pairing                                                               | Method                                               | 128 × 128                          |                                                        |                                              | 256 × 256                                              |                                              |                                              | 512 × 512                                    |                                              |                                                 |
|-----------------------------------------------------------------------|------------------------------------------------------|------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|-------------------------------------------------|
|                                                                       |                                                      | $\varepsilon=10^{-3}$              | $\varepsilon=h^2$                                      | $\varepsilon=10^{-8}$                        | $\varepsilon=10^{-3}$                                  | $\varepsilon=h^2$                            | $\varepsilon=10^{-8}$                        | $\varepsilon=10^{-3}$                        | $\varepsilon=h^2$                            | $\varepsilon=10^{-8}$                           |
| {NO, Jacobi}                                                          | Solver only                                          | 232.1 ms                           | 326.6 ms                                               | 623.3 ms                                     | $\geq 2.52 \text{ s}^{(12)}$                           | $\geq 2.52 \text{ s}^{(12)}$                 | $\geq 2.52 \text{ s}^{(12)}$                 | $\geq 2.48 \text{ s}^{(16)}$                 | $\geq 2.48 \text{ s}^{(16)}$                 | $\geq 2.48 \text{ s}^{(16)}$                    |
|                                                                       | HINTS ( $\tau=15$ )                                  | 976 $\mu\text{s}$                  | 136.8 ms                                               | 1.94 ms                                      | 1.97 ms                                                | 1.97 ms                                      | 1.97 ms                                      | 296 ms                                       | 7.11 ms                                      | $\geq 437.5 \text{ ms}^{(7)}$                   |
|                                                                       | Best fixed schedule                                  | 671 $\mu\text{s}$ $\tau=25, t_0=0$ | 708 $\mu\text{s}$ $\tau=25, t_0=0$                     | 65.3 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 1.04 ms $\tau=50, t_0=0$                               | 1.55 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 104.0 ms $\tau=50, t_0=0$                    | 2.90 ms $\tau=50, t_0=0$                     | 18.4 ms $\text{ms}_{\text{acc}}\text{-shot}$ | $\geq 308.5 \text{ ms}^{(7)}$ $\tau=50, t_0=0$  |
|                                                                       | Best residual-decay rule                             | 690 $\mu\text{s}$ $\theta=0.98$    | 704 $\mu\text{s}$ $\theta=0.98$                        | 85.8 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 1.04 ms $\theta=0.98$                                  | 1.53 ms $\theta=0.98$                        | 157.7 ms $\theta=0.98$                       | 2.85 ms $\theta=0.98$                        | 19.0 ms $\text{ms}_{\text{acc}}\text{-shot}$ | $\geq 594.6 \text{ ms}^{(7)}$ $\theta=0.98$     |
|                                                                       | One-shot schedule                                    | 684 $\mu\text{s}$                  | 704 $\mu\text{s}$                                      | 65.3 ms                                      | 1.04 ms                                                | 1.55 ms                                      | 710.7 ms                                     | 2.88 ms                                      | 18.4 ms                                      | $\geq 2.48 \text{ s}^{(16)}$                    |
|                                                                       | Greedy oracle (Alg. 1)                               | 683 $\mu\text{s}$                  | 707 $\mu\text{s}$                                      | 44.0 ms                                      | 1.05 ms                                                | 1.54 ms                                      | 82.2 ms                                      | 2.97 ms                                      | 20.0 ms                                      | $\geq 278.5 \text{ ms}^{(7)}$                   |
|                                                                       | Cost-aware oracle                                    | 686 $\mu\text{s}$                  | 710 $\mu\text{s}$                                      | 45.9 ms                                      | 1.04 ms                                                | 1.51 ms                                      | 81.9 ms                                      | 2.94 ms                                      | 24.6 ms                                      | $\geq 294.2 \text{ ms}^{(7)}$                   |
|                                                                       | Learned router (ours)                                | 703 $\mu\text{s}$                  | 723 $\mu\text{s}$                                      | <b>45.6 ms</b>                               | 1.05 ms                                                | 1.57 ms                                      | <b>82.9 ms</b>                               | 2.95 ms                                      | 20.6 ms                                      | <b><math>\geq 277.0 \text{ ms}^{(7)}</math></b> |
| {NO, Jacobi (0.67)}                                                   | Solver only                                          | 356.7 ms                           | 802.6 ms                                               | 958.5 ms                                     | $\geq 2.52 \text{ s}^{(12)}$                           | $\geq 2.52 \text{ s}^{(12)}$                 | $\geq 2.52 \text{ s}^{(12)}$                 | $\geq 2.42 \text{ s}^{(16)}$                 | $\geq 2.42 \text{ s}^{(16)}$                 | $\geq 2.42 \text{ s}^{(16)}$                    |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.01 ms                            | 1.01 ms                                                | 2.01 ms                                      | 1.89 ms                                                | 1.92 ms                                      | 9.35 ms                                      | 7.48 ms                                      | 42.3 ms                                      | 137.8 ms                                        |
|                                                                       | Best fixed schedule                                  | 705 $\mu\text{s}$ $\tau=25, t_0=0$ | 734 $\mu\text{s}$ $\tau=25, t_0=0$                     | 1.77 ms $\tau=15, t_0=0$                     | 1.02 ms $\tau=10, t_0=0$                               | 1.71 ms $\tau=10, t_0=0$                     | 6.74 ms $\tau=50, t_0=0$                     | 2.80 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 25.1 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 119.3 ms $\tau=50$                              |
|                                                                       | Best residual-decay rule                             | 704 $\mu\text{s}$ $\theta=0.98$    | 721 $\mu\text{s}$ $\theta=0.98$                        | 2.76 ms $\theta=0.98$                        | 1.00 ms $\theta=0.98$                                  | 1.74 ms $\theta=0.98$                        | 6.77 ms $\theta=0.98$                        | 2.85 ms $\theta=0.98$                        | 25.5 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 85.0 ms $\theta=0.98$                           |
|                                                                       | One-shot schedule                                    | 712 $\mu\text{s}$                  | 724 $\mu\text{s}$                                      | 78.7 ms                                      | 1.02 ms                                                | 1.76 ms                                      | 838.2 ms                                     | 2.80 ms                                      | 25.1 ms                                      | $\geq 2.40 \text{ s}^{(16)}$                    |
|                                                                       | Greedy oracle (Alg. 1)                               | 700 $\mu\text{s}$                  | 727 $\mu\text{s}$                                      | 1.79 ms                                      | 1.02 ms                                                | 1.70 ms                                      | 6.77 ms                                      | 2.88 ms                                      | 25.5 ms                                      | 85.3 ms                                         |
|                                                                       | Cost-aware oracle                                    | 701 $\mu\text{s}$                  | 716 $\mu\text{s}$                                      | 2.06 ms                                      | 1.02 ms                                                | 1.73 ms                                      | 6.77 ms                                      | 2.94 ms                                      | 31.0 ms                                      | 91.2 ms                                         |
|                                                                       | Learned router (ours)                                | 736 $\mu\text{s}$                  | 763 $\mu\text{s}$                                      | 2.14 ms                                      | 1.04 ms                                                | 1.76 ms                                      | 6.89 ms                                      | 2.87 ms                                      | 25.5 ms                                      | <b>76.2 ms</b>                                  |
| {NO, GS}                                                              | Solver only                                          | 328.2 ms                           | 461.2 ms                                               | 877.2 ms                                     | 5.32 s                                                 | 8.58 s                                       | $\geq 9.39 \text{ s}^{(12)}$                 | $\geq 8.50 \text{ s}^{(16)}$                 | $\geq 8.50 \text{ s}^{(16)}$                 | $\geq 8.50 \text{ s}^{(16)}$                    |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.53 ms                            | 1.58 ms                                                | 4.56 ms                                      | 4.48 ms                                                | 4.99 ms                                      | 11.9 ms                                      | 17.4 ms                                      | 42.5 ms                                      | 135.6 ms                                        |
|                                                                       | Best fixed schedule                                  | 689 $\mu\text{s}$ $\tau=15, t_0=0$ | 712 $\mu\text{s}$ $\tau=15, t_0=0$                     | 2.01 ms $\tau=15, t_0=0$                     | 1.01 ms $\tau=10, t_0=0$                               | 2.28 ms $\tau=15, t_0=0$                     | 10.5 ms $\tau=15, t_0=0$                     | 2.85 ms $\tau=50, t_0=0$                     | 39.0 ms $\tau=50, t_0=0$                     | 125.0 ms $\tau=50, t_0=0$                       |
|                                                                       | Best residual-decay rule                             | 696 $\mu\text{s}$ $\theta=0.98$    | 713 $\mu\text{s}$ $\theta=0.98$                        | 2.78 ms $\theta=0.98$                        | 971 $\mu\text{s}$ $\theta=0.98$                        | 2.25 ms $\theta=0.98$                        | 9.96 ms $\theta=0.98$                        | 2.86 ms $\theta=0.98$                        | 39.3 ms $\theta=0.98$                        | 121.0 ms $\theta=0.98$                          |
|                                                                       | One-shot schedule                                    | 699 $\mu\text{s}$                  | 715 $\mu\text{s}$                                      | 114.2 ms                                     | 1.01 ms                                                | 2.29 ms                                      | 1.40 s                                       | 2.91 ms                                      | 38.8 ms                                      | $\geq 8.52 \text{ s}^{(16)}$                    |
|                                                                       | Greedy oracle (Alg. 1)                               | 693 $\mu\text{s}$                  | 733 $\mu\text{s}$                                      | 2.52 ms                                      | 1.01 ms                                                | 2.26 ms                                      | 9.99 ms                                      | 2.88 ms                                      | 39.7 ms                                      | 121.2 ms                                        |
|                                                                       | Cost-aware oracle                                    | 696 $\mu\text{s}$                  | 728 $\mu\text{s}$                                      | 2.08 ms                                      | 1.00 ms                                                | 2.26 ms                                      | 9.44 ms                                      | 2.97 ms                                      | 38.9 ms                                      | 122.1 ms                                        |
|                                                                       | Learned router (ours)                                | 721 $\mu\text{s}$                  | 758 $\mu\text{s}$                                      | 2.13 ms                                      | 1.03 ms                                                | 2.32 ms                                      | 9.79 ms                                      | 2.90 ms                                      | 39.4 ms                                      | 122.7 ms                                        |
| {NO, SymGS}                                                           | Solver only                                          | 247.5 ms                           | 348.0 ms                                               | 661.8 ms                                     | 4.32 s                                                 | 6.96 s                                       | 11.59 s                                      | $\geq 13.74 \text{ s}^{(16)}$                | $\geq 13.74 \text{ s}^{(16)}$                | $\geq 13.74 \text{ s}^{(16)}$                   |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.97 ms                            | 1.99 ms                                                | 4.02 ms                                      | 6.61 ms                                                | 6.81 ms                                      | 12.9 ms                                      | 26.8 ms                                      | 36.4 ms                                      | 104.6 ms                                        |
|                                                                       | Best fixed schedule                                  | 690 $\mu\text{s}$ $\tau=15, t_0=0$ | 744 $\mu\text{s}$ $\tau=15, t_0=0$                     | 1.89 ms $\tau=15, t_0=0$                     | 992 $\mu\text{s}$ $\text{ms}_{\text{acc}}\text{-shot}$ | 2.00 ms $\tau=15, t_0=0$                     | 8.28 ms $\tau=15, t_0=0$                     | 3.04 ms $\tau=50, t_0=0$                     | 33.0 ms $\tau=50, t_0=0$                     | 99.3 ms $\tau=50, t_0=0$                        |
|                                                                       | Best residual-decay rule                             | 691 $\mu\text{s}$ $\theta=0.98$    | 734 $\mu\text{s}$ $\theta=0.98$                        | 3.33 ms $\theta=0.98$                        | 998 $\mu\text{s}$ $\theta=0.98$                        | 2.05 ms $\theta=0.98$                        | 9.55 ms $\theta=0.98$                        | 2.80 ms $\theta=0.98$                        | 31.6 ms $\theta=0.98$                        | 99.0 ms $\theta=0.98$                           |
|                                                                       | One-shot schedule                                    | 684 $\mu\text{s}$                  | 725 $\mu\text{s}$                                      | 81.7 ms                                      | 992 $\mu\text{s}$                                      | 2.03 ms                                      | 1.13 s                                       | 2.97 ms                                      | 32.3 ms                                      | $\geq 13.70 \text{ s}^{(16)}$                   |
|                                                                       | Greedy oracle (Alg. 1)                               | 694 $\mu\text{s}$                  | 744 $\mu\text{s}$                                      | 1.91 ms                                      | 988 $\mu\text{s}$                                      | 2.14 ms                                      | 2.96 ms                                      | 2.96 ms                                      | 32.0 ms                                      | 88.1 ms                                         |
|                                                                       | Cost-aware oracle                                    | 687 $\mu\text{s}$                  | 752 $\mu\text{s}$                                      | 2.02 ms                                      | 995 $\mu\text{s}$                                      | 2.04 ms                                      | 8.22 ms                                      | 2.90 ms                                      | 32.2 ms                                      | 99.5 ms                                         |
|                                                                       | Learned router (ours)                                | 714 $\mu\text{s}$                  | 793 $\mu\text{s}$                                      | 2.08 ms                                      | 1.00 ms                                                | 2.02 ms                                      | 8.58 ms                                      | 2.92 ms                                      | 32.1 ms                                      | 101.5 ms                                        |
| {NO, SOR (1.5)}                                                       | Solver only                                          | 110.1 ms                           | 155.2 ms                                               | 295.3 ms                                     | 1.77 s                                                 | 2.85 s                                       | 4.74 s                                       | $\geq 8.59 \text{ s}^{(16)}$                 | $\geq 8.59 \text{ s}^{(16)}$                 | $\geq 8.59 \text{ s}^{(16)}$                    |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.55 ms                            | 3.06 ms                                                | 4.60 ms                                      | 4.35 ms                                                | 8.60 ms                                      | 12.8 ms                                      | 17.5 ms                                      | 36.1 ms                                      | 63.7 ms                                         |
|                                                                       | Best fixed schedule                                  | 697 $\mu\text{s}$ $\tau=15, t_0=0$ | 733 $\mu\text{s}$ $\text{ms}_{\text{acc}}\text{-shot}$ | 2.72 ms $\tau=15, t_0=0$                     | 995 $\mu\text{s}$ $\text{ms}_{\text{acc}}\text{-shot}$ | 2.08 ms $\tau=15, t_0=0$                     | 9.31 ms $\tau=15, t_0=0$                     | 2.95 ms $\tau=50, t_0=0$                     | 19.5 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 62.0 ms $\tau=50, t_0=0$                        |
|                                                                       | Best residual-decay rule                             | 692 $\mu\text{s}$ $\theta=0.98$    | 728 $\mu\text{s}$ $\theta=0.98$                        | 4.08 ms $\theta=0.98$                        | 982 $\mu\text{s}$ $\theta=0.98$                        | 2.10 ms $\theta=0.98$                        | 9.89 ms $\theta=0.98$                        | 2.85 ms $\theta=0.98$                        | 19.3 ms $\theta=0.98$                        | 57.6 ms $\theta=0.98$                           |
|                                                                       | One-shot schedule                                    | 687 $\mu\text{s}$                  | 733 $\mu\text{s}$                                      | 58.0 ms                                      | 995 $\mu\text{s}$                                      | 2.17 ms                                      | 6.45 ms                                      | 2.89 ms                                      | 19.5 ms                                      | $\geq 8.07 \text{ s}^{(16)}$                    |
|                                                                       | Greedy oracle (Alg. 1)                               | 698 $\mu\text{s}$                  | 731 $\mu\text{s}$                                      | 3.02 ms                                      | 1000 $\mu\text{s}$                                     | 2.11 ms                                      | 8.78 ms                                      | 2.87 ms                                      | 19.9 ms                                      | 57.8 ms                                         |
|                                                                       | Cost-aware oracle                                    | 691 $\mu\text{s}$                  | 728 $\mu\text{s}$                                      | 2.75 ms                                      | 981 $\mu\text{s}$                                      | 2.06 ms                                      | 8.86 ms                                      | 2.91 ms                                      | 19.8 ms                                      | 58.3 ms                                         |
|                                                                       | Learned router (ours)                                | 723 $\mu\text{s}$                  | 777 $\mu\text{s}$                                      | 2.86 ms                                      | 1.03 ms                                                | 2.13 ms                                      | 9.35 ms                                      | 2.94 ms                                      | 19.8 ms                                      | 59.9 ms                                         |
| {NO, Multigrid}                                                       | Solver only                                          | 698 $\mu\text{s}$                  | 1.01 ms                                                | 2.00 ms                                      | 2.47 ms                                                | 3.74 ms                                      | 7.28 ms                                      | 10.7 ms                                      | 21.0 ms                                      | 31.8 ms                                         |
|                                                                       | HINTS ( $\tau=15$ )                                  | 699 $\mu\text{s}$                  | 1.01 ms                                                | 2.00 ms                                      | 2.46 ms                                                | 3.71 ms                                      | 7.30 ms                                      | 10.8 ms                                      | 21.1 ms                                      | 32.3 ms                                         |
|                                                                       | Best fixed schedule                                  | 699 $\mu\text{s}$ $\tau=50$        | 1.01 ms $\tau=5$                                       | 1.72 ms $\tau=10, t_0=0$                     | 978 $\mu\text{s}$ $\tau=50, t_0=0$                     | 2.20 ms $\tau=5, t_0=0$                      | 5.88 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 2.93 ms $\tau=5, t_0=0$                      | 13.3 ms $\tau=10, t_0=0$                     | 24.2 ms $\text{ms}_{\text{acc}}\text{-shot}$    |
|                                                                       | Best residual-decay rule                             | 700 $\mu\text{s}$ $\theta=0.9$     | 851 $\mu\text{s}$ $\theta=0.95$                        | 1.71 ms $\theta=0.95$                        | 972 $\mu\text{s}$ $\theta=0.95$                        | 2.22 ms $\theta=0.95$                        | 5.82 ms $\theta=0.95$                        | 2.85 ms $\theta=0.98$                        | 13.2 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 24.0 ms $\theta=0.9$                            |
|                                                                       | One-shot schedule                                    | 705 $\mu\text{s}$                  | 872 $\mu\text{s}$                                      | 1.73 ms                                      | 986 $\mu\text{s}$                                      | 2.26 ms                                      | 5.88 ms                                      | 3.11 ms                                      | 13.5 ms                                      | 24.2 ms                                         |
|                                                                       | Greedy oracle (Alg. 1)                               | 697 $\mu\text{s}$                  | 855 $\mu\text{s}$                                      | 1.73 ms                                      | 981 $\mu\text{s}$                                      | 2.29 ms                                      | 6.10 ms                                      | 2.89 ms                                      | 13.2 ms                                      | 23.7 ms                                         |
|                                                                       | Cost-aware oracle                                    | 696 $\mu\text{s}$                  | 887 $\mu\text{s}$                                      | 1.71 ms                                      | 988 $\mu\text{s}$                                      | 2.30 ms                                      | 5.93 ms                                      | 3.16 ms                                      | 13.6 ms                                      | 24.4 ms                                         |
|                                                                       | Learned router (ours)                                | 722 $\mu\text{s}$                  | 880 $\mu\text{s}$                                      | 1.70 ms                                      | 993 $\mu\text{s}$                                      | 2.29 ms                                      | 5.95 ms                                      | 3.06 ms                                      | 13.3 ms                                      | 23.4 ms                                         |
| Classical baselines without corrector (same stencil, same accounting) |                                                      |                                    |                                                        |                                              |                                                        |                                              |                                              |                                              |                                              |                                                 |
| Pairing                                                               | Multigrid V(2,2) alone (point GS)                    | 688 $\mu\text{s}$                  | 1.01 ms                                                | 2.00 ms                                      | 2.65 ms                                                | 4.07 ms                                      | 7.70 ms                                      | 10.9 ms                                      | 21.0 ms                                      | 32.3 ms                                         |
|                                                                       | CG                                                   | 4.51 ms                            | 5.39 ms                                                | 7.59 ms                                      | 25.4 ms                                                | 32.5 ms                                      | 43.2 ms                                      | 215.9 ms                                     | 309.2 ms                                     | 374.8 ms                                        |
|                                                                       | PCG (SymGS)                                          | 6.08 ms                            | 9.28 ms                                                | 15.8 ms                                      | 49.9 ms                                                | 81.9 ms                                      | 121.8 ms                                     | 411.0 ms                                     | 758.7 ms                                     | 1.01 s                                          |
|                                                                       | PCG (multigrid)                                      | 708 $\mu\text{s}$                  | 982 $\mu\text{s}$                                      | 1.70 ms                                      | 2.60 ms                                                | 3.82 ms                                      | 6.24 ms                                      | 10.6 ms                                      | 16.3 ms                                      | 25.9 ms                                         |
|                                                                       | FFT direct solve (exact; constant coefficients only) | 235 $\mu\text{s}$                  | 235 $\mu\text{s}$                                      | 235 $\mu\text{s}$                            | 1.06 ms                                                | 1.06 ms                                      | 1.06 ms                                      | 5.20 ms                                      | 5.20 ms                                      | 5.20 ms                                         |

Table 5: Poisson: paired median speedup [bootstrap 95% interval] of the learned router at  $\varepsilon = h^2$  and  $10^{-8}$  over HINTS ( $\tau=15$ ), over the strongest fixed schedule for that cell, over multigrid alone and over multigrid-preconditioned CG, with the two-sided Wilcoxon  $p$ -value (“slower” marks a median ratio below one) and the bootstrap 95% interval; bold: paired ratio  $\geq 1.1\times$  and significant after a Holm correction over all tests of the table.

| Pairing      | Method                   | $\varepsilon=10^{-3}$              | $\varepsilon=h^2$                  | $\varepsilon=10^{-8}$                        | $\varepsilon=10^{-3}$        | $\varepsilon=h^2$                            | $\varepsilon=10^{-8}$        | $\varepsilon=10^{-3}$        | $\varepsilon=h^2$                            | $\varepsilon=10^{-8}$                           |
|--------------|--------------------------|------------------------------------|------------------------------------|----------------------------------------------|------------------------------|----------------------------------------------|------------------------------|------------------------------|----------------------------------------------|-------------------------------------------------|
| {NO, Jacobi} | Solver only              | 232.1 ms                           | 326.6 ms                           | 623.3 ms                                     | $\geq 2.52 \text{ s}^{(12)}$ | $\geq 2.52 \text{ s}^{(12)}$                 | $\geq 2.52 \text{ s}^{(12)}$ | $\geq 2.48 \text{ s}^{(16)}$ | $\geq 2.48 \text{ s}^{(16)}$                 | $\geq 2.48 \text{ s}^{(16)}$                    |
|              | HINTS ( $\tau=15$ )      | 976 $\mu\text{s}$                  | 136.8 ms                           | 1.94 ms                                      | 1.97 ms                      | 1.97 ms                                      | 1.97 ms                      | 296 ms                       | 7.11 ms                                      | $\geq 437.5 \text{ ms}^{(7)}$                   |
|              | Best fixed schedule      | 671 $\mu\text{s}$ $\tau=25, t_0=0$ | 708 $\mu\text{s}$ $\tau=25, t_0=0$ | 65.3 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 1.04 ms $\tau=50, t_0=0$     | 1.55 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 104.0 ms $\tau=50, t_0=0$    | 2.90 ms $\tau=50, t_0=0$     | 18.4 ms $\text{ms}_{\text{acc}}\text{-shot}$ | $\geq 308.5 \text{ ms}^{(7)}$ $\tau=50, t_0=0$  |
|              | Best residual-decay rule | 690 $\mu\text{s}$ $\theta=0.98$    | 704 $\mu\text{s}$ $\theta=0.98$    | 85.8 ms $\text{ms}_{\text{acc}}\text{-shot}$ | 1.04 ms $\theta=0.98$        | 1.53 ms $\theta=0.98$                        | 157.7 ms $\theta=0.98$       | 2.85 ms $\theta=0.98$        | 19.0 ms $\text{ms}_{\text{acc}}\text{-shot}$ | $\geq 594.6 \text{ ms}^{(7)}$ $\theta=0.98$     |
|              | One-shot schedule        | 684 $\mu\text{s}$                  | 704 $\mu\text{s}$                  | 65.3 ms                                      | 1.04 ms                      | 1.55 ms                                      | 710.7 ms                     | 2.88 ms                      | 18.4 ms                                      | $\geq 2.48 \text{ s}^{(16)}$                    |
|              | Greedy oracle (Alg. 1)   | 683 $\mu\text{s}$                  | 707 $\mu\text{s}$                  | 44.0 ms                                      | 1.05 ms                      | 1.54 ms                                      | 82.2 ms                      | 2.97 ms                      | 20.0 ms                                      | $\geq 278.5 \text{ ms}^{(7)}$                   |
|              | Cost-aware oracle        | 686 $\mu\text{s}$                  | 710 $\mu\text{s}$                  | 45.9 ms                                      | 1.04 ms                      | 1.51 ms                                      | 81.9 ms                      | 2.94 ms                      | 24.6 ms                                      | $\geq 294.2 \text{ ms}^{(7)}$                   |
|              | Learned router (ours)    | 703 $\mu\text{s}$                  | 723 $\mu\text{s}$                  | <b>45.6 ms</b>                               | 1.05 ms                      | 1.57 ms                                      | <b>82.9 ms</b>               | 2.95 ms                      | 20.6 ms                                      | <b><math>\geq 277.0 \text{ ms}^{(7)}</math></b> |

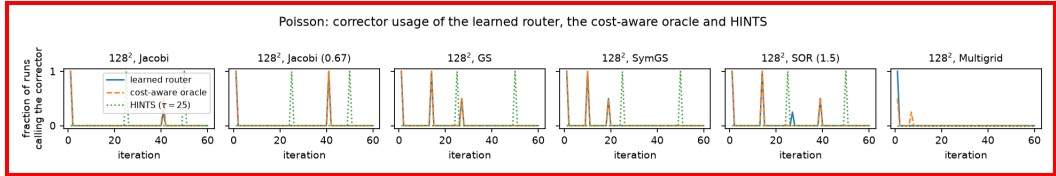


Figure 5: Poisson: fraction of test runs that execute a corrector call at each iteration, for the learned router, the cost-aware oracle and HINTS ( $\tau=15$ ), for every pairing (columns) and grid (rows; all test instances of each grid).

Table 6: Poisson: median iterations (and corrector calls) to  $\varepsilon = h^2$  for HINTS ( $\tau=15$ ), the cost-aware oracle and the learned router on every grid.

| Pairing             | 128 $\times$ 128 |        |        | 256 $\times$ 256 |        |        | 512 $\times$ 512 |         |         |
|---------------------|------------------|--------|--------|------------------|--------|--------|------------------|---------|---------|
|                     | HINTS-15         | oracle | router | HINTS-15         | oracle | router | HINTS-15         | oracle  | router  |
| {NO, Jacobi}        | 15 (1)           | 2 (1)  | 2 (1)  | 15 (1)           | 8 (1)  | 8 (1)  | 70 (4)           | 67 (1)  | 67 (1)  |
| {NO, Jacobi (0.67)} | 15 (1)           | 2 (1)  | 2 (1)  | 15 (1)           | 12 (1) | 12 (1) | 105 (6)          | 100 (1) | 100 (1) |
| {NO, GS}            | 16 (1)           | 2 (1)  | 2 (1)  | 17 (1)           | 6 (1)  | 6 (1)  | 37 (2)           | 36 (1)  | 36 (1)  |
| {NO, SymGS}         | 15 (1)           | 2 (1)  | 2 (1)  | 16 (1)           | 4 (1)  | 4 (1)  | 20 (1)           | 18 (1)  | 18 (1)  |
| {NO, SOR (1.5)}     | 30 (2)           | 2 (1)  | 2 (1)  | 30 (2)           | 6 (1)  | 6 (1)  | 30 (2)           | 18 (1)  | 18 (1)  |
| {NO, Multigrid}     | 3 (0)            | 2 (1)  | 2 (1)  | 3 (0)            | 2 (1)  | 2 (1)  | 4 (0)            | 3 (1)   | 3 (1)   |

Table 7: Poisson: measured cost of one classical iteration  $c_{\text{solver}}$  and of one corrector call  $c_{\text{NO}}$  (residual evaluation included), and the resulting macro-action size  $m_{\text{solver}}$  (sweeps per decision; the corrector is always applied once).

| Pairing             | 128 $\times$ 128    |                   |                     | 256 $\times$ 256    |                   |                     | 512 $\times$ 512    |                 |                     |
|---------------------|---------------------|-------------------|---------------------|---------------------|-------------------|---------------------|---------------------|-----------------|---------------------|
|                     | $c_{\text{solver}}$ | $c_{\text{NO}}$   | $m_{\text{solver}}$ | $c_{\text{solver}}$ | $c_{\text{NO}}$   | $m_{\text{solver}}$ | $c_{\text{solver}}$ | $c_{\text{NO}}$ | $m_{\text{solver}}$ |
| {NO, Jacobi}        | 21 $\mu\text{s}$    | 672 $\mu\text{s}$ | 32                  | 63 $\mu\text{s}$    | 945 $\mu\text{s}$ | 15                  | 471 $\mu\text{s}$   | 2.62 ms         | 6                   |
| {NO, Jacobi (0.67)} | 21 $\mu\text{s}$    | 672 $\mu\text{s}$ | 32                  | 63 $\mu\text{s}$    | 945 $\mu\text{s}$ | 15                  | 467 $\mu\text{s}$   | 2.62 ms         | 6                   |
| {NO, GS}            | 61 $\mu\text{s}$    | 672 $\mu\text{s}$ | 11                  | 249 $\mu\text{s}$   | 945 $\mu\text{s}$ | 4                   | 1.22 ms             | 2.62 ms         | 2                   |
| {NO, SymGS}         | 92 $\mu\text{s}$    | 672 $\mu\text{s}$ | 7                   | 398 $\mu\text{s}$   | 945 $\mu\text{s}$ | 2                   | 1.85 ms             | 2.62 ms         | 1                   |
| {NO, SOR (1.5)}     | 61 $\mu\text{s}$    | 672 $\mu\text{s}$ | 11                  | 247 $\mu\text{s}$   | 945 $\mu\text{s}$ | 4                   | 1.22 ms             | 2.62 ms         | 2                   |
| {NO, Multigrid}     | 326 $\mu\text{s}$   | 672 $\mu\text{s}$ | 2                   | 1.24 ms             | 945 $\mu\text{s}$ | 1                   | 5.51 ms             | 2.62 ms         | 1                   |

Table 8: Poisson,  $128 \times 128$ : iteration-based metrics of the main text—relative error after  $T = 300$  iterations and AUC over the  $T$  iterations (mean and standard error over 64 instances;  $p$ : one-sided paired  $t$ -test that the method’s AUC exceeds the router’s). Bold: router better than both solver-only and HINTS.

| Method                | AUC                       | $p$          |
|-----------------------|---------------------------|--------------|
| {NO, Jacobi}          |                           |              |
| Solver only           | 2.69e+02 (7.2e-01)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.39e+01 (2.7e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>6.08e-03 (1.5e-03)</b> | -            |
| Cost-aware oracle     | 6.08e-03 (1.5e-03)        | -            |
| {NO, Jacobi (0.67)}   |                           |              |
| Solver only           | 2.78e+02 (5.7e-01)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.39e+01 (2.4e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.13e-03 (5.0e-04)</b> | -            |
| Cost-aware oracle     | 2.13e-03 (5.0e-04)        | -            |
| {NO, GS}              |                           |              |
| Solver only           | 2.44e+02 (1.0e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.38e+01 (6.1e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>1.71e-03 (4.1e-04)</b> | -            |
| Cost-aware oracle     | 1.71e-03 (4.1e-04)        | -            |
| {NO, SymGS}           |                           |              |
| Solver only           | 2.05e+02 (1.2e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.37e+01 (1.1e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>1.26e-03 (3.0e-04)</b> | -            |
| Cost-aware oracle     | 1.26e-03 (3.0e-04)        | -            |
| {NO, SOR (1.5)}       |                           |              |
| Solver only           | 1.75e+02 (1.2e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.35e+01 (1.4e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.79e-03 (6.7e-04)</b> | -            |
| Cost-aware oracle     | 2.80e-03 (6.7e-04)        | -            |

Table 9: Poisson,  $128 \times 128$ : five independent router-training seeds per pairing—mean and standard deviation over seeds of the median work-unit time to  $h^2$ , the share of test instances on which the seed router’s decision sequence to  $h^2$  coincides with the main router’s, the range over seeds of the paired speedup over HINTS ( $\tau=15$ ), and the number of seeds significantly faster ( $p < 0.01$ ) than both HINTS ( $\tau=15$ ) and the best fixed period.

| $N$              | Pairing             | time to $h^2$ over 5 seeds | identical decisions | speedup vs. HINTS-15 | seeds with $p < 0.01$ |
|------------------|---------------------|----------------------------|---------------------|----------------------|-----------------------|
| 128 <sup>2</sup> | {NO, Jacobi}        | 0.71 $\pm$ 0.00 ms         | 100%                | 1.41–1.41 $\times$   | 0/5                   |
| 128 <sup>2</sup> | {NO, Jacobi (0.67)} | 0.71 $\pm$ 0.00 ms         | 100%                | 1.39–1.39 $\times$   | 0/5                   |
| 128 <sup>2</sup> | {NO, GS}            | 0.73 $\pm$ 0.00 ms         | 100%                | 2.16–2.16 $\times$   | 0/5                   |
| 128 <sup>2</sup> | {NO, SymGS}         | 0.74 $\pm$ 0.00 ms         | 100%                | 2.73–2.73 $\times$   | 0/5                   |
| 128 <sup>2</sup> | {NO, SOR (1.5)}     | 0.73 $\pm$ 0.00 ms         | 100%                | 3.74–3.74 $\times$   | 0/5                   |

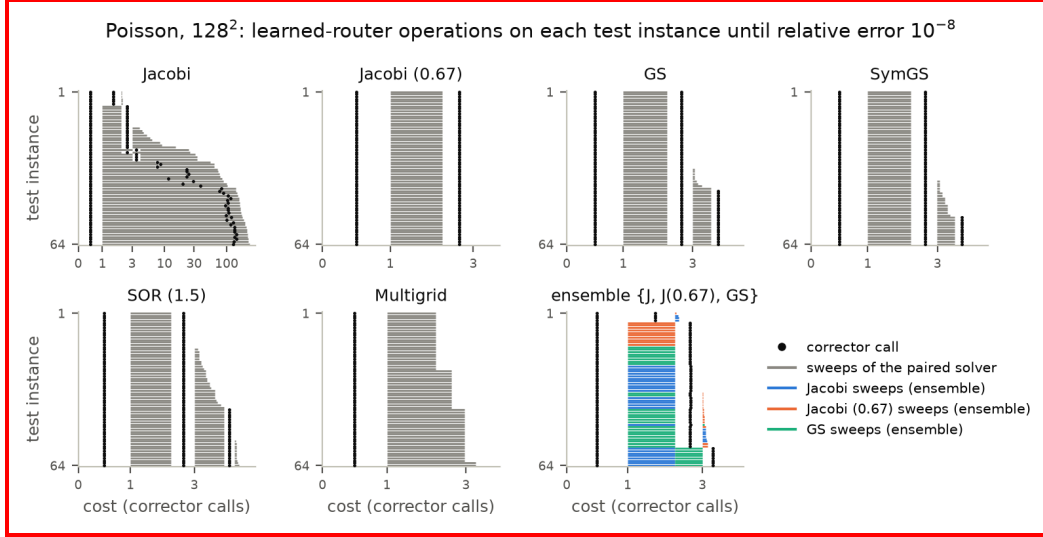


Figure 6: Poisson,  $128 \times 128$ : operations of the learned router on each of the 64 test instances (rows, sorted by cost) until the relative error first falls below  $10^{-8}$ , for every pairing and for the ensemble  $\{\text{Jacobi}, \text{Jacobi (0.67)}, \text{GS}\}$ , as executed in the timed replays. Dots mark corrector calls and lines the solver sweeps between them (coloured by solver in the ensemble); the horizontal axis is the cumulative cost in units of one corrector call, linear below one call and logarithmic above.

### 1249 E.3 Convection–diffusion

1250 The convection–diffusion results (Tables 10 to 15 and fig. 7) mirror the Poisson ones, the same  
 1251 two regimes with the nonsymmetric operator changing only the details: the undamped-Jacobi  
 1252 pairing reaches  $10^{-8}$  only after thousands of sweeps for every policy, and multigrid-preconditioned  
 1253 BiCGSTAB, unlike multigrid-preconditioned CG on the Poisson equation, is slower than the router at  
 1254  $h^2$  in every pairing (Table 11). At  $256^2$  the picture is again the Poisson one: at  $10^{-8}$  the router is  
 1255 significantly faster than the strongest fixed schedule in the Jacobi ( $1.28\times$ , 31 of 32 instances) and  
 1256 damped-Jacobi ( $1.06\times$ ) pairings and within 2% of it in the other four, 1–3% behind it at  $h^2$  and  
 1257  $10^{-6}$ , and  $1.2\text{--}3.4\times$  faster than HINTS ( $\tau=15$ ) throughout; multigrid alone is  $1.4\text{--}2.4\times$  slower than  
 1258 the router at  $h^2$  in every pairing and faster only at  $10^{-8}$  in the fast-smoother pairings, and in the  
 1259 multigrid pairing, where a cycle is dearer than a corrector call and is scored per unit of cost, the  
 1260 router is  $1.2\text{--}1.5\times$  faster than multigrid alone and  $1.3\text{--}2.2\times$  faster than multigrid-preconditioned  
 1261 BiCGSTAB (Table 11); the two nested ensembles are  $1.24\times$  and  $1.26\times$  faster than the best single-  
 1262 member router on every instance (Table 30). At  $512^2$  (16 instances, cap 8000 operations) the router  
 1263 is significantly faster than the strongest fixed schedule at  $10^{-8}$  in the damped-Jacobi pairing ( $1.12\times$ ,  
 1264 14 of 16 instances) and faster on 7 of the 9 Jacobi instances that any policy reaches within the cap  
 1265 ( $1.12\times$ ,  $p = 0.07$ ), within 3% of it in the other four pairings at every tolerance except SOR ( $0.97\times$ ),  
 1266 and  $1.06\text{--}1.8\times$  faster than HINTS ( $\tau=15$ ); as on the Poisson equation, multigrid alone is the best  
 1267 classical method at this grid in the Gauss–Seidel, SymGS and SOR pairings, while routing over  
 1268  $\{\text{NO}, \text{multigrid}\}$  stays  $1.3\times$  faster than multigrid alone at  $10^{-8}$  (Table 11).

Table 10: Convection–diffusion: median wall-clock time to relative error  $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$  on the  $128^2$ ,  $256^2$  and  $512^2$  grids for every pairing and policy, and for the classical baselines without corrector (conventions as in Table 4).

| Pairing                                                               | Method                   | 128 × 128                           |                                         |                                                | 256 × 256                                      |                                                |                                                 | 512 × 512                                      |                                                |                                                 |
|-----------------------------------------------------------------------|--------------------------|-------------------------------------|-----------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------|------------------------------------------------|-------------------------------------------------|
|                                                                       |                          | $\varepsilon=10^{-3}$               | $\varepsilon=h^2$                       | $\varepsilon=10^{-8}$                          | $\varepsilon=10^{-3}$                          | $\varepsilon=h^2$                              | $\varepsilon=10^{-8}$                           | $\varepsilon=10^{-3}$                          | $\varepsilon=h^2$                              | $\varepsilon=10^{-8}$                           |
| {NO, Jacobi}                                                          | Solver only              | 227 ms                              | 323 ms                                  | 619 ms                                         | $\geq 2.52 \text{ s}^{131}$                    | $\geq 2.52 \text{ s}^{132}$                    | $\geq 2.52 \text{ s}^{132}$                     | $\geq 2.43 \text{ s}^{136}$                    | $\geq 2.43 \text{ s}^{136}$                    | $\geq 2.43 \text{ s}^{136}$                     |
|                                                                       | HINTS ( $\tau=15$ )      | 976 $\mu\text{s}$                   | 986 $\mu\text{s}$                       | 201.2 ms                                       | 1.97 ms                                        | 1.99 ms                                        | 442.1 ms                                        | 3.25 ms                                        | 39.9 ms                                        | $\geq 1.69 \text{ s}^{17}$                      |
|                                                                       | Best fixed schedule      | 678 $\mu\text{s}_{\text{best-sch}}$ | 712 $\mu\text{s}_{\text{best-sch}}$     | 67.8 ms <sub>best-sch</sub>                    | 1.06 ms <sub><math>\tau=10, t_0=0</math></sub> | 1.84 ms <sub>best-sch</sub>                    | 270.4 ms <sub><math>\tau=50, t_0=0</math></sub> | 2.81 ms <sub><math>\tau=50, t_0=0</math></sub> | 28.0 ms <sub>best-sch</sub>                    | $\geq 1.11 \text{ s}^{17, \tau=50}$             |
|                                                                       | Best residual-decay rule | 677 $\mu\text{s}_{\text{best-rd}}$  | 706 $\mu\text{s}_{\text{best-rd}}$      | 126.7 ms <sub>best-rd</sub>                    | 1.06 ms <sub>best-rd</sub>                     | 1.83 ms <sub>best-rd</sub>                     | 416.3 ms <sub>best-rd</sub>                     | 2.87 ms <sub>best-rd</sub>                     | 28.5 ms <sub>best-rd</sub>                     | $\geq 2.77 \text{ s}^{17, \tau=50}$             |
|                                                                       | One-shot schedule        | 678 $\mu\text{s}$                   | 712 $\mu\text{s}$                       | 67.8 ms                                        | 1.08 ms                                        | 1.84 ms                                        | 686.8 ms                                        | 2.88 ms                                        | 28.0 ms                                        | $\geq 2.34 \text{ s}^{16}$                      |
|                                                                       | Greedy oracle (Alg. 1)   | 675 $\mu\text{s}$                   | 720 $\mu\text{s}$                       | 65.3 ms                                        | 1.04 ms                                        | 1.81 ms                                        | 210.5 ms                                        | 2.81 ms                                        | 25.0 ms                                        | $\geq 948.2 \text{ ms}^{17}$                    |
|                                                                       | Cost-aware oracle        | 689 $\mu\text{s}$                   | 716 $\mu\text{s}$                       | 62.8 ms                                        | 1.08 ms                                        | 1.85 ms                                        | 210.3 ms                                        | 2.83 ms                                        | 26.6 ms                                        | $\geq 932.7 \text{ ms}^{17}$                    |
|                                                                       | Learned router (ours)    | 717 $\mu\text{s}$                   | 746 $\mu\text{s}$                       | <b>66.4 ms</b>                                 | 1.09 ms                                        | 1.90 ms                                        | <b>212.1 ms</b>                                 | 2.85 ms                                        | 25.4 ms                                        | $\geq 1.05 \text{ s}^{17}$                      |
| {NO, Jacobi (0.67)}                                                   | Solver only              | 354.4 ms                            | 502.4 ms                                | 964.9 ms                                       | $\geq 2.48 \text{ s}^{132}$                    | $\geq 2.48 \text{ s}^{132}$                    | $\geq 2.48 \text{ s}^{132}$                     | $\geq 2.48 \text{ s}^{136}$                    | $\geq 2.48 \text{ s}^{136}$                    | $\geq 2.48 \text{ s}^{136}$                     |
|                                                                       | HINTS ( $\tau=15$ )      | 1.03 ms                             | 1.03 ms                                 | 2.05 ms                                        | 1.88 ms                                        | 2.12 ms                                        | 10.0 ms                                         | 6.97 ms                                        | 62.6 ms                                        | 163.2 ms                                        |
|                                                                       | Best fixed schedule      | 729 $\mu\text{s}_{\text{best-sch}}$ | 768 $\mu\text{s}_{\text{best-sch}}$     | 1.86 ms <sub><math>\tau=10, t_0=0</math></sub> | 1.06 ms <sub>best-sch</sub>                    | 2.13 ms <sub>best-sch</sub>                    | 7.33 ms <sub><math>\tau=50, t_0=0</math></sub>  | 2.84 ms <sub>best-sch</sub>                    | 37.5 ms <sub>best-sch</sub>                    | 124.8 ms <sub><math>\tau=50, t_0=0</math></sub> |
|                                                                       | Best residual-decay rule | 721 $\mu\text{s}_{\text{best-rd}}$  | 753 $\mu\text{s}_{\text{best-rd}}$      | 2.88 ms <sub>best-rd</sub>                     | 1.03 ms <sub>best-rd</sub>                     | 2.11 ms <sub>best-rd</sub>                     | 7.22 ms <sub>best-rd</sub>                      | 2.81 ms <sub>best-rd</sub>                     | 36.0 ms <sub>best-rd</sub>                     | 100.4 ms <sub>best-rd</sub>                     |
|                                                                       | One-shot schedule        | 735 $\mu\text{s}$                   | 766 $\mu\text{s}$                       | 64.2 ms                                        | 1.06 ms                                        | 2.13 ms                                        | 818.6 ms                                        | 2.84 ms                                        | 37.5 ms                                        | $\geq 2.39 \text{ s}^{16}$                      |
|                                                                       | Greedy oracle (Alg. 1)   | 726 $\mu\text{s}$                   | 762 $\mu\text{s}$                       | 1.88 ms                                        | 1.04 ms                                        | 2.14 ms                                        | 7.24 ms                                         | 2.84 ms                                        | 42.4 ms                                        | 110.0 ms                                        |
|                                                                       | Cost-aware oracle        | 728 $\mu\text{s}$                   | 764 $\mu\text{s}$                       | 2.15 ms                                        | 1.03 ms                                        | 2.12 ms                                        | 7.16 ms                                         | 2.81 ms                                        | 39.0 ms                                        | 98.2 ms                                         |
|                                                                       | Learned router (ours)    | 764 $\mu\text{s}$                   | 813 $\mu\text{s}$                       | 2.21 ms                                        | 1.07 ms                                        | 2.16 ms                                        | 7.34 ms                                         | 2.92 ms                                        | 44.9 ms                                        | 120.7 ms                                        |
| {NO, GS}                                                              | Solver only              | 2739 ms                             | 388.2 ms                                | 742.4 ms                                       | 4.79 s                                         | 7.82 s                                         | $\geq 9.57 \text{ s}^{132}$                     | $\geq 8.57 \text{ s}^{136}$                    | $\geq 8.57 \text{ s}^{136}$                    | $\geq 8.57 \text{ s}^{136}$                     |
|                                                                       | HINTS ( $\tau=15$ )      | 1.58 ms                             | 1.88 ms                                 | 4.70 ms                                        | 4.48 ms                                        | 7.89 ms                                        | 13.1 ms                                         | 18.3 ms                                        | 58.1 ms                                        | 154.0 ms                                        |
|                                                                       | Best fixed schedule      | 726 $\mu\text{s}_{\text{best-sch}}$ | 810 $\mu\text{s}_{\text{best-sch}}$     | 2.56 ms <sub><math>\tau=10, t_0=0</math></sub> | 1.03 ms <sub>best-sch</sub>                    | 2.81 ms <sub>best-sch</sub>                    | 10.7 ms <sub><math>\tau=50, t_0=0</math></sub>  | 2.89 ms <sub>best-sch</sub>                    | 54.6 ms <sub><math>\tau=50, t_0=0</math></sub> | 141.1 ms <sub><math>\tau=50, t_0=0</math></sub> |
|                                                                       | Best residual-decay rule | 736 $\mu\text{s}_{\text{best-rd}}$  | 805 $\mu\text{s}_{\text{best-rd}}$      | 2.97 ms <sub>best-rd</sub>                     | 1.02 ms <sub>best-rd</sub>                     | 2.82 ms <sub>best-rd</sub>                     | 10.1 ms <sub>best-rd</sub>                      | 2.85 ms <sub>best-rd</sub>                     | 52.8 ms <sub>best-rd</sub>                     | 132.0 ms <sub>best-rd</sub>                     |
|                                                                       | One-shot schedule        | 730 $\mu\text{s}$                   | 794 $\mu\text{s}$                       | 149.7 ms                                       | 1.03 ms                                        | 2.81 ms                                        | 1.74 s                                          | 2.89 ms                                        | 53.0 ms                                        | $\geq 8.56 \text{ s}^{13}$                      |
|                                                                       | Greedy oracle (Alg. 1)   | 741 $\mu\text{s}$                   | 793 $\mu\text{s}$                       | 2.76 ms                                        | 1.01 ms                                        | 2.85 ms                                        | 10.5 ms                                         | 2.86 ms                                        | 51.8 ms                                        | 133.3 ms                                        |
|                                                                       | Cost-aware oracle        | 741 $\mu\text{s}$                   | 824 $\mu\text{s}$                       | 2.32 ms                                        | 1.01 ms                                        | 2.76 ms                                        | 9.83 ms                                         | 2.91 ms                                        | 53.1 ms                                        | 135.2 ms                                        |
|                                                                       | Learned router (ours)    | 752 $\mu\text{s}$                   | 853 $\mu\text{s}$                       | 2.40 ms                                        | 1.03 ms                                        | 2.83 ms                                        | 9.96 ms                                         | 2.93 ms                                        | 52.7 ms                                        | 140.1 ms                                        |
| {NO, SymGS}                                                           | Solver only              | 239 s                               | 339.8 ms                                | 652.5 ms                                       | 4.20 s                                         | 6.84 s                                         | 11.47 s                                         | $\geq 13.69 \text{ s}^{136}$                   | $\geq 13.69 \text{ s}^{136}$                   | $\geq 13.69 \text{ s}^{136}$                    |
|                                                                       | HINTS ( $\tau=15$ )      | 2.05 ms                             | 2.15 ms                                 | 4.33 ms                                        | 6.64 ms                                        | 7.49 ms                                        | 13.6 ms                                         | 24.6 ms                                        | 46.9 ms                                        | 119.2 ms                                        |
|                                                                       | Best fixed schedule      | 761 $\mu\text{s}_{\text{best-sch}}$ | 807 $\mu\text{s}_{\text{best-sch}}$     | 2.14 ms <sub><math>\tau=10, t_0=0</math></sub> | 998 $\mu\text{s}_{\text{best-sch}}$            | 2.65 ms <sub><math>\tau=10, t_0=0</math></sub> | 8.98 ms <sub><math>\tau=50, t_0=0</math></sub>  | 2.93 ms <sub>best-sch</sub>                    | 44.1 ms <sub><math>\tau=50, t_0=0</math></sub> | 111.9 ms <sub><math>\tau=50, t_0=0</math></sub> |
|                                                                       | Best residual-decay rule | 753 $\mu\text{s}_{\text{best-rd}}$  | 824 $\mu\text{s}_{\text{best-rd}}$      | 3.78 ms <sub>best-rd</sub>                     | 1.00 ms <sub>best-rd</sub>                     | 2.60 ms <sub>best-rd</sub>                     | 10.4 ms <sub>best-rd</sub>                      | 2.74 ms <sub>best-rd</sub>                     | 42.6 ms <sub>best-rd</sub>                     | 110.9 ms <sub>best-rd</sub>                     |
|                                                                       | One-shot schedule        | 759 $\mu\text{s}$                   | 822 $\mu\text{s}$                       | 118.2 ms                                       | 998 $\mu\text{s}$                              | 2.70 ms                                        | 1.34 s                                          | 2.77 ms                                        | 42.8 ms                                        | $\geq 13.72 \text{ s}^{13}$                     |
|                                                                       | Greedy oracle (Alg. 1)   | 761 $\mu\text{s}$                   | 812 $\mu\text{s}$                       | 2.68 ms                                        | 1.04 ms                                        | 2.72 ms                                        | 9.00 ms                                         | 2.84 ms                                        | 42.2 ms                                        | 109.5 ms                                        |
|                                                                       | Cost-aware oracle        | 766 $\mu\text{s}$                   | 830 $\mu\text{s}$                       | 2.25 ms                                        | 1.02 ms                                        | 2.68 ms                                        | 8.88 ms                                         | 2.94 ms                                        | 42.5 ms                                        | 111.9 ms                                        |
|                                                                       | Learned router (ours)    | 783 $\mu\text{s}$                   | 864 $\mu\text{s}$                       | 2.33 ms                                        | 1.05 ms                                        | 2.72 ms                                        | 9.08 ms                                         | 2.86 ms                                        | 42.8 ms                                        | 112.5 ms                                        |
| {NO, SOR (1.5)}                                                       | Solver only              | 63.4 ms                             | 89.8 ms                                 | 172.2 ms                                       | 1.27 s                                         | 2.22 s                                         | 3.72 s                                          | $\geq 8.44 \text{ s}^{136}$                    | $\geq 8.44 \text{ s}^{136}$                    | $\geq 8.44 \text{ s}^{136}$                     |
|                                                                       | HINTS ( $\tau=15$ )      | 1.59 ms                             | 3.14 ms                                 | 4.71 ms                                        | 4.57 ms                                        | 8.81 ms                                        | 13.0 ms                                         | 17.0 ms                                        | 34.0 ms                                        | 61.8 ms                                         |
|                                                                       | Best fixed schedule      | 731 $\mu\text{s}_{\text{best-sch}}$ | 876 $\mu\text{s}_{\text{best-sch}}$     | 3.39 ms <sub><math>\tau=10, t_0=0</math></sub> | 1.00 ms <sub>best-sch</sub>                    | 2.46 ms <sub><math>\tau=10, t_0=0</math></sub> | 8.66 ms <sub><math>\tau=50, t_0=0</math></sub>  | 2.79 ms <sub>best-sch</sub>                    | 23.0 ms <sub><math>\tau=50, t_0=0</math></sub> | 58.3 ms <sub><math>\tau=50, t_0=0</math></sub>  |
|                                                                       | Best residual-decay rule | 727 $\mu\text{s}_{\text{best-rd}}$  | 867 $\mu\text{s}_{\text{best-rd}}$      | 3.72 ms <sub>best-rd</sub>                     | 997 $\mu\text{s}_{\text{best-rd}}$             | 2.33 ms <sub>best-rd</sub>                     | 9.87 ms <sub>best-rd</sub>                      | 2.76 ms <sub>best-rd</sub>                     | 22.9 ms <sub>best-rd</sub>                     | 55.8 ms <sub>best-rd</sub>                      |
|                                                                       | One-shot schedule        | 727 $\mu\text{s}$                   | 869 $\mu\text{s}$                       | 47.5 ms                                        | 1.00 ms                                        | 2.54 ms                                        | 720.4 ms                                        | 2.91 ms                                        | 23.6 ms                                        | $\geq 8.22 \text{ s}^{13}$                      |
|                                                                       | Greedy oracle (Alg. 1)   | 727 $\mu\text{s}$                   | 852 $\mu\text{s}$                       | 3.31 ms                                        | 984 $\mu\text{s}$                              | 2.46 ms                                        | 9.59 ms                                         | 2.90 ms                                        | 24.0 ms                                        | 60.5 ms                                         |
|                                                                       | Cost-aware oracle        | 728 $\mu\text{s}$                   | 851 $\mu\text{s}$                       | 3.06 ms                                        | 1.02 ms                                        | 2.48 ms                                        | 8.91 ms                                         | 2.81 ms                                        | 22.7 ms                                        | 56.4 ms                                         |
|                                                                       | Learned router (ours)    | 750 $\mu\text{s}$                   | 906 $\mu\text{s}$                       | 3.20 ms                                        | 1.05 ms                                        | 2.59 ms                                        | 9.25 ms                                         | 2.94 ms                                        | 23.3 ms                                        | 59.7 ms                                         |
| {NO, Multigrid}                                                       | Solver only              | 714 $\mu\text{s}$                   | 1.04 ms                                 | 2.01 ms                                        | 2.44 ms                                        | 4.34 ms                                        | 7.18 ms                                         | 10.8 ms                                        | 21.0 ms                                        | 31.6 ms                                         |
|                                                                       | HINTS ( $\tau=15$ )      | 714 $\mu\text{s}$                   | 1.05 ms                                 | 2.04 ms                                        | 2.45 ms                                        | 4.26 ms                                        | 7.24 ms                                         | 10.7 ms                                        | 20.9 ms                                        | 31.9 ms                                         |
|                                                                       | Best fixed schedule      | 713 $\mu\text{s}_{\text{best-sch}}$ | 1.05 ms <sub><math>\tau=10</math></sub> | 1.74 ms <sub><math>\tau=15, t_0=0</math></sub> | 996 $\mu\text{s}_{\text{best-sch}}$            | 2.34 ms <sub>best-sch</sub>                    | 5.81 ms <sub><math>\tau=50</math></sub>         | 2.92 ms <sub>best-sch</sub>                    | 13.6 ms <sub>best-sch</sub>                    | 24.2 ms <sub><math>\tau=25, t_0=0</math></sub>  |
|                                                                       | Best residual-decay rule | 736 $\mu\text{s}_{\text{best-rd}}$  | 1.05 ms <sub>best-rd</sub>              | 1.76 ms <sub>best-rd</sub>                     | 1.01 ms <sub>best-rd</sub>                     | 2.26 ms <sub>best-rd</sub>                     | 6.01 ms <sub>best-rd</sub>                      | 2.87 ms <sub>best-rd</sub>                     | 13.7 ms <sub>best-rd</sub>                     | 24.4 ms <sub>best-rd</sub>                      |
|                                                                       | One-shot schedule        | 743 $\mu\text{s}$                   | 1.04 ms                                 | 1.76 ms                                        | 991 $\mu\text{s}$                              | 2.34 ms                                        | 6.08 ms                                         | 2.93 ms                                        | 13.6 ms                                        | 24.1 ms                                         |
|                                                                       | Greedy oracle (Alg. 1)   | 730 $\mu\text{s}$                   | 1.06 ms                                 | 1.76 ms                                        | 976 $\mu\text{s}$                              | 2.27 ms                                        | 6.10 ms                                         | 2.99 ms                                        | 13.6 ms                                        | 24.6 ms                                         |
|                                                                       | Cost-aware oracle        | 713 $\mu\text{s}$                   | 1.01 ms                                 | 1.89 ms                                        | 1.01 ms                                        | 2.28 ms                                        | 5.98 ms                                         | 2.87 ms                                        | 13.5 ms                                        | 24.3 ms                                         |
|                                                                       | Learned router (ours)    | 756 $\mu\text{s}$                   | 1.09 ms                                 | 1.83 ms                                        | 1.01 ms                                        | 2.37 ms                                        | 6.05 ms                                         | 3.03 ms                                        | 13.7 ms                                        | 24.5 ms                                         |
| Classical baselines without corrector (same stencil, same accounting) |                          |                                     |                                         |                                                |                                                |                                                |                                                 |                                                |                                                |                                                 |
| Multigrid V(2,2) alone (point GS)                                     |                          |                                     |                                         |                                                |                                                |                                                |                                                 |                                                |                                                |                                                 |
| BiCGSTAB                                                              |                          |                                     |                                         |                                                |                                                |                                                |                                                 |                                                |                                                |                                                 |
| BiCGSTAB (multigrid)                                                  |                          |                                     |                                         |                                                |                                                |                                                |                                                 |                                                |                                                |                                                 |
| GMRES(20)                                                             |                          |                                     |                                         |                                                |                                                |                                                |                                                 |                                                |                                                |                                                 |
| FFT direct solve (exact; constant coefficients only)                  |                          |                                     |                                         |                                                |                                                |                                                |                                                 |                                                |                                                |                                                 |

Table 11: Convection–diffusion: paired median speedup [bootstrap 95% interval] of the learned router at  $\varepsilon = h^2$  and  $10^{-8}$  over HINTS ( $\tau=15$ ), the best fixed period, multigrid alone and multigrid-preconditioned BiCGSTAB, with two-sided Wilcoxon  $p$ -values (conventions as in Table 5).

| Pairing             | $\varepsilon=h^2$ | $\varepsilon=10^{-8}$ | $\varepsilon=h^2$ | $\varepsilon=10^{-8}$ | $\varepsilon=h^2$ | $\varepsilon=10^{-8}$ | $\varepsilon=h^2$ | $\varepsilon=10^{-8}$ | $\varepsilon=h^2$ | $\varepsilon=10^{-8}$ |
|---------------------|-------------------|-----------------------|-------------------|-----------------------|-------------------|-----------------------|-------------------|-----------------------|-------------------|-----------------------|
| {NO, Jacobi}        | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   |
| {NO, Jacobi (0.67)} | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   |
| {NO, GS}            | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   |
| {NO, SymGS}         | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   |
| {NO, SOR (1.5)}     | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   |
| {NO, Multigrid}     | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   | 1.0               | 1.0                   |

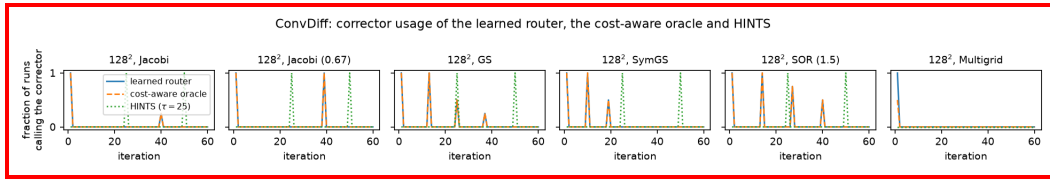


Figure 7: Convection–diffusion: corrector usage per iteration of the learned router, the cost-aware oracle and HINTS ( $\tau=15$ ) for every pairing (columns) and grid (rows).

Table 12: Convection–diffusion: median iterations (and corrector calls) to  $\varepsilon = h^2$  on every grid.

| Pairing             | 128 × 128 |        |        | 256 × 256 |        |        | 512 × 512 |         |         |
|---------------------|-----------|--------|--------|-----------|--------|--------|-----------|---------|---------|
|                     | HINTS-15  | oracle | router | HINTS-15  | oracle | router | HINTS-15  | oracle  | router  |
| {NO, Jacobi}        | 15 (1)    | 2 (1)  | 2 (1)  | 15 (1)    | 13 (1) | 13 (1) | 94 (6)    | 90 (1)  | 90 (1)  |
| {NO, Jacobi (0.67)} | 15 (1)    | 3 (1)  | 3 (1)  | 19 (1)    | 19 (1) | 19 (1) | 142 (9)   | 134 (1) | 134 (1) |
| {NO, GS}            | 20 (1)    | 2 (1)  | 2 (1)  | 28 (2)    | 8 (1)  | 8 (1)  | 49 (3)    | 47 (1)  | 47 (1)  |
| {NO, SymGS}         | 16 (1)    | 2 (1)  | 2 (1)  | 17 (1)    | 5 (1)  | 5 (1)  | 27 (1)    | 25 (1)  | 25 (1)  |
| {NO, SOR (1.5)}     | 30 (2)    | 4 (1)  | 4 (1)  | 30 (2)    | 7 (1)  | 7 (1)  | 30 (2)    | 22 (1)  | 22 (1)  |
| {NO, Multigrid}     | 3 (0)     | 2 (0)  | 2 (1)  | 4 (0)     | 2 (1)  | 2 (1)  | 4 (0)     | 3 (1)   | 3 (1)   |

Table 13: Convection–diffusion: measured per-iteration costs and macro-action sizes.

| Pairing             | 128 × 128           |                   |                     | 256 × 256           |                   |                     | 512 × 512           |                 |                     |
|---------------------|---------------------|-------------------|---------------------|---------------------|-------------------|---------------------|---------------------|-----------------|---------------------|
|                     | $c_{\text{solver}}$ | $c_{\text{NO}}$   | $m_{\text{solver}}$ | $c_{\text{solver}}$ | $c_{\text{NO}}$   | $m_{\text{solver}}$ | $c_{\text{solver}}$ | $c_{\text{NO}}$ | $m_{\text{solver}}$ |
| {NO, Jacobi}        | 21 $\mu\text{s}$    | 626 $\mu\text{s}$ | 30                  | 63 $\mu\text{s}$    | 937 $\mu\text{s}$ | 15                  | 489 $\mu\text{s}$   | 2.64 ms         | 5                   |
| {NO, Jacobi (0.67)} | 19 $\mu\text{s}$    | 626 $\mu\text{s}$ | 34                  | 63 $\mu\text{s}$    | 937 $\mu\text{s}$ | 15                  | 498 $\mu\text{s}$   | 2.64 ms         | 5                   |
| {NO, GS}            | 54 $\mu\text{s}$    | 626 $\mu\text{s}$ | 12                  | 231 $\mu\text{s}$   | 937 $\mu\text{s}$ | 4                   | 1.26 ms             | 2.64 ms         | 2                   |
| {NO, SymGS}         | 81 $\mu\text{s}$    | 626 $\mu\text{s}$ | 8                   | 373 $\mu\text{s}$   | 937 $\mu\text{s}$ | 3                   | 1.85 ms             | 2.64 ms         | 1                   |
| {NO, SOR (1.5)}     | 54 $\mu\text{s}$    | 626 $\mu\text{s}$ | 12                  | 230 $\mu\text{s}$   | 937 $\mu\text{s}$ | 4                   | 1.26 ms             | 2.64 ms         | 2                   |
| {NO, Multigrid}     | 288 $\mu\text{s}$   | 626 $\mu\text{s}$ | 2                   | 1.19 ms             | 937 $\mu\text{s}$ | 1                   | 5.70 ms             | 2.64 ms         | 1                   |

Table 14: Convection–diffusion, 128 × 128: iteration-based metrics (conventions as in Table 8).

| Method                | AUC                       | $p$          |
|-----------------------|---------------------------|--------------|
| {NO, Jacobi}          |                           |              |
| Solver only           | 2.60e+02 (1.4e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.39e+01 (5.7e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>1.38e-02 (3.2e-03)</b> | -            |
| Cost-aware oracle     | 1.38e-02 (3.2e-03)        | -            |
| {NO, Jacobi (0.67)}   |                           |              |
| Solver only           | 2.71e+02 (1.1e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.39e+01 (5.1e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>4.83e-03 (1.0e-03)</b> | -            |
| Cost-aware oracle     | 4.83e-03 (1.0e-03)        | -            |
| {NO, GS}              |                           |              |
| Solver only           | 2.24e+02 (1.8e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.37e+01 (1.4e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>3.76e-03 (8.1e-04)</b> | -            |
| Cost-aware oracle     | 3.76e-03 (8.1e-04)        | -            |
| {NO, SymGS}           |                           |              |
| Solver only           | 1.89e+02 (2.2e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.35e+01 (2.2e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.89e-03 (6.3e-04)</b> | -            |
| Cost-aware oracle     | 2.89e-03 (6.3e-04)        | -            |
| {NO, SOR (1.5)}       |                           |              |
| Solver only           | 1.21e+02 (1.5e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.31e+01 (3.6e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>5.86e-03 (1.3e-03)</b> | -            |
| Cost-aware oracle     | 5.86e-03 (1.3e-03)        | -            |

Table 15: Convection–diffusion, 128 × 128: five-seed router-training trials (conventions as in Table 9).

| $N$              | Pairing             | time to $h^2$ over 5 seeds | identical decisions | speedup vs. HINTS-15 | seeds with $p < 0.01$ |
|------------------|---------------------|----------------------------|---------------------|----------------------|-----------------------|
| 128 <sup>2</sup> | {NO, Jacobi}        | 0.68 ± 0.00 ms             | 100%                | 1.44–1.44×           | 0/5                   |
| 128 <sup>2</sup> | {NO, Jacobi (0.67)} | 0.69 ± 0.00 ms             | 100%                | 1.31–1.31×           | 0/5                   |
| 128 <sup>2</sup> | {NO, GS}            | 0.73 ± 0.00 ms             | 100%                | 2.31–2.31×           | 0/5                   |
| 128 <sup>2</sup> | {NO, SymGS}         | 0.73 ± 0.00 ms             | 100%                | 2.54–2.54×           | 0/5                   |
| 128 <sup>2</sup> | {NO, SOR (1.5)}     | 0.78 ± 0.00 ms             | 100%                | 3.54–3.54×           | 0/5                   |

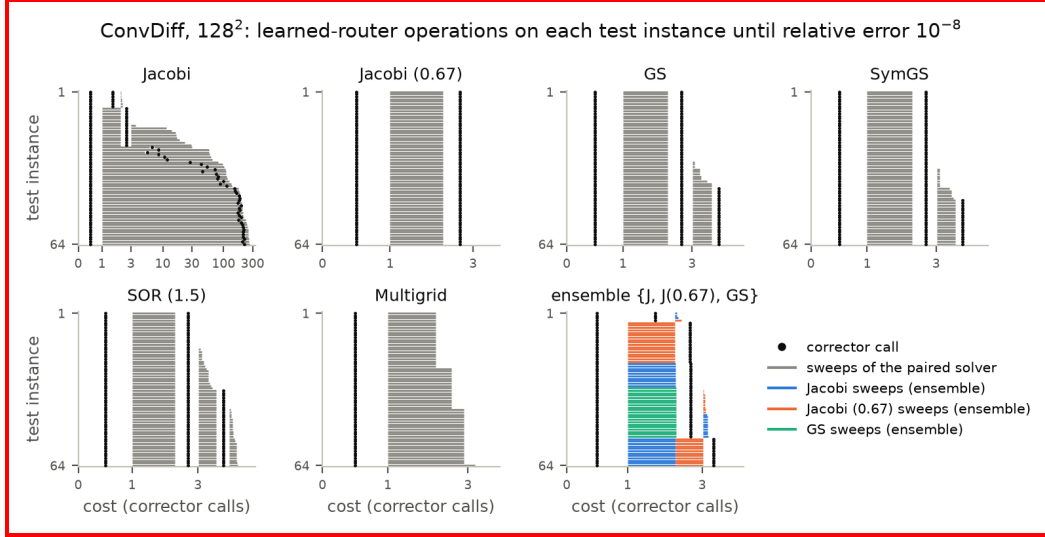


Figure 8: Convection–diffusion,  $128 \times 128$ : operations of the learned router on each of the 64 test instances (rows, sorted by cost) until the relative error first falls below  $10^{-8}$ , for every pairing and for the ensemble  $\{\text{Jacobi}, \text{Jacobi (0.67)}, \text{GS}\}$ , as executed in the timed replays. Dots mark corrector calls and lines the solver sweeps between them (coloured by solver in the ensemble); the horizontal axis is the cumulative cost in units of one corrector call, linear below one call and logarithmic above.

#### 1269 E.4 Anisotropic diffusion

1270 The anisotropic equation is where the two components of the hybrid are most complementary and  
 1271 where the point smoothers are weakest: they damp the  $x_1$ -oscillatory modes only by  $(1 - \epsilon)/(1 + \epsilon) \approx$   
 1272  $0.98$  per sweep (none of the five point smoothers alone reaches  $h^2$  within the iteration cap at  $128^2$ ,  
 1273 Table 16), whereas the corrector, whose sensor grid keeps full resolution along  $x_1$ , removes exactly  
 1274 those modes together with the smooth ones in a single call and leaves the smoother only the  $x_2$ -  
 1275 oscillatory band, which it damps at the isotropic rate. The corrector’s  $N \times N/4$  band is thus the spectral  
 1276 counterpart of semi-coarsening, and the honest classical comparison is the method a practitioner  
 1277 would use on this problem: line Gauss–Seidel along  $x_2$  (which contracts the  $x_1$ -oscillatory modes  
 1278 by  $0.35$ – $0.45$  per sweep, measured on the modes  $(N/2, 0)$  and  $(N/4, 0)$  at  $128^2$ , where point Jacobi  
 1279 contracts them by  $0.98$ ) and line-smoothed multigrid, which regains grid-independent convergence  
 1280 (asymptotically  $\approx 0.04$  per cycle on one  $128^2$  instance, against  $\approx 0.86$  with point smoothing). Both  
 1281 are included, line GS as the sixth pairing  $\{\text{NO}, \text{line GS}\}$  and, together with point-smoothed multigrid  
 1282 and the CG variants, as classical baselines timed in the same session; against point-smoothed multigrid  
 1283 and multigrid-preconditioned CG the router is faster in every pairing at both tolerances except the  
 1284 Jacobi pairing at  $10^{-8}$  (Table 17). For the point-smoother pairings the router reaches  $h^2$  in the time  
 1285 of one corrector call plus a few sweeps, as does the one-shot schedule, while HINTS ( $\tau=15$ ) is  
 1286  $1.4$ – $2.8\times$  slower there and the best fixed schedule, one call at  $t = 0$  followed by sweeps, is  $4$ – $5\%$   
 1287 faster than the router, the cost of the router’s per-step decision (Table 17). At  $10^{-8}$  the router is  
 1288 faster than the best fixed schedule in the Jacobi pairing ( $1.7\times$ ), at parity in the GS, SymGS and  
 1289 damped-Jacobi pairings, and slower in the SOR ( $0.95\times$ ) and line-GS ( $0.83\times$ ) pairings, the latter  
 1290 being one whose best schedule has a period shorter than a macro-action; the five-seed trials for  
 1291 this equation are **[pending]** (Table 21). The  $512^2$  grid is not run: the anisotropic corrector keeps  
 1292 full resolution along  $x_1$  ( $N \times N/4$  sensors), so at  $512^2$  its least-squares branch would be a dense  
 1293  $65,536 \times 65,536$  matrix (17 GB in single precision), which the current design cannot hold; a coarser  
 1294  $x_1$  resolution would leave the point smoothers modes they cannot damp (Section E.7 discusses exactly  
 1295 this regime).

Table 16: Anisotropic diffusion: median wall-clock time to relative error  $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$  on the  $128^2$  (64 instances, six pairings including line GS) and  $256^2$  (16 instances) grids for every policy, and for the classical baselines without corrector (conventions as in Table 4; the point-smoother-only runs are capped at 60,000 and 40,000 iterations; censored runs are marked †).

| Pairing                                                               | Method                                               | 128 × 128                           |                                     |                                    | 256 × 256                          |                                    |                                    |
|-----------------------------------------------------------------------|------------------------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
|                                                                       |                                                      | $\varepsilon=10^{-3}$               | $\varepsilon=h^2$                   | $\varepsilon=10^{-8}$              | $\varepsilon=10^{-3}$              | $\varepsilon=h^2$                  | $\varepsilon=10^{-8}$              |
| {NO, Jacobi}                                                          | Solver only                                          | $\geq 1.23 \text{ s}^{164}$         | $\geq 1.23 \text{ s}^{164}$         | $\geq 1.23 \text{ s}^{164}$        | $\geq 2.53 \text{ s}^{116}$        | $\geq 2.53 \text{ s}^{116}$        | $\geq 2.53 \text{ s}^{116}$        |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.04 ms                             | 1.04 ms                             | 53.1 ms                            | 9.74 ms                            | 9.74 ms                            | 19.5 ms                            |
|                                                                       | Best fixed schedule                                  | 722 $\mu\text{s}_{\tau=50, t_0=0}$  | 729 $\mu\text{s}_{\tau=50, t_0=0}$  | 26.8 $\text{ms}_{\tau=50, t_0=0}$  | 8.82 $\text{ms}_{\text{one-shot}}$ | 8.82 $\text{ms}_{\text{one-shot}}$ | 18.7 $\text{ms}_{\tau=15, t_0=0}$  |
|                                                                       | Best residual-decay rule                             | 721 $\mu\text{s}_{\text{one-shot}}$ | 732 $\mu\text{s}_{\text{one-shot}}$ | 32.6 $\text{ms}_{\text{one-shot}}$ | 8.79 $\text{ms}_{\text{one-shot}}$ | 8.81 $\text{ms}_{\text{one-shot}}$ | 25.9 $\text{ms}_{\text{one-shot}}$ |
|                                                                       | One-shot schedule                                    | 730 $\mu\text{s}$                   | 738 $\mu\text{s}$                   | $\geq 1.23 \text{ s}^{151}$        | 8.82 ms                            | 8.82 ms                            | $\geq 2.54 \text{ s}^{116}$        |
|                                                                       | Greedy oracle (Alg. 1)                               | 722 $\mu\text{s}$                   | 744 $\mu\text{s}$                   | 17.2 ms                            | 8.78 ms                            | 8.80 ms                            | 18.8 ms                            |
|                                                                       | Cost-aware oracle                                    | 717 $\mu\text{s}$                   | 720 $\mu\text{s}$                   | 17.5 ms                            | 8.82 ms                            | 8.83 ms                            | 25.8 ms                            |
|                                                                       | Learned router (ours)                                | 759 $\mu\text{s}$                   | 764 $\mu\text{s}$                   | <b>17.0 ms</b>                     | 8.82 ms                            | 8.82 ms                            | 18.9 ms                            |
| {NO, Jacobi (0.67)}                                                   | Solver only                                          | $\geq 1.21 \text{ s}^{164}$         | $\geq 1.21 \text{ s}^{164}$         | $\geq 1.21 \text{ s}^{164}$        | $\geq 2.53 \text{ s}^{116}$        | $\geq 2.53 \text{ s}^{116}$        | $\geq 2.53 \text{ s}^{116}$        |
|                                                                       | HINTS ( $\tau=15$ )                                  | 970 $\mu\text{s}$                   | 970 $\mu\text{s}$                   | 2.00 ms                            | 9.69 ms                            | 9.69 ms                            | 19.3 ms                            |
|                                                                       | Best fixed schedule                                  | 668 $\mu\text{s}_{\tau=15, t_0=0}$  | 681 $\mu\text{s}_{\tau=25, t_0=0}$  | 1.92 $\text{ms}_{\tau=25, t_0=0}$  | 8.73 $\text{ms}_{\tau=50, t_0=0}$  | 8.74 $\text{ms}_{\tau=50, t_0=0}$  | 19.0 $\text{ms}_{\tau=25, t_0=0}$  |
|                                                                       | Best residual-decay rule                             | 665 $\mu\text{s}_{\text{one-shot}}$ | 679 $\mu\text{s}_{\text{one-shot}}$ | 2.57 $\text{ms}_{\text{one-shot}}$ | 8.74 $\text{ms}_{\text{one-shot}}$ | 8.75 $\text{ms}_{\text{one-shot}}$ | 25.7 $\text{ms}_{\text{one-shot}}$ |
|                                                                       | One-shot schedule                                    | 683 $\mu\text{s}$                   | 688 $\mu\text{s}$                   | $\geq 1.22 \text{ s}^{159}$        | 8.77 ms                            | 8.77 ms                            | $\geq 2.53 \text{ s}^{116}$        |
|                                                                       | Greedy oracle (Alg. 1)                               | 672 $\mu\text{s}$                   | 684 $\mu\text{s}$                   | 1.89 ms                            | 8.74 ms                            | 8.77 ms                            | 18.6 ms                            |
|                                                                       | Cost-aware oracle                                    | 668 $\mu\text{s}$                   | 680 $\mu\text{s}$                   | 1.97 ms                            | 8.76 ms                            | 8.82 ms                            | 22.2 ms                            |
|                                                                       | Learned router (ours)                                | 695 $\mu\text{s}$                   | 700 $\mu\text{s}$                   | 2.03 ms                            | 8.77 ms                            | 8.80 ms                            | 18.7 ms                            |
| {NO, GS}                                                              | Solver only                                          | $\geq 3.40 \text{ s}^{164}$         | $\geq 3.40 \text{ s}^{164}$         | $\geq 3.40 \text{ s}^{164}$        | $\geq 9.40 \text{ s}^{116}$        | $\geq 9.40 \text{ s}^{116}$        | $\geq 9.40 \text{ s}^{116}$        |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.54 ms                             | 1.54 ms                             | 7.06 ms                            | 12.5 ms                            | 12.5 ms                            | 25.1 ms                            |
|                                                                       | Best fixed schedule                                  | 666 $\mu\text{s}_{\tau=10, t_0=0}$  | 672 $\mu\text{s}_{\tau=10, t_0=0}$  | 2.30 $\text{ms}_{\tau=15, t_0=0}$  | 8.71 $\text{ms}_{\tau=10, t_0=0}$  | 8.74 $\text{ms}_{\tau=10, t_0=0}$  | 21.3 $\text{ms}_{\tau=10, t_0=0}$  |
|                                                                       | Best residual-decay rule                             | 669 $\mu\text{s}_{\text{one-shot}}$ | 674 $\mu\text{s}_{\text{one-shot}}$ | 3.29 $\text{ms}_{\text{one-shot}}$ | 8.71 $\text{ms}_{\text{one-shot}}$ | 8.74 $\text{ms}_{\text{one-shot}}$ | 33.8 $\text{ms}_{\text{one-shot}}$ |
|                                                                       | One-shot schedule                                    | 687 $\mu\text{s}$                   | 699 $\mu\text{s}$                   | $\geq 3.37 \text{ s}^{143}$        | 8.76 ms                            | 8.77 ms                            | $\geq 9.40 \text{ s}^{116}$        |
|                                                                       | Greedy oracle (Alg. 1)                               | 670 $\mu\text{s}$                   | 688 $\mu\text{s}$                   | 2.95 ms                            | 8.73 ms                            | 8.76 ms                            | 29.0 ms                            |
|                                                                       | Cost-aware oracle                                    | 665 $\mu\text{s}$                   | 678 $\mu\text{s}$                   | 2.33 ms                            | 8.74 ms                            | 8.76 ms                            | 23.2 ms                            |
|                                                                       | Learned router (ours)                                | 695 $\mu\text{s}$                   | 710 $\mu\text{s}$                   | 2.52 ms                            | 8.75 ms                            | 8.76 ms                            | 23.0 ms                            |
| {NO, SymGS}                                                           | Solver only                                          | $\geq 5.05 \text{ s}^{164}$         | $\geq 5.05 \text{ s}^{164}$         | $\geq 5.05 \text{ s}^{164}$        | $\geq 15.43 \text{ s}^{116}$       | $\geq 15.43 \text{ s}^{116}$       | $\geq 15.43 \text{ s}^{116}$       |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.98 ms                             | 1.98 ms                             | 3.98 ms                            | 14.6 ms                            | 14.6 ms                            | 29.2 ms                            |
|                                                                       | Best fixed schedule                                  | 669 $\mu\text{s}_{\tau=50, t_0=0}$  | 677 $\mu\text{s}_{\tau=50, t_0=0}$  | 2.24 $\text{ms}_{\tau=10, t_0=0}$  | 8.72 $\text{ms}_{\tau=10, t_0=0}$  | 8.73 $\text{ms}_{\tau=10, t_0=0}$  | 21.2 $\text{ms}_{\tau=10, t_0=0}$  |
|                                                                       | Best residual-decay rule                             | 668 $\mu\text{s}_{\text{one-shot}}$ | 679 $\mu\text{s}_{\text{one-shot}}$ | 3.41 $\text{ms}_{\text{one-shot}}$ | 8.71 $\text{ms}_{\text{one-shot}}$ | 8.74 $\text{ms}_{\text{one-shot}}$ | 33.9 $\text{ms}_{\text{one-shot}}$ |
|                                                                       | One-shot schedule                                    | 683 $\mu\text{s}$                   | 700 $\mu\text{s}$                   | $\geq 3.21 \text{ s}^{149}$        | 8.75 ms                            | 8.77 ms                            | $\geq 15.44 \text{ s}^{116}$       |
|                                                                       | Greedy oracle (Alg. 1)                               | 666 $\mu\text{s}$                   | 678 $\mu\text{s}$                   | 2.69 ms                            | 8.73 ms                            | 8.76 ms                            | 28.4 ms                            |
|                                                                       | Cost-aware oracle                                    | 669 $\mu\text{s}$                   | 683 $\mu\text{s}$                   | 2.15 ms                            | 8.71 ms                            | 8.73 ms                            | 22.8 ms                            |
|                                                                       | Learned router (ours)                                | 701 $\mu\text{s}$                   | 714 $\mu\text{s}$                   | 2.24 ms                            | 8.75 ms                            | 8.75 ms                            | 22.9 ms                            |
| {NO, SOR (1.5)}                                                       | Solver only                                          | $\geq 3.39 \text{ s}^{164}$         | $\geq 3.39 \text{ s}^{164}$         | $\geq 3.39 \text{ s}^{164}$        | $\geq 9.56 \text{ s}^{116}$        | $\geq 9.56 \text{ s}^{116}$        | $\geq 9.56 \text{ s}^{116}$        |
|                                                                       | HINTS ( $\tau=15$ )                                  | 1.53 ms                             | 1.53 ms                             | 4.79 ms                            | 12.5 ms                            | 12.5 ms                            | 26.7 ms                            |
|                                                                       | Best fixed schedule                                  | 656 $\mu\text{s}_{\tau=10, t_0=0}$  | 665 $\mu\text{s}_{\tau=10, t_0=0}$  | 2.67 $\text{ms}_{\tau=15, t_0=0}$  | 8.72 $\text{ms}_{\tau=10, t_0=0}$  | 8.74 $\text{ms}_{\tau=10, t_0=0}$  | 23.8 $\text{ms}_{\tau=10, t_0=0}$  |
|                                                                       | Best residual-decay rule                             | 658 $\mu\text{s}_{\text{one-shot}}$ | 664 $\mu\text{s}_{\text{one-shot}}$ | 3.56 $\text{ms}_{\text{one-shot}}$ | 8.72 $\text{ms}_{\text{one-shot}}$ | 8.74 $\text{ms}_{\text{one-shot}}$ | 34.0 $\text{ms}_{\text{one-shot}}$ |
|                                                                       | One-shot schedule                                    | 667 $\mu\text{s}$                   | 678 $\mu\text{s}$                   | $\geq 1.51 \text{ s}^{146}$        | 8.74 ms                            | 8.79 ms                            | $\geq 9.59 \text{ s}^{116}$        |
|                                                                       | Greedy oracle (Alg. 1)                               | 658 $\mu\text{s}$                   | 663 $\mu\text{s}$                   | 3.38 ms                            | 8.73 ms                            | 8.74 ms                            | 30.0 ms                            |
|                                                                       | Cost-aware oracle                                    | 656 $\mu\text{s}$                   | 664 $\mu\text{s}$                   | 2.68 ms                            | 8.72 ms                            | 8.75 ms                            | 25.7 ms                            |
|                                                                       | Learned router (ours)                                | 687 $\mu\text{s}$                   | 690 $\mu\text{s}$                   | 2.95 ms                            | 8.77 ms                            | 8.81 ms                            | 25.9 ms                            |
| {NO, Line GS}                                                         | Solver only                                          | 248.6 ms                            | 349.2 ms                            | 662.9 ms                           | 3.67 s                             | 5.90 s                             | 9.80 s                             |
|                                                                       | HINTS ( $\tau=15$ )                                  | 2.00 ms                             | 2.00 ms                             | 3.99 ms                            | 13.8 ms                            | 13.8 ms                            | 27.5 ms                            |
|                                                                       | Best fixed schedule                                  | 652 $\mu\text{s}_{\tau=10, t_0=0}$  | 656 $\mu\text{s}_{\tau=10, t_0=0}$  | 1.67 $\text{ms}_{\tau=5, t_0=0}$   | 8.72 $\text{ms}_{\tau=15, t_0=0}$  | 8.73 $\text{ms}_{\tau=15, t_0=0}$  | 18.8 $\text{ms}_{\tau=15, t_0=0}$  |
|                                                                       | Best residual-decay rule                             | 655 $\mu\text{s}_{\text{one-shot}}$ | 658 $\mu\text{s}_{\text{one-shot}}$ | 3.33 $\text{ms}_{\text{one-shot}}$ | 8.71 $\text{ms}_{\text{one-shot}}$ | 8.73 $\text{ms}_{\text{one-shot}}$ | 33.9 $\text{ms}_{\text{one-shot}}$ |
|                                                                       | One-shot schedule                                    | 656 $\mu\text{s}$                   | 660 $\mu\text{s}$                   | 48.5 ms                            | 8.74 ms                            | 8.76 ms                            | 1.16 s                             |
|                                                                       | Greedy oracle (Alg. 1)                               | 654 $\mu\text{s}$                   | 657 $\mu\text{s}$                   | 1.48 ms                            | 8.72 ms                            | 8.74 ms                            | 17.8 ms                            |
|                                                                       | Cost-aware oracle                                    | 652 $\mu\text{s}$                   | 657 $\mu\text{s}$                   | 1.96 ms                            | 8.73 ms                            | 8.74 ms                            | 25.8 ms                            |
|                                                                       | Learned router (ours)                                | 682 $\mu\text{s}$                   | 686 $\mu\text{s}$                   | 2.03 ms                            | 8.76 ms                            | 8.76 ms                            | <b>18.0 ms</b>                     |
| Classical baselines without corrector (same stencil, same accounting) |                                                      |                                     |                                     |                                    |                                    |                                    |                                    |
|                                                                       | Multigrid V(2,2) alone (point GS)                    | 2.70 ms                             | 8.19 ms                             | 40.1 ms                            | 9.85 ms                            | 48.8 ms                            | 158.3 ms                           |
|                                                                       | Multigrid V(2,2) alone (line GS)                     | 1.05 ms                             | 1.06 ms                             | 2.55 ms                            | 3.83 ms                            | 5.64 ms                            | 9.33 ms                            |
|                                                                       | CG                                                   | 17.6 ms                             | 20.0 ms                             | 25.7 ms                            | 97.0 ms                            | 117.3 ms                           | 145.0 ms                           |
|                                                                       | PCG (SymGS)                                          | 23.5 ms                             | 26.6 ms                             | 49.8 ms                            | 186.6 ms                           | 223.6 ms                           | 381.1 ms                           |
|                                                                       | PCG (multigrid)                                      | 1.05 ms                             | 2.08 ms                             | 6.95 ms                            | 3.91 ms                            | 11.3 ms                            | 26.4 ms                            |
|                                                                       | FFT direct solve (exact; constant coefficients only) | 245 $\mu\text{s}$                   | 245 $\mu\text{s}$                   | 245 $\mu\text{s}$                  | 1.07 ms                            | 1.07 ms                            | 1.07 ms                            |

Table 17: Anisotropic diffusion: paired median speedup [bootstrap 95% interval] of the learned router at  $\varepsilon = h^2$  and  $10^{-8}$  over HINTS ( $\tau=15$ ) and the best fixed period, with two-sided Wilcoxon  $p$ -values and bootstrap intervals; the classical baselines (point- and line-smoothed multigrid, CG variants) are timed in the same session.

| Pairing                          | 128 × 128                 |                                  |                       |                          |                              |                            |                       |                        |                        |                          | 256 × 256                 |                                  |                       |                          |                              |                            |                       |                        |                        |                          |
|----------------------------------|---------------------------|----------------------------------|-----------------------|--------------------------|------------------------------|----------------------------|-----------------------|------------------------|------------------------|--------------------------|---------------------------|----------------------------------|-----------------------|--------------------------|------------------------------|----------------------------|-----------------------|------------------------|------------------------|--------------------------|
|                                  | 128 <sup>2</sup> , Jacobi | 128 <sup>2</sup> , Jacobi (0.67) | 128 <sup>2</sup> , GS | 128 <sup>2</sup> , SymGS | 128 <sup>2</sup> , SOR (1.5) | 128 <sup>2</sup> , Line GS | 128 <sup>2</sup> , CG | 128 <sup>2</sup> , PCG | 128 <sup>2</sup> , FFT | 128 <sup>2</sup> , HINTS | 256 <sup>2</sup> , Jacobi | 256 <sup>2</sup> , Jacobi (0.67) | 256 <sup>2</sup> , GS | 256 <sup>2</sup> , SymGS | 256 <sup>2</sup> , SOR (1.5) | 256 <sup>2</sup> , Line GS | 256 <sup>2</sup> , CG | 256 <sup>2</sup> , PCG | 256 <sup>2</sup> , FFT | 256 <sup>2</sup> , HINTS |
| 128 <sup>2</sup> , Jacobi        | 1.04                      | 1.04                             | 53.1                  | 9.74                     | 9.74                         | 19.5                       | 25.9                  | 25.4                   | 1.04                   | 1.04                     | 9.69                      | 9.69                             | 19.3                  | 19.0                     | 25.7                         | 25.7                       | 25.7                  | 25.7                   | 25.7                   | 19.3                     |
| 128 <sup>2</sup> , Jacobi (0.67) | 970                       | 970                              | 2.00                  | 8.73                     | 8.74                         | 19.0                       | 25.7                  | 25.7                   | 970                    | 970                      | 9.69                      | 9.69                             | 19.3                  | 19.0                     | 25.7                         | 25.7                       | 25.7                  | 25.7                   | 25.7                   | 19.3                     |
| 128 <sup>2</sup> , GS            | 722                       | 729                              | 26.8                  | 8.82                     | 8.82                         | 18.7                       | 25.9                  | 25.4                   | 722                    | 729                      | 8.82                      | 8.82                             | 18.8                  | 25.8                     | 25.8                         | 25.8                       | 25.8                  | 25.8                   | 25.8                   | 18.9                     |
| 128 <sup>2</sup> , SymGS         | 722                       | 744                              | 17.2                  | 8.78                     | 8.80                         | 18.8                       | 25.9                  | 25.4                   | 722                    | 744                      | 8.78                      | 8.80                             | 18.8                  | 25.8                     | 25.8                         | 25.8                       | 25.8                  | 25.8                   | 25.8                   | 18.9                     |
| 128 <sup>2</sup> , SOR (1.5)     | 717                       | 720                              | 17.5                  | 8.82                     | 8.83                         | 25.8                       | 25.9                  | 25.4                   | 717                    | 720                      | 8.82                      | 8.83                             | 25.8                  | 25.8                     | 25.8                         | 25.8                       | 25.8                  | 25.8                   | 25.8                   | 18.9                     |
| 128 <sup>2</sup> , Line GS       | 759                       | 764                              | 17.0                  | 8.82                     | 8.82                         | 18.9                       | 25.9                  | 25.4                   | 759                    | 764                      | 8.82                      | 8.82                             | 18.9                  | 25.8                     | 25.8                         | 25.8                       | 25.8                  | 25.8                   | 25.8                   | 18.9                     |
| 128 <sup>2</sup> , CG            | 17.6                      | 20.0                             | 25.7                  | 97.0                     | 117.3                        | 145.0                      | 381.1                 | 26.4                   | 17.6                   | 20.0                     | 25.7                      | 97.0                             | 117.3                 | 145.0                    | 381.1                        | 26.4                       | 17.6                  | 20.0                   | 25.7                   | 97.0                     |
| 128 <sup>2</sup> , PCG           | 23.5                      | 26.6                             | 49.8                  | 186.6                    | 223.6                        | 381.1                      | 26.4                  | 23.5                   | 23.5                   | 26.6                     | 49.8                      | 186.6                            | 223.6                 | 381.1                    | 26.4                         | 23.5                       | 23.5                  | 26.6                   | 49.8                   | 186.6                    |
| 128 <sup>2</sup> , FFT           | 245                       | 245                              | 245                   | 1.07                     | 1.07                         | 1.07                       | 1.07                  | 245                    | 245                    | 245                      | 1.07                      | 1.07                             | 1.07                  | 1.07                     | 1.07                         | 1.07                       | 1.07                  | 1.07                   | 1.07                   | 1.07                     |

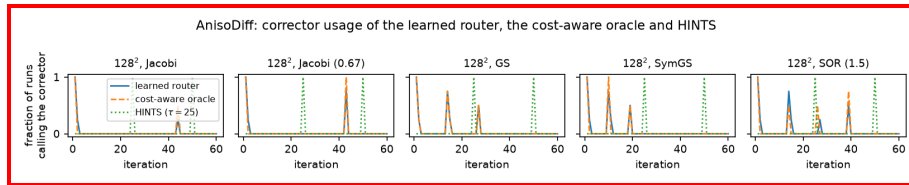


Figure 9: Anisotropic diffusion: corrector usage per iteration of the learned router, the cost-aware oracle and HINTS ( $\tau=15$ ) for every pairing (columns) and grid (rows).

Table 18: Anisotropic diffusion: median iterations (and corrector calls) to  $\varepsilon = h^2$ .

| Pairing             | 128 $\times$ 128 |        |        | 256 $\times$ 256 |        |        |
|---------------------|------------------|--------|--------|------------------|--------|--------|
|                     | HINTS-15         | oracle | router | HINTS-15         | oracle | router |
| {NO, Jacobi}        | 15 (1)           | 1 (1)  | 1 (1)  | 15 (1)           | 1 (1)  | 1 (1)  |
| {NO, Jacobi (0.67)} | 15 (1)           | 1 (1)  | 1 (1)  | 15 (1)           | 1 (1)  | 1 (1)  |
| {NO, GS}            | 15 (1)           | 1 (1)  | 1 (1)  | 15 (1)           | 1 (1)  | 1 (1)  |
| {NO, SymGS}         | 15 (1)           | 1 (1)  | 1 (1)  | 15 (1)           | 1 (1)  | 1 (1)  |
| {NO, SOR (1.5)}     | 15 (1)           | 1 (1)  | 1 (1)  | 15 (1)           | 1 (1)  | 1 (1)  |
| {NO, Line GS}       | 15 (1)           | 1 (1)  | 1 (1)  | 15 (1)           | 1 (1)  | 1 (1)  |

Table 19: Anisotropic diffusion: measured per-iteration costs and macro-action sizes.

| Pairing             | 128 $\times$ 128    |                   |                     | 256 $\times$ 256    |                 |                     |
|---------------------|---------------------|-------------------|---------------------|---------------------|-----------------|---------------------|
|                     | $c_{\text{solver}}$ | $c_{\text{NO}}$   | $m_{\text{solver}}$ | $c_{\text{solver}}$ | $c_{\text{NO}}$ | $m_{\text{solver}}$ |
| {NO, Jacobi}        | 21 $\mu\text{s}$    | 628 $\mu\text{s}$ | 30                  | 68 $\mu\text{s}$    | 8.74 ms         | 129                 |
| {NO, Jacobi (0.67)} | 21 $\mu\text{s}$    | 628 $\mu\text{s}$ | 30                  | 67 $\mu\text{s}$    | 8.74 ms         | 130                 |
| {NO, GS}            | 54 $\mu\text{s}$    | 628 $\mu\text{s}$ | 12                  | 258 $\mu\text{s}$   | 8.74 ms         | 34                  |
| {NO, SymGS}         | 82 $\mu\text{s}$    | 628 $\mu\text{s}$ | 8                   | 415 $\mu\text{s}$   | 8.74 ms         | 21                  |
| {NO, SOR (1.5)}     | 54 $\mu\text{s}$    | 628 $\mu\text{s}$ | 12                  | 257 $\mu\text{s}$   | 8.74 ms         | 34                  |
| {NO, Line GS}       | 84 $\mu\text{s}$    | 628 $\mu\text{s}$ | 7                   | 345 $\mu\text{s}$   | 8.74 ms         | 25                  |

Table 20: Anisotropic diffusion,  $128 \times 128$ : iteration-based metrics (conventions as in Table 8).

| Method                | AUC                       | $p$          |
|-----------------------|---------------------------|--------------|
| {NO, Jacobi}          |                           |              |
| Solver only           | 2.99e+02 (4.6e-02)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.40e+01 (1.3e-04)        | $< 10^{-10}$ |
| Learned router (ours) | <b>7.15e-04 (2.0e-04)</b> | -            |
| Cost-aware oracle     | 7.16e-04 (2.0e-04)        | -            |
| {NO, Jacobi (0.67)}   |                           |              |
| Solver only           | 3.00e+02 (3.2e-02)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.40e+01 (8.1e-05)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.94e-04 (7.6e-05)</b> | -            |
| Cost-aware oracle     | 2.95e-04 (7.6e-05)        | -            |
| {NO, GS}              |                           |              |
| Solver only           | 2.99e+02 (8.4e-02)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.40e+01 (2.4e-04)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.04e-04 (5.4e-05)</b> | -            |
| Cost-aware oracle     | 2.05e-04 (5.4e-05)        | -            |
| {NO, SymGS}           |                           |              |
| Solver only           | 2.97e+02 (1.5e-01)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.40e+01 (4.6e-04)        | $< 10^{-10}$ |
| Learned router (ours) | <b>1.31e-04 (3.4e-05)</b> | -            |
| Cost-aware oracle     | 1.31e-04 (3.4e-05)        | -            |
| {NO, SOR (1.5)}       |                           |              |
| Solver only           | 2.96e+02 (2.1e-01)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.40e+01 (6.7e-04)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.97e-04 (7.9e-05)</b> | -            |
| Cost-aware oracle     | 3.01e-04 (8.0e-05)        | -            |
| {NO, Line GS}         |                           |              |
| Solver only           | 2.09e+02 (1.4e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.37e+01 (1.4e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>7.84e-05 (2.1e-05)</b> | -            |
| Cost-aware oracle     | 7.82e-05 (2.1e-05)        | -            |

Table 21: Anisotropic diffusion,  $128 \times 128$ : five-seed router-training trials (conventions as in Table 9).

| $N$              | Pairing             | time to $h^2$ over 5 seeds | identical decisions | speedup vs. HINTS-15 | seeds with $p < 0.01$ |
|------------------|---------------------|----------------------------|---------------------|----------------------|-----------------------|
| 128 <sup>2</sup> | {NO, Jacobi}        | $0.65 \pm 0.00$ ms         | 100%                | 1.44–1.44×           | 0/5                   |
| 128 <sup>2</sup> | {NO, Jacobi (0.67)} | $0.65 \pm 0.00$ ms         | 100%                | 1.44–1.44×           | 0/5                   |
| 128 <sup>2</sup> | {NO, GS}            | $0.65 \pm 0.00$ ms         | 100%                | 2.15–2.15×           | 0/5                   |
| 128 <sup>2</sup> | {NO, SymGS}         | $0.65 \pm 0.00$ ms         | 100%                | 2.73–2.73×           | 0/5                   |
| 128 <sup>2</sup> | {NO, SOR (1.5)}     | $0.65 \pm 0.00$ ms         | 100%                | 2.14–2.14×           | 0/5                   |
| 128 <sup>2</sup> | {NO, Line GS}       | $0.65 \pm 0.00$ ms         | 100%                | 2.79–2.79×           | 0/5                   |

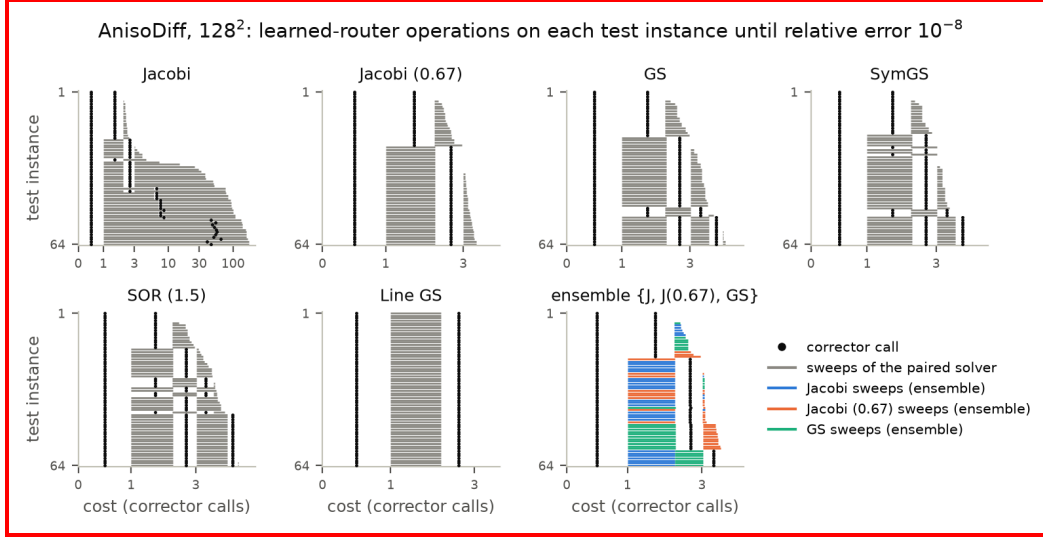


Figure 10: Anisotropic diffusion,  $128 \times 128$ : operations of the learned router on each of the 64 test instances (rows, sorted by cost) until the relative error first falls below  $10^{-8}$ , for every pairing and for the ensemble  $\{\text{Jacobi}, \text{Jacobi (0.67)}, \text{GS}\}$ , as executed in the timed replays. Dots mark corrector calls and lines the solver sweeps between them (coloured by solver in the ensemble); the horizontal axis is the cumulative cost in units of one corrector call, linear below one call and logarithmic above.

## 1296 E.5 Variable-coefficient diffusion

1297 The three equations above have constant coefficients on a periodic domain, so their discrete operators  
1298 are diagonalised by the FFT: the exact band-limited inverse that the corrector approximates has  
1299 a closed form, and a direct FFT solve is available, which at  $128^2$  takes  $1/3.25 \times -1/3.08 \times$  of  
1300 the best router’s time (Tables 4 and 10). To evaluate the hybrid where neither exists we add  
1301  $-\nabla \cdot (a(\mathbf{x})\nabla u) = f$  on the same periodic domain with a fixed, smooth, positive coefficient field  
1302  $a(\mathbf{x}) = \exp(\kappa z(\mathbf{x}))$ , where  $z$  is a band-limited random field ( $|k| \leq 4$ , one fixed realisation, scaled by  
1303 its maximum on a fixed  $1024^2$  reference grid, so that  $a$  is one grid-independent function sampled  
1304 on every grid) and  $\kappa = \frac{1}{2} \ln 10$ , so that the contrast  $\max a / \min a$  is at most 10 by construction  
1305 (9.99 measured on the grids used); the forcing distribution is unchanged. The discretisation is  
1306 the conservative 5-point scheme with face-averaged coefficients (symmetric positive semidefinite,  
1307 constants in its null space); reference solutions come from a sparse LU factorisation with one pinned  
1308 unknown, which is also reported as a direct-solve baseline (its solve time with the factorisation  
1309 cached, the factorisation itself costing  $\approx 0.5$  s at  $128^2$ ). Geometric multigrid rediscratises the coarse  
1310 operators with block-averaged coefficients and contracts the error by  $\approx 0.1$  per  $V(2,2)$ -cycle. The  
1311 corrector is the same least-squares DeepONet as before—a linear map on the  $64 \times 64$  band—but fitted  
1312 to (residual, error) pairs of *this* operator, whose band-limited inverse  $R(\mathcal{L}_h)^{-1}P$  is not diagonal in  
1313 the Fourier basis and has no closed form. It is nevertheless a fixed  $4096 \times 3969$  matrix that could be  
1314 assembled directly from 4096 sparse solves, one per sensor basis function; the least-squares fit on  
1315 random pairs recovers it to the same  $\approx 10^{-6}$  as on the constant-coefficient equations. What changes  
1316 is that this matrix is no longer the inverse of the band-restricted error: a variable coefficient couples  
1317 the band to the modes outside it, so an out-of-band residual leaves band error that the corrector does  
1318 not see, and a band correction leaves out-of-band error behind. That coupling, not fit error, is why the  
1319 corrector is called repeatedly along a solve and why the router’s decisions matter beyond the first call;  
1320 what this experiment tests is routing on an operator without Fourier structure (no shared eigenbasis  
1321 with the smoothers), not that the correction must be learned. The coefficient field is fixed rather than  
1322 sampled, so this experiment tests a corrector learned for one operator without an analytic inverse, not  
1323 generalisation across coefficient fields; the latter would require the coefficient-conditioned branch  
1324 of the original DeepONet and is left to future work. Tables 22 to 27 and fig. 11 report the same  
1325 quantities as for the other equations on the  $128^2$  and  $256^2$  grids.

Table 22: Variable-coefficient diffusion: median wall-clock time to relative error  $\varepsilon \in \{10^{-3}, h^2, 10^{-8}\}$  on the  $128^2$  and  $256^2$  grids for every pairing and policy, and for the classical baselines without corrector, including the sparse direct solve (conventions as in Table 4).

| Pairing                                                                      | Method                                        | 128 × 128                     |                                             |                                                |
|------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------|---------------------------------------------|------------------------------------------------|
|                                                                              |                                               | $\varepsilon=10^{-3}$         | $\varepsilon=h^2$                           | $\varepsilon=10^{-8}$                          |
| {NO, Jacobi}                                                                 | Solver only                                   | 453.9 ms                      | 654.4 ms                                    | 1.28 s                                         |
|                                                                              | HINTS ( $\tau=15$ )                           | 1.24 ms                       | 1.24 ms                                     | 179.2 ms                                       |
|                                                                              | Best fixed schedule                           | 706 $\mu s_{\tau=25, t_0=0}$  | 737 $\mu s_{\tau=25, t_0=0}$                | 103.0 ms <sub>one-shot</sub>                   |
|                                                                              | Best residual-decay rule                      | 709 $\mu s_{\theta=0.95}$     | 728 $\mu s_{\theta=0.95}$                   | 149.8 ms <sub><math>\theta=0.9</math></sub>    |
|                                                                              | One-shot schedule                             | 719 $\mu s$                   | 732 $\mu s$                                 | 103.0 ms                                       |
|                                                                              | Greedy oracle (Alg. 1)                        | 706 $\mu s$                   | 728 $\mu s$                                 | 78.7 ms                                        |
|                                                                              | Cost-aware oracle                             | 707 $\mu s$                   | 742 $\mu s$                                 | 79.0 ms                                        |
|                                                                              | Learned router (ours)                         | 743 $\mu s$                   | 784 $\mu s$                                 | <b>79.4 ms</b>                                 |
| {NO, Jacobi (0.67)}                                                          | Solver only                                   | 668.1 ms                      | 957.4 ms                                    | 1.86 s                                         |
|                                                                              | HINTS ( $\tau=15$ )                           | 1.20 ms                       | 1.20 ms                                     | 2.37 ms                                        |
|                                                                              | Best fixed schedule                           | 677 $\mu s_{\tau=5, t_0=0}$   | 705 $\mu s_{\tau=50, t_0=0}$                | 2.04 ms <sub><math>\tau=10</math></sub>        |
|                                                                              | Best residual-decay rule                      | 676 $\mu s_{\theta=0.98}$     | 708 $\mu s_{\theta=0.95}$                   | 2.92 ms <sub><math>\theta=0.9</math></sub>     |
|                                                                              | One-shot schedule                             | 678 $\mu s$                   | 727 $\mu s$                                 | 102.4 ms                                       |
|                                                                              | Greedy oracle (Alg. 1)                        | 680 $\mu s$                   | 723 $\mu s$                                 | 2.05 ms                                        |
|                                                                              | Cost-aware oracle                             | 679 $\mu s$                   | 734 $\mu s$                                 | 2.13 ms                                        |
|                                                                              | Learned router (ours)                         | 709 $\mu s$                   | 752 $\mu s$                                 | 2.20 ms                                        |
| {NO, GS}                                                                     | Solver only                                   | 722.9 ms                      | 1.03 s                                      | 2.01 s                                         |
|                                                                              | HINTS ( $\tau=15$ )                           | 2.47 ms                       | 2.58 ms                                     | 7.33 ms                                        |
|                                                                              | Best fixed schedule                           | 680 $\mu s_{\tau=5, t_0=0}$   | 770 $\mu s_{\tau=10, t_0=0}$                | 2.76 ms <sub><math>\tau=10, t_0=0</math></sub> |
|                                                                              | Best residual-decay rule                      | 672 $\mu s_{\theta=0.98}$     | 751 $\mu s_{\theta=0.98}$                   | 4.47 ms <sub><math>\theta=0.9</math></sub>     |
|                                                                              | One-shot schedule                             | 674 $\mu s$                   | 756 $\mu s$                                 | 255.5 ms                                       |
|                                                                              | Greedy oracle (Alg. 1)                        | 674 $\mu s$                   | 740 $\mu s$                                 | 3.34 ms                                        |
|                                                                              | Cost-aware oracle                             | 679 $\mu s$                   | 772 $\mu s$                                 | 3.09 ms                                        |
|                                                                              | Learned router (ours)                         | 699 $\mu s$                   | 792 $\mu s$                                 | 3.14 ms                                        |
| {NO, SymGS}                                                                  | Solver only                                   | 581.8 ms                      | 833.8 ms                                    | 1.62 s                                         |
|                                                                              | HINTS ( $\tau=15$ )                           | 3.49 ms                       | 3.59 ms                                     | 7.12 ms                                        |
|                                                                              | Best fixed schedule                           | 676 $\mu s_{\tau=10, t_0=0}$  | 780 $\mu s_{\text{one-shot}}$               | 2.56 ms <sub><math>\tau=5, t_0=0</math></sub>  |
|                                                                              | Best residual-decay rule                      | 674 $\mu s_{\theta=0.98}$     | 778 $\mu s_{\theta=0.98}$                   | 5.49 ms <sub><math>\theta=0.9</math></sub>     |
|                                                                              | One-shot schedule                             | 682 $\mu s$                   | 780 $\mu s$                                 | 182.3 ms                                       |
|                                                                              | Greedy oracle (Alg. 1)                        | 681 $\mu s$                   | 789 $\mu s$                                 | 2.55 ms                                        |
|                                                                              | Cost-aware oracle                             | 680 $\mu s$                   | 804 $\mu s$                                 | 2.56 ms                                        |
|                                                                              | Learned router (ours)                         | 703 $\mu s$                   | 831 $\mu s$                                 | 2.57 ms                                        |
| {NO, SOR (1.5)}                                                              | Solver only                                   | 243.4 ms                      | 347.3 ms                                    | 674.2 ms                                       |
|                                                                              | HINTS ( $\tau=15$ )                           | 2.44 ms                       | 4.90 ms                                     | 7.33 ms                                        |
|                                                                              | Best fixed schedule                           | 673 $\mu s_{\tau=15, t_0=0}$  | 813 $\mu s_{\tau=10, t_0=0}$                | 4.28 ms <sub><math>\tau=15, t_0=0</math></sub> |
|                                                                              | Best residual-decay rule                      | 677 $\mu s_{\theta=0.98}$     | 808 $\mu s_{\theta=0.95}$                   | 5.97 ms <sub><math>\theta=0.9</math></sub>     |
|                                                                              | One-shot schedule                             | 675 $\mu s$                   | 820 $\mu s$                                 | 119.4 ms                                       |
|                                                                              | Greedy oracle (Alg. 1)                        | 682 $\mu s$                   | 806 $\mu s$                                 | 4.62 ms                                        |
|                                                                              | Cost-aware oracle                             | 677 $\mu s$                   | 805 $\mu s$                                 | 4.29 ms                                        |
|                                                                              | Learned router (ours)                         | 708 $\mu s$                   | 848 $\mu s$                                 | 4.40 ms                                        |
| {NO, Multigrid}                                                              | Solver only                                   | 1.94 ms                       | 2.56 ms                                     | 4.46 ms                                        |
|                                                                              | HINTS ( $\tau=15$ )                           | 1.92 ms                       | 2.53 ms                                     | 4.46 ms                                        |
|                                                                              | Best fixed schedule                           | 681 $\mu s_{\text{one-shot}}$ | 998 $\mu s_{\text{one-shot}}$               | 2.66 ms <sub><math>\tau=10, t_0=0</math></sub> |
|                                                                              | Best residual-decay rule                      | 679 $\mu s_{\theta=0.98}$     | 1.01 ms <sub><math>\theta=0.95</math></sub> | 2.66 ms <sub><math>\theta=0.95</math></sub>    |
|                                                                              | One-shot schedule                             | 681 $\mu s$                   | 998 $\mu s$                                 | 2.65 ms                                        |
|                                                                              | Greedy oracle (Alg. 1)                        | 685 $\mu s$                   | 1.00 ms                                     | 2.66 ms                                        |
|                                                                              | Cost-aware oracle                             | 682 $\mu s$                   | 1.03 ms                                     | 2.65 ms                                        |
|                                                                              | Learned router (ours)                         | 707 $\mu s$                   | 1.06 ms                                     | 2.73 ms                                        |
| <i>Classical baselines without corrector (same stencil, same accounting)</i> |                                               |                               |                                             |                                                |
|                                                                              | Multigrid V(2,2) alone (point GS)             | 2.01 ms                       | 2.66 ms                                     | 4.61 ms                                        |
|                                                                              | CG                                            | 13.9 ms                       | 19.7 ms                                     | 35.8 ms                                        |
|                                                                              | PCG (SymGS)                                   | 17.2 ms                       | 24.7 ms                                     | 41.3 ms                                        |
|                                                                              | PCG (multigrid)                               | 1.36 ms                       | 2.01 ms                                     | 3.91 ms                                        |
|                                                                              | Sparse LU direct solve (cached factorisation) | 932 $\mu s$                   | 932 $\mu s$                                 | 932 $\mu s$                                    |

Table 23: Variable-coefficient diffusion: paired median speedup [bootstrap 95% interval] of the learned router at  $\varepsilon = h^2$  and  $10^{-8}$  over HINTS ( $\tau=15$ ), the best fixed schedule, the one-shot schedule, multigrid alone and multigrid-preconditioned CG, with Holm-corrected two-sided Wilcoxon significance (conventions as in Table 5).

| Pairing             | $\varepsilon$ | 128 × 128                            |                                                    |                                                    |                                                    |                                            |                                             |
|---------------------|---------------|--------------------------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|--------------------------------------------|---------------------------------------------|
|                     |               | vs. HINTS-15                         | vs. best schedule                                  | vs. best decay rule                                | vs. multigrid                                      | vs. MG-Keylor                              | vs. sparse LU                               |
| {NO, Jacobi}        | $h^2$         | 1.65 × [1.60, 1.71] ( $< 10^{-10}$ ) | 0.96 × [0.95, 0.98] (downer, 0.015)                | 0.96 × [0.94, 0.98] (downer, 0.016)                | 3.26 × [2.93, 3.50] ( $2 \times 10^{-5}$ )         | 2.55 × [2.32, 2.65] ( $7 \times 10^{-5}$ ) | 1.17 × [1.01, 1.22] (0.377)                 |
|                     | $10^{-8}$     | 2.19 × [1.98, 2.22] ( $< 10^{-10}$ ) | 1.19 × [1.05, 2.75] ( $5 \times 10^{-3}$ )         | 1.82 × [1.76, 1.86] ( $< 10^{-10}$ )               | 0.06 × [0.03, 0.16] (downer, $5 \times 10^{-4}$ )  | 0.05 × [0.02, 0.14] (downer, $10^{-4}$ )   | 0.01 × [0.01, 0.03] (downer, $< 10^{-10}$ ) |
| {NO, Jacobi (0.67)} | $h^2$         | 1.59 × [1.39, 1.60] ( $< 10^{-10}$ ) | 0.95 × [0.94, 0.96] (downer, $2 \times 10^{-10}$ ) | 0.95 × [0.95, 0.96] (downer, $< 10^{-10}$ )        | 3.40 × [2.97, 3.87] ( $< 10^{-10}$ )               | 2.57 × [2.29, 2.72] ( $< 10^{-10}$ )       | 1.23 × [1.06, 1.28] ( $6 \times 10^{-3}$ )  |
|                     | $10^{-8}$     | 1.08 × [1.05, 1.08] ( $10^{-9}$ )    | 0.94 × [0.93, 0.96] (downer, $8 \times 10^{-4}$ )  | 1.33 × [1.32, 1.35] ( $< 10^{-10}$ )               | 45.5 × [40.4, 52.8] ( $< 10^{-10}$ )               | 2.84 × [2.61, 2.86] ( $< 10^{-10}$ )       | 1.73 × [1.70, 1.76] ( $< 10^{-10}$ )        |
| {NO, GS}            | $h^2$         | 3.19 × [2.51, 3.56] ( $< 10^{-10}$ ) | 0.96 × [0.95, 0.97] (downer, $9 \times 10^{-4}$ )  | 0.96 × [0.95, 0.96] (downer, $3 \times 10^{-10}$ ) | 2.94 × [2.51, 3.40] ( $< 10^{-10}$ )               | 2.44 × [1.88, 2.66] ( $< 10^{-10}$ )       | 1.16 × [0.98, 1.32] (0.630)                 |
|                     | $10^{-8}$     | 2.21 × [1.92, 2.76] ( $< 10^{-10}$ ) | 1.03 × [1.00, 1.06] (0.039)                        | 1.67 × [1.57, 1.72] ( $< 10^{-10}$ )               | 76.8 × [59.7, 104] ( $< 10^{-10}$ )                | 1.44 × [1.19, 1.72] ( $< 10^{-10}$ )       | 1.19 × [1.06, 1.49] ( $9 \times 10^{-3}$ )  |
| {NO, SymGS}         | $h^2$         | 4.32 × [2.57, 4.83] ( $< 10^{-10}$ ) | 0.96 × [0.95, 0.97] (downer, $2 \times 10^{-4}$ )  | 0.96 × [0.94, 0.97] (downer, $5 \times 10^{-4}$ )  | 0.96 × [0.95, 0.97] (downer, $2 \times 10^{-4}$ )  | 2.83 × [2.51, 3.42] ( $< 10^{-10}$ )       | 2.25 × [1.88, 2.66] ( $< 10^{-10}$ )        |
|                     | $10^{-8}$     | 2.78 × [2.26, 3.01] ( $< 10^{-10}$ ) | 0.96 × [0.97, 0.99] (downer, $3 \times 10^{-4}$ )  | 2.07 × [1.93, 2.18] ( $< 10^{-10}$ )               | 60.6 × [48.1, 89.8] ( $< 10^{-10}$ )               | 1.69 × [1.46, 1.85] ( $< 10^{-10}$ )       | 1.42 × [1.18, 1.61] ( $7 \times 10^{-3}$ )  |
| {NO, SOR (1.5)}     | $h^2$         | 5.11 × [3.92, 6.66] ( $< 10^{-10}$ ) | 0.95 × [0.94, 0.96] (downer, $3 \times 10^{-10}$ ) | 0.97 × [0.96, 0.98] (downer, $2 \times 10^{-4}$ )  | 0.96 × [0.95, 0.97] (downer, $3 \times 10^{-4}$ )  | 2.73 × [2.66, 3.40] ( $< 10^{-10}$ )       | 2.34 × [1.48, 2.66] ( $2 \times 10^{-3}$ )  |
|                     | $10^{-8}$     | 1.66 × [1.25, 1.80] ( $< 10^{-10}$ ) | 0.97 × [0.97, 0.99] (downer, 0.286)                | 1.42 × [1.36, 1.51] ( $< 10^{-10}$ )               | 26.9 × [18.7, 32.3] ( $< 10^{-10}$ )               | 1.03 × [0.83, 1.14] (0.446)                | 0.84 × [0.64, 1.01] (downer, 0.017)         |
| {NO, Multigrid}     | $h^2$         | 2.42 × [1.87, 3.57] ( $< 10^{-10}$ ) | 0.96 × [0.95, 0.97] (downer, $< 10^{-10}$ )        | 0.97 × [0.96, 0.98] (downer, $10^{-10}$ )          | 0.96 × [0.95, 0.97] (downer, $< 10^{-10}$ )        | 2.30 × [1.83, 3.55] ( $< 10^{-10}$ )       | 1.97 × [1.38, 2.65] ( $< 10^{-10}$ )        |
|                     | $10^{-8}$     | 1.62 × [1.57, 1.82] ( $< 10^{-10}$ ) | 0.97 × [0.97, 0.98] (downer, $< 10^{-10}$ )        | 0.98 × [0.97, 0.98] (downer, $3 \times 10^{-4}$ )  | 0.97 × [0.97, 0.98] (downer, $4 \times 10^{-10}$ ) | 1.63 × [1.56, 1.76] ( $< 10^{-10}$ )       | 1.38 × [1.25, 1.43] ( $< 10^{-10}$ )        |

(figure pending)

Figure 11: Variable-coefficient diffusion: corrector usage per iteration of the learned router, the cost-aware oracle and HINTS ( $\tau=15$ ) for every pairing (columns) and grid (rows).

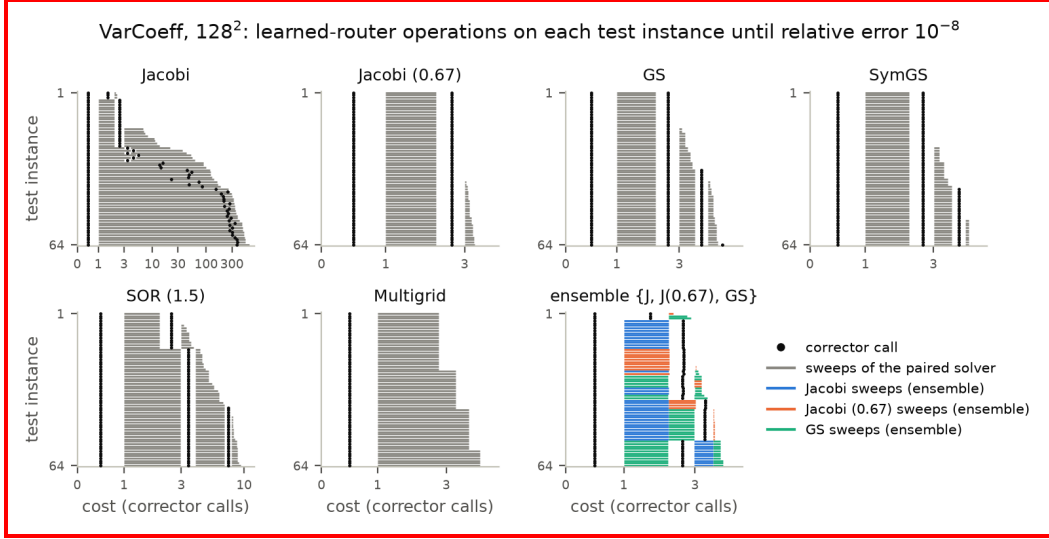


Figure 12: Variable-coefficient diffusion,  $128 \times 128$ : operations of the learned router on each of the 64 test instances (rows, sorted by cost) until the relative error first falls below  $10^{-8}$ , for every pairing and for the ensemble  $\{\text{Jacobi}, \text{Jacobi (0.67)}, \text{GS}\}$ , as executed in the timed replays. Dots mark corrector calls and lines the solver sweeps between them (coloured by solver in the ensemble); the horizontal axis is the cumulative cost in units of one corrector call, linear below one call and logarithmic above.

Table 24: Variable-coefficient diffusion: median iterations (and corrector calls) to  $\varepsilon = h^2$ .

| Pairing                               | $128 \times 128$ |        |        |
|---------------------------------------|------------------|--------|--------|
|                                       | HINTS-15         | oracle | router |
| $\{\text{NO}, \text{Jacobi}\}$        | 15 (1)           | 2 (1)  | 2 (1)  |
| $\{\text{NO}, \text{Jacobi (0.67)}\}$ | 15 (1)           | 2 (1)  | 2 (1)  |
| $\{\text{NO}, \text{GS}\}$            | 16 (1)           | 2 (1)  | 2 (1)  |
| $\{\text{NO}, \text{SymGS}\}$         | 16 (1)           | 2 (1)  | 2 (1)  |
| $\{\text{NO}, \text{SOR (1.5)}\}$     | 30 (2)           | 2 (1)  | 2 (1)  |
| $\{\text{NO}, \text{Multigrid}\}$     | 4 (0)            | 2 (1)  | 2 (1)  |

Table 25: Variable-coefficient diffusion: measured per-iteration costs and macro-action sizes.

| Pairing             | $128 \times 128$    |                   |                     |
|---------------------|---------------------|-------------------|---------------------|
|                     | $c_{\text{solver}}$ | $c_{\text{NO}}$   | $m_{\text{solver}}$ |
| {NO, Jacobi}        | $37 \mu\text{s}$    | $749 \mu\text{s}$ | 20                  |
| {NO, Jacobi (0.67)} | $36 \mu\text{s}$    | $749 \mu\text{s}$ | 21                  |
| {NO, GS}            | $121 \mu\text{s}$   | $749 \mu\text{s}$ | 6                   |
| {NO, SymGS}         | $192 \mu\text{s}$   | $749 \mu\text{s}$ | 4                   |
| {NO, SOR (1.5)}     | $122 \mu\text{s}$   | $749 \mu\text{s}$ | 6                   |
| {NO, Multigrid}     | $619 \mu\text{s}$   | $749 \mu\text{s}$ | 1                   |

Table 26: Variable-coefficient diffusion,  $128 \times 128$ : iteration-based metrics (conventions as in Table 8).

| Method                | AUC                       | $p$          |
|-----------------------|---------------------------|--------------|
| {NO, Jacobi}          |                           |              |
| Solver only           | 2.66e+02 (9.5e-01)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.39e+01 (3.7e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>6.91e-03 (1.7e-03)</b> | -            |
| Cost-aware oracle     | <i>6.91e-03 (1.7e-03)</i> | -            |
| {NO, Jacobi (0.67)}   |                           |              |
| Solver only           | 2.76e+02 (7.5e-01)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.39e+01 (3.1e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>2.49e-03 (5.9e-04)</b> | -            |
| Cost-aware oracle     | <i>2.49e-03 (5.9e-04)</i> | -            |
| {NO, GS}              |                           |              |
| Solver only           | 2.40e+02 (1.3e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.38e+01 (8.1e-03)        | $< 10^{-10}$ |
| Learned router (ours) | <b>1.93e-03 (4.6e-04)</b> | -            |
| Cost-aware oracle     | <i>1.93e-03 (4.6e-04)</i> | -            |
| {NO, SymGS}           |                           |              |
| Solver only           | 2.01e+02 (1.6e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.36e+01 (1.4e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>1.45e-03 (3.5e-04)</b> | -            |
| Cost-aware oracle     | <i>1.45e-03 (3.5e-04)</i> | -            |
| {NO, SOR (1.5)}       |                           |              |
| Solver only           | 1.71e+02 (1.7e+00)        | $< 10^{-10}$ |
| HINTS ( $\tau=15$ )   | 1.35e+01 (1.9e-02)        | $< 10^{-10}$ |
| Learned router (ours) | <b>3.27e-03 (7.9e-04)</b> | -            |
| Cost-aware oracle     | <i>3.27e-03 (7.9e-04)</i> | -            |

Table 27: Variable-coefficient diffusion: five-seed router-training trials (conventions as in Table 9).

| $N$     | Pairing             | time to $h^2$ over 5 seeds | identical decisions | speedup vs. HINTS-15 | seeds with $p < 0.01$ |
|---------|---------------------|----------------------------|---------------------|----------------------|-----------------------|
| $128^2$ | {NO, Jacobi}        | $0.80 \pm 0.00$ ms         | 100%                | 1.65–1.65 $\times$   | 0/5                   |
| $128^2$ | {NO, Jacobi (0.67)} | $0.80 \pm 0.00$ ms         | 100%                | 1.60–1.60 $\times$   | 0/5                   |
| $128^2$ | {NO, GS}            | $0.85 \pm 0.00$ ms         | 100%                | 3.00–3.00 $\times$   | 0/5                   |
| $128^2$ | {NO, SymGS}         | $0.88 \pm 0.00$ ms         | 100%                | 4.07–4.07 $\times$   | 0/5                   |
| $128^2$ | {NO, SOR (1.5)}     | $0.91 \pm 0.00$ ms         | 100%                | 4.79–4.79 $\times$   | 0/5                   |

## 1326 E.6 Decision granularity

1327 The cost-aware router decides once per macro-action, and the macro-action of a cheap solver is as  
 1328 expensive as one corrector call—32 Jacobi or 11 Gauss–Seidel sweeps at  $128^2$  with the compiled

1329 kernels. Below truncation level the best fixed schedules at  $10^{-8}$  are  $\tau=15$  (first call at 0) (7),  $\tau=10$   
1330 (first call at 0) (7),  $\tau=5$  (first call at 0) (4), one-shot (3),  $\tau=50$  (first call at 0) (1),  $\tau=25$  (first call at  
1331 0) (1),  $\tau=10$  (1) (pairings in parentheses), the periodic ones among which call the corrector more  
1332 often than a cost-equalised macro-action of 11 Gauss–Seidel or 32 Jacobi sweeps allows the router  
1333 to, a plausible cause of the router’s losses in several pairings at  $10^{-8}$  (Tables 5, 11 and 17). The  
1334 granularity is a free parameter of the construction of Section 4: the unit of cost can be any fraction of  
1335 a corrector call, actions dearer than the unit (here the corrector itself) being compared per unit of cost.  
1336 Table 28 repeats the  $128^2$  study with the unit set to one quarter of a corrector call (about a quarter  
1337 of those sweeps per decision; the corrector is then the one action dearer than the unit and is scored per  
1338 unit of cost), routers retrained with the same recipe, on the same confirmatory instances, for every  
1339 pairing and for the nested two-member Jacobi ensemble.

Table 28: Decision granularity at  $128 \times 128$ : median time to  $h^2$  and  $10^{-8}$  of the cost-aware oracle and the learned router with the unit of cost equal to one corrector call (main results) and to one quarter of it, and, in both cases, the router’s paired work-unit ratio at  $10^{-8}$  against the main run’s best fixed schedule on the same instances (the ablation runs a reduced schedule family and reuses the main run’s fixed-schedule work units).

| Equation                       | Pairing       | unit = one corrector call |              |                  |                                              | unit = corrector call / 4 |              |                  |                                             |
|--------------------------------|---------------|---------------------------|--------------|------------------|----------------------------------------------|---------------------------|--------------|------------------|---------------------------------------------|
|                                |               | oracle $h^2$              | router $h^2$ | router $10^{-8}$ | vs. best schedule ( $10^{-8}$ )              | oracle $h^2$              | router $h^2$ | router $10^{-8}$ | vs. best schedule ( $10^{-8}$ )             |
| Poisson                        | Jacobi        | 710 $\mu$ s               | 723 $\mu$ s  | 45.6 ms          | <b>1.19</b> $\times$ ( $7 \times 10^{-7}$ )  | 821 $\mu$ s               | 856 $\mu$ s  | 49.2 ms          | <b>1.17</b> $\times$ ( $8 \times 10^{-6}$ ) |
|                                | Jacobi (0.67) | 716 $\mu$ s               | 763 $\mu$ s  | 2.14 ms          | 0.85 $\times$ (slower, $8 \times 10^{-10}$ ) | 714 $\mu$ s               | 757 $\mu$ s  | 1.88 ms          | 0.98 $\times$ (slower, 0.588)               |
|                                | GS            | 728 $\mu$ s               | 758 $\mu$ s  | 2.13 ms          | 0.93 $\times$ (slower, $10^{-9}$ )           | 710 $\mu$ s               | 756 $\mu$ s  | 2.22 ms          | 1.05 $\times$ ( $5 \times 10^{-4}$ )        |
|                                | SymGS         | 752 $\mu$ s               | 793 $\mu$ s  | 2.08 ms          | 0.86 $\times$ (slower, $4 \times 10^{-4}$ )  | 744 $\mu$ s               | 792 $\mu$ s  | 1.97 ms          | 0.99 $\times$ (slower, 0.215)               |
|                                | SOR (1.5)     | 728 $\mu$ s               | 777 $\mu$ s  | 2.86 ms          | 0.99 $\times$ (slower, 0.482)                | 810 $\mu$ s               | 853 $\mu$ s  | 3.03 ms          | 0.98 $\times$ (slower, 0.221)               |
|                                | Multigrid     | 867 $\mu$ s               | 880 $\mu$ s  | 1.76 ms          | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 1.02 ms                   | 1.07 ms      | 2.48 ms          | 0.80 $\times$ (slower, $< 10^{-10}$ )       |
| Convection–diffusion           | Jacobi        | 716 $\mu$ s               | 746 $\mu$ s  | 66.4 ms          | 1.02 $\times$ ( $2 \times 10^{-4}$ )         | 741 $\mu$ s               | 759 $\mu$ s  | 66.5 ms          | 1.00 $\times$ (slower, 0.003)               |
|                                | Jacobi (0.67) | 764 $\mu$ s               | 813 $\mu$ s  | 2.21 ms          | 0.87 $\times$ (slower, 0.009)                | 743 $\mu$ s               | 790 $\mu$ s  | 1.93 ms          | 0.98 $\times$ (slower, 0.176)               |
|                                | GS            | 824 $\mu$ s               | 853 $\mu$ s  | 2.40 ms          | 0.91 $\times$ (slower, $5 \times 10^{-5}$ )  | 750 $\mu$ s               | 812 $\mu$ s  | 2.78 ms          | 1.05 $\times$ ( $10^{-7}$ )                 |
|                                | SymGS         | 830 $\mu$ s               | 864 $\mu$ s  | 2.33 ms          | 0.82 $\times$ (slower, $10^{-8}$ )           | 785 $\mu$ s               | 832 $\mu$ s  | 2.14 ms          | 0.98 $\times$ (slower, 0.009)               |
|                                | SOR (1.5)     | 851 $\mu$ s               | 906 $\mu$ s  | 3.20 ms          | 0.99 $\times$ (slower, 0.784)                | 803 $\mu$ s               | 847 $\mu$ s  | 3.47 ms          | 0.98 $\times$ (slower, 0.556)               |
|                                | Multigrid     | 1.01 ms                   | 1.09 ms      | 1.83 ms          | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 1.01 ms                   | 1.06 ms      | 2.07 ms          | 0.85 $\times$ (slower, $4 \times 10^{-5}$ ) |
| Anisotropic diffusion          | Jacobi        | 720 $\mu$ s               | 764 $\mu$ s  | 17.0 ms          | <b>1.57</b> $\times$ ( $< 10^{-10}$ )        | 727 $\mu$ s               | 756 $\mu$ s  | 17.9 ms          | <b>1.53</b> $\times$ ( $< 10^{-10}$ )       |
|                                | Jacobi (0.67) | 680 $\mu$ s               | 700 $\mu$ s  | 2.03 ms          | 0.99 $\times$ (slower, 0.411)                | 726 $\mu$ s               | 753 $\mu$ s  | 2.11 ms          | 0.98 $\times$ (slower, 0.041)               |
|                                | GS            | 678 $\mu$ s               | 710 $\mu$ s  | 2.52 ms          | 1.01 $\times$ (0.021)                        | 716 $\mu$ s               | 742 $\mu$ s  | 2.52 ms          | 1.02 $\times$ (0.017)                       |
|                                | SymGS         | 683 $\mu$ s               | 714 $\mu$ s  | 2.24 ms          | 1.03 $\times$ ( $3 \times 10^{-6}$ )         | 706 $\mu$ s               | 746 $\mu$ s  | 2.42 ms          | 1.12 $\times$ (0.119)                       |
|                                | SOR (1.5)     | 664 $\mu$ s               | 690 $\mu$ s  | 2.95 ms          | 0.98 $\times$ (slower, 0.145)                | 709 $\mu$ s               | 736 $\mu$ s  | 3.17 ms          | 0.98 $\times$ (slower, 0.805)               |
|                                | Line GS       | 657 $\mu$ s               | 686 $\mu$ s  | 2.03 ms          | 0.86 $\times$ (slower, $< 10^{-10}$ )        | 708 $\mu$ s               | 734 $\mu$ s  | 1.64 ms          | <b>1.10</b> $\times$ ( $< 10^{-10}$ )       |
| Variable-coefficient diffusion | Jacobi        | 742 $\mu$ s               | 784 $\mu$ s  | 79.4 ms          | <b>1.19</b> $\times$ ( $7 \times 10^{-7}$ )  | 744 $\mu$ s               | 794 $\mu$ s  | 82.0 ms          | <b>1.17</b> $\times$ ( $7 \times 10^{-6}$ ) |
|                                | Jacobi (0.67) | 734 $\mu$ s               | 752 $\mu$ s  | 2.20 ms          | 0.97 $\times$ (slower, $9 \times 10^{-4}$ )  | 856 $\mu$ s               | 904 $\mu$ s  | 2.37 ms          | 1.05 $\times$ ( $6 \times 10^{-6}$ )        |
|                                | GS            | 772 $\mu$ s               | 792 $\mu$ s  | 3.14 ms          | 1.05 $\times$ ( $4 \times 10^{-4}$ )         | 763 $\mu$ s               | 806 $\mu$ s  | 3.39 ms          | 1.04 $\times$ ( $5 \times 10^{-4}$ )        |
|                                | SymGS         | 804 $\mu$ s               | 831 $\mu$ s  | 2.57 ms          | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 806 $\mu$ s               | 825 $\mu$ s  | 2.70 ms          | 0.98 $\times$ (slower, 0.215)               |
|                                | SOR (1.5)     | 805 $\mu$ s               | 848 $\mu$ s  | 4.40 ms          | 0.99 $\times$ (slower, 0.216)                | 835 $\mu$ s               | 869 $\mu$ s  | 4.58 ms          | 0.98 $\times$ (slower, 0.204)               |
|                                | Multigrid     | 1.03 ms                   | 1.06 ms      | 2.73 ms          | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 980 $\mu$ s               | 1.02 ms      | 2.72 ms          | 0.99 $\times$ (slower, $< 10^{-10}$ )       |

## 1340 E.7 Solver ensembles

1341 **When can a second classical solver help?** The cost-aware oracle is a per-decision comparison  
1342 of macro-actions, so a router over  $\text{NO} \cup \mathcal{W}$  can only beat the best pairwise router of its members  
1343 if, along the way to the tolerance, the error passes through regimes in which *different* members  
1344 give the largest reduction per unit cost. We screened this ceiling exhaustively before training any  
1345 router: for every subset  $\mathcal{W}$  of up to three members of {Jacobi, Jacobi (0.67), GS, SymGS, SOR (1.5),  
1346 multigrid} (multigrid on the isotropic equations only), every equation and grid, we ran the cost-aware  
1347 oracle over  $\text{NO} \cup \mathcal{W}$  on the test instances and compared its work-unit time to that of the oracle over  
1348 each member alone (Table 29). Two facts emerged in the development screening (**[provisional]**).  
1349 At  $\varepsilon = h^2$  no ensemble beat the best single solver anywhere: one corrector call followed by one  
1350 or two macro-actions of the cheapest smoother already reaches truncation-level accuracy, so there  
1351 is no second regime to exploit. At  $\varepsilon = 10^{-8}$ , after the corrector has removed the smooth band,  
1352 the smoother has to damp the whole band  $|k| > 31$ , whose sub-bands favour different solvers:  
1353 undamped Jacobi is the cheapest per unit cost on the modes near the band edge (it contracts them by  
1354  $\cos(2\pi \cdot 32/N)$ -like factors that damped Jacobi cannot match), but it does not damp the near-Nyquist  
1355 modes at all, which damped Jacobi and GS remove in a few sweeps. In the development screening  
1356 (**[provisional]**: Table 29 is computed with the development-run macro-action sizes; the confirmatory  
1357 screening with the compiled kernels’ sizes is pending) the oracle over {Jacobi, Jacobi (0.67)}—a  
1358 *nonstationary* Jacobi iteration whose damping parameter is chosen per macro-action—was therefore  
1359 faster than every single solver, by 13–28% on four of the six (equation, grid) settings at  $128^2$  and  
1360  $256^2$ , and adding a third member (GS) changed nothing, i.e., the gain saturates. Whether this ceiling  
1361 survives the coarser macro-actions of the compiled kernels is exactly what the nested-ensemble  
1362 runs below measure. At  $512^2$  (development screening) multigrid dominated every subset (its cycle

removes all sub-bands at once), so the oracle over any ensemble containing it equalled the oracle over {NO, multigrid}: adding members neither helped nor hurt. The “difficulty” that matters for ensembles is thus not the equation but the width of the spectrum left to the classical members after the corrector, which grows with the grid (fixed band) and with the tolerance.

**Nested ensembles.** Table 30 evaluates the learned router over the nested sets {Jacobi}, {Jacobi (0.67)}, {Jacobi, Jacobi (0.67)} and {Jacobi, Jacobi (0.67), GS} on every equation at  $128^2$  and  $256^2$ , and at  $128^2$  also over the four ensembles of Section 7 ({Jacobi, GS} up to all five solvers); the single-member routers are the pairwise routers of Sections E.2 to E.4, evaluated in the same session on the same instances, and all members keep the macro-action sizes of their pairwise runs. The table reports live and work-unit times to  $h^2$ ,  $10^{-6}$  and  $10^{-8}$  for the router and its oracle, and the paired speedup at  $10^{-8}$  over the best single-member router (and of the ensemble oracle over the best single-member oracle) with its  $p$ -value; work units are the primary measure, since they are determined by the decision sequence alone. The  $128^2$  cells and the Poisson and convection–diffusion  $256^2$  cells are final under the revised protocol; the other  $256^2$  cells are **[pending]**. *Separation:* at  $\varepsilon = 10^{-8}$  the ensemble router is faster than every single-member router by at least 10% and significantly so (bold in the table) in 0 of the 20 ensemble cells, and the ensemble *oracle* is  $0.95 \times$ – $1.02 \times$  of the best single-member oracle in work units (the greedy rule itself finds no second regime to exploit). That differed from the development runs, in which the two-member Jacobi set was 8–15% faster than damped Jacobi alone on convection–diffusion and anisotropic diffusion (Table 33), and the difference is plausibly the decision granularity. With the numpy sweeps a Jacobi macro-action was about ten sweeps, and the oracle alternated undamped Jacobi—the cheapest member on the band-edge modes  $|k| \approx 32$ , which it contracts by about 0.5 per sweep at  $128^2$  but which leaves the Nyquist modes untouched—with damped Jacobi, which removes them. With the compiled sweeps a damped-Jacobi macro-action is 32 sweeps: it contracts the band-edge modes by  $0.665^{32} \approx 2 \times 10^{-6}$  and the Nyquist modes by  $0.34^{32} \approx 1 \times 10^{-15}$  (single-mode factors at  $128^2$ ), which suggests that the band the corrector leaves is removed within one or two macro-actions, leaving a nonstationary schedule little to gain; the granularity ablation tests this hypothesis. Section E.6 repeats these cells with a four times finer unit and tests directly whether the gain returns **[pending]**; the  $256^2$  grids, where the band left to the smoothers is twice as wide and needs several macro-actions, are the other place it could, and there it does: on the Poisson and convection–diffusion equations at  $256^2$  the nested ensembles are  $1.23$ – $1.26 \times$  faster than the best single-member router at  $10^{-8}$  on every instance (work units and live time), their oracles  $1.26$ – $1.28 \times$  faster than the best single-member oracle, i.e., the greedy rule finds the nonstationary Jacobi schedule that the screening predicted; over all equations at  $256^2$  the ensemble routers are significantly faster than the best single-member router in **[pending]** of **[pending]** cells (ratios **[pending]**–**[pending]** at  $10^{-8}$ , **[pending]**–**[pending]** at  $h^2$ ; oracles **[pending]**–**[pending]**). *Saturation:* in the first run, adding GS to the Jacobi pair changed the oracle’s schedule by at most a few percent, because GS never won a cost-aware comparison against the Jacobi pair. *Monotonicity:* the ensemble routers’ ratios against the best single-member router in work units are  $1.00 \times$ – $1.00 \times$  at  $h^2$  and  $0.74 \times$ – $1.00 \times$  at  $10^{-8}$ , with 9 of the 20 cells more than 5% slower (Table 30 lists them; the router’s imitation of its oracle degrades among near-tied members while the oracles themselves stay within a few percent of the best single oracle). A greedy rule is not monotone in its action set in general, and its learned imitation is less so; the table reports how far the losses go.

Table 29: Oracle-level screening of ensembles (**[provisional: development-run costs and instances]**): for every equation and grid, the fastest single solver and the fastest ensemble of up to three members under the cost-aware oracle (median work-unit time over 16 test instances), at  $\varepsilon = h^2$  and  $10^{-8}$ ; the ratio is the speedup of the ensemble over the best single solver, and the last column is the ratio for {Jacobi, Jacobi (0.67)}.

| Equation              | $N$     | $\varepsilon = h^2$         |                                      |               | $\varepsilon = 10^{-8}$ |                               |               | {Jacobi, Jacobi (0.67)} |
|-----------------------|---------|-----------------------------|--------------------------------------|---------------|-------------------------|-------------------------------|---------------|-------------------------|
|                       |         | best single                 | best ensemble                        | ratio         | best single             | best ensemble                 | ratio         |                         |
| Poisson               | $128^2$ | Jacobi (740 $\mu$ s)        | {Jacobi, SOR(1.5)} (740 $\mu$ s)     | $1.00 \times$ | Multigrid (1.97 ms)     | {Jacobi, Multigrid} (1.97 ms) | $1.00 \times$ | $0.97 \times$           |
| Convection–diffusion  | $128^2$ | Jacobi (0.67) (733 $\mu$ s) | {Jacobi, Jacobi(0.67)} (733 $\mu$ s) | $1.00 \times$ | Multigrid (1.74 ms)     | {Jacobi, Multigrid} (1.74 ms) | $1.00 \times$ | $0.92 \times$           |
| Anisotropic diffusion | $128^2$ | Jacobi (642 $\mu$ s)        | {Jacobi, Jacobi(0.67)} (642 $\mu$ s) | $1.00 \times$ | Line GS (1.86 ms)       | {Jacobi, LineGS} (1.86 ms)    | $1.00 \times$ | $0.94 \times$           |

Table 30: Nested ensembles: median time to  $\varepsilon \in \{h^2, 10^{-6}, 10^{-8}\}$  of the learned router (live and work units) and of the cost-aware oracle (work units) over  $\text{NO} \cup \mathcal{W}$ , for the single members and the nested two- and three-member sets, all evaluated in one session on the same instances (64 at  $128^2$ , 32 at  $256^2$ , 16 for anisotropic diffusion at  $256^2$ ). The last columns give the paired speedup at  $10^{-8}$  over the best single-member router (resp. oracle) with the two-sided Wilcoxon  $p$ -value; bold:  $\geq 1.1\times$  and significant after a Holm correction over all tests of the table.

| Equation                       | $N$     | $\mathcal{W}$                               | $\varepsilon = h^2$ |                   | $\varepsilon = 10^{-6}$ |                              | $\varepsilon = 10^{-8}$      |                                | $\varepsilon = 10^{-8}$ , vs. best single solver |                                |
|--------------------------------|---------|---------------------------------------------|---------------------|-------------------|-------------------------|------------------------------|------------------------------|--------------------------------|--------------------------------------------------|--------------------------------|
|                                |         |                                             | router              | oracle (WU)       | router                  | oracle (WU)                  | router                       | oracle (WU)                    | router                                           | oracle                         |
| Poisson                        | $128^2$ | {Jacobi}                                    | 896 $\mu\text{s}$   | 707 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.70 ms                      | 1.49 ms                      | 1.48 ms                        | 44.9 ms                                          | 45.8 ms                        |
|                                |         | (GS)                                        | 809 $\mu\text{s}$   | 727 $\mu\text{s}$ | 717 $\mu\text{s}$       | 1.61 ms                      | 1.53 ms                      | 1.69 ms                        | 2.48 ms                                          | 2.08 ms                        |
|                                |         | Sym(GS)                                     | 966 $\mu\text{s}$   | 743 $\mu\text{s}$ | 751 $\mu\text{s}$       | 1.33 ms                      | 1.07 ms                      | 1.15 ms                        | 2.38 ms                                          | 2.02 ms                        |
|                                |         | Jacobi(0.67)                                | 880 $\mu\text{s}$   | 707 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.05 ms                      | 874 $\mu\text{s}$            | 886 $\mu\text{s}$              | 2.39 ms                                          | 2.04 ms                        |
|                                |         | (SOR(1.5))                                  | 918 $\mu\text{s}$   | 727 $\mu\text{s}$ | 717 $\mu\text{s}$       | 1.76 ms                      | 1.68 ms                      | 1.69 ms                        | 3.03 ms                                          | 2.82 ms                        |
|                                |         | Jacobi, (GS)                                | 860 $\mu\text{s}$   | 708 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.33 ms                      | 1.22 ms                      | 1.39 ms                        | 2.74 ms                                          | 2.18 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 897 $\mu\text{s}$   | 708 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.09 ms                      | 887 $\mu\text{s}$            | 886 $\mu\text{s}$              | 2.71 ms                                          | 2.05 ms                        |
|                                |         | Jacobi, GS, Sym(GS)                         | 792 $\mu\text{s}$   | 708 $\mu\text{s}$ | 713 $\mu\text{s}$       | 1.34 ms                      | 1.24 ms                      | 1.15 ms                        | 2.82 ms                                          | 2.70 ms                        |
|                                |         | Jacobi, Jacobi(0.67), (GS)                  | 877 $\mu\text{s}$   | 728 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.13 ms                      | 967 $\mu\text{s}$            | 886 $\mu\text{s}$              | 2.53 ms                                          | 2.06 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67)           | 913 $\mu\text{s}$   | 707 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.14 ms                      | 906 $\mu\text{s}$            | 886 $\mu\text{s}$              | 2.73 ms                                          | 2.08 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), SOR(1.5) | 885 $\mu\text{s}$   | 707 $\mu\text{s}$ | 697 $\mu\text{s}$       | 1.41 ms                      | 1.25 ms                      | 886 $\mu\text{s}$              | 2.86 ms                                          | 2.51 ms                        |
|                                |         |                                             | 1.53 ms             | 1.47 ms           | 1.46 ms                 | $\geq 2.50 \text{ ms}^{(5)}$ | $\geq 2.46 \text{ ms}^{(5)}$ | $\geq 82.8 \text{ ms}^{(15)}$  | $\geq 82.6 \text{ ms}^{(15)}$                    | –                              |
|                                |         | Jacobi(0.67)                                | 1.79 ms             | 1.73 ms           | 1.72 ms                 | 3.31 ms                      | 3.20 ms                      | 3.18 ms                        | 6.94 ms                                          | 6.75 ms                        |
|                                |         | (GS)                                        | 2.43 ms             | 2.25 ms           | 2.22 ms                 | 5.43 ms                      | 5.10 ms                      | 4.75 ms                        | 10.2 ms                                          | 9.71 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 1.60 ms             | 1.48 ms           | 1.46 ms                 | 2.57 ms                      | 2.46 ms                      | 2.45 ms                        | 5.64 ms                                          | 5.45 ms                        |
| Correction-diffusion           | $128^2$ | Jacobi, Jacobi(0.67), GS                    | 1.56 ms             | 1.48 ms           | 1.46 ms                 | 2.57 ms                      | 2.46 ms                      | 2.45 ms                        | 5.65 ms                                          | 5.45 ms                        |
|                                |         | {Jacobi}                                    | 914 $\mu\text{s}$   | 682 $\mu\text{s}$ | 672 $\mu\text{s}$       | 3.85 ms                      | 2.95 ms                      | 2.91 ms                        | 91.6 ms                                          | 65.8 ms                        |
|                                |         | (GS)                                        | 904 $\mu\text{s}$   | 731 $\mu\text{s}$ | 721 $\mu\text{s}$       | 2.42 ms                      | 1.93 ms                      | 1.91 ms                        | 2.72 ms                                          | 2.07 ms                        |
|                                |         | Sym(GS)                                     | 978 $\mu\text{s}$   | 731 $\mu\text{s}$ | 721 $\mu\text{s}$       | 1.81 ms                      | 1.58 ms                      | 1.59 ms                        | 2.58 ms                                          | 1.98 ms                        |
|                                |         | Jacobi(0.67)                                | 936 $\mu\text{s}$   | 687 $\mu\text{s}$ | 677 $\mu\text{s}$       | 1.13 ms                      | 854 $\mu\text{s}$            | 863 $\mu\text{s}$              | 2.59 ms                                          | 1.91 ms                        |
|                                |         | (SOR(1.5))                                  | 1.07 ms             | 785 $\mu\text{s}$ | 775 $\mu\text{s}$       | 2.41 ms                      | 1.93 ms                      | 1.91 ms                        | 3.36 ms                                          | 2.79 ms                        |
|                                |         | Jacobi, (GS)                                | 969 $\mu\text{s}$   | 692 $\mu\text{s}$ | 688 $\mu\text{s}$       | 1.52 ms                      | 1.40 ms                      | 1.36 ms                        | 2.61 ms                                          | 2.03 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 955 $\mu\text{s}$   | 689 $\mu\text{s}$ | 677 $\mu\text{s}$       | 1.23 ms                      | 874 $\mu\text{s}$            | 863 $\mu\text{s}$              | 2.79 ms                                          | 1.92 ms                        |
|                                |         | Jacobi, GS, Sym(GS)                         | 878 $\mu\text{s}$   | 682 $\mu\text{s}$ | 721 $\mu\text{s}$       | 1.42 ms                      | 1.36 ms                      | 1.59 ms                        | 3.10 ms                                          | 2.57 ms                        |
|                                |         | Jacobi, Jacobi(0.67), GS                    | 926 $\mu\text{s}$   | 732 $\mu\text{s}$ | 677 $\mu\text{s}$       | 1.48 ms                      | 1.32 ms                      | 863 $\mu\text{s}$              | 2.62 ms                                          | 1.94 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67)           | 935 $\mu\text{s}$   | 692 $\mu\text{s}$ | 677 $\mu\text{s}$       | 1.68 ms                      | 1.37 ms                      | 863 $\mu\text{s}$              | 2.58 ms                                          | 2.02 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), SOR(1.5) | 981 $\mu\text{s}$   | 693 $\mu\text{s}$ | 677 $\mu\text{s}$       | 1.71 ms                      | 1.43 ms                      | 863 $\mu\text{s}$              | 2.62 ms                                          | 2.00 ms                        |
|                                |         |                                             | 1.87 ms             | 1.75 ms           | 1.74 ms                 | $\geq 2.95 \text{ ms}^{(8)}$ | $\geq 2.78 \text{ ms}^{(8)}$ | $\geq 162.4 \text{ ms}^{(16)}$ | $\geq 161.6 \text{ ms}^{(16)}$                   | $\geq 160.8 \text{ ms}^{(16)}$ |
|                                |         | Jacobi(0.67)                                | 2.29 ms             | 2.13 ms           | 2.12 ms                 | 3.81 ms                      | 3.62 ms                      | 3.60 ms                        | 7.61 ms                                          | 7.24 ms                        |
|                                |         | (GS)                                        | 2.92 ms             | 2.61 ms           | 2.59 ms                 | 6.31 ms                      | 5.64 ms                      | 5.37 ms                        | 10.3 ms                                          | 8.35 ms                        |
| Anisotropic diffusion          | $128^2$ | Jacobi, Jacobi(0.67), GS                    | 1.89 ms             | 1.75 ms           | 1.74 ms                 | 2.98 ms                      | 2.79 ms                      | 2.78 ms                        | 6.31 ms                                          | 5.67 ms                        |
|                                |         | Jacobi, GS, Sym(GS)                         | 1.79 ms             | 1.75 ms           | 1.74 ms                 | 2.85 ms                      | 2.79 ms                      | 2.78 ms                        | 5.86 ms                                          | 5.71 ms                        |
|                                |         | {Jacobi}                                    | 887 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.33 ms                      | 1.07 ms                      | 854 $\mu\text{s}$              | 22.0 ms                                          | 17.2 ms                        |
|                                |         | (GS)                                        | 884 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.47 ms                      | 1.14 ms                      | 1.13 ms                        | 2.89 ms                                          | 2.29 ms                        |
|                                |         | Sym(GS)                                     | 881 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.36 ms                      | 1.06 ms                      | 1.05 ms                        | 2.61 ms                                          | 2.06 ms                        |
|                                |         | Jacobi(0.67)                                | 807 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.29 ms                      | 958 $\mu\text{s}$            | 947 $\mu\text{s}$              | 2.63 ms                                          | 1.92 ms                        |
|                                |         | (SOR(1.5))                                  | 878 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.72 ms                      | 1.28 ms                      | 1.27 ms                        | 3.35 ms                                          | 2.67 ms                        |
|                                |         | Jacobi, (GS)                                | 882 $\mu\text{s}$   | 653 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.30 ms                      | 1.03 ms                      | 854 $\mu\text{s}$              | 2.76 ms                                          | 2.32 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 873 $\mu\text{s}$   | 653 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.23 ms                      | 949 $\mu\text{s}$            | 937 $\mu\text{s}$              | 2.52 ms                                          | 1.89 ms                        |
|                                |         | Jacobi, GS, Sym(GS)                         | 899 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.39 ms                      | 1.11 ms                      | 968 $\mu\text{s}$              | 2.66 ms                                          | 2.13 ms                        |
|                                |         | Jacobi, Jacobi(0.67), GS                    | 896 $\mu\text{s}$   | 653 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.21 ms                      | 896 $\mu\text{s}$            | 927 $\mu\text{s}$              | 2.57 ms                                          | 1.96 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67)           | 875 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.25 ms                      | 947 $\mu\text{s}$            | 927 $\mu\text{s}$              | 2.55 ms                                          | 1.98 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), SOR(1.5) | 883 $\mu\text{s}$   | 652 $\mu\text{s}$ | 642 $\mu\text{s}$       | 1.19 ms                      | 922 $\mu\text{s}$            | 948 $\mu\text{s}$              | 2.58 ms                                          | 2.03 ms                        |
|                                |         |                                             | 8.79 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | $\geq 18.7 \text{ ms}^{(7)}$                     | $\geq 18.9 \text{ ms}^{(7)}$   |
|                                |         | Jacobi(0.67)                                | 8.84 ms             | 8.80 ms           | 8.79 ms                 | 17.5 ms                      | 17.5 ms                      | 17.5 ms                        | 18.7 ms                                          | 18.8 ms                        |
| Anisotropic diffusion          | $256^2$ | (GS)                                        | 8.81 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | 23.1 ms                                          | 23.5 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 8.82 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 16.3 ms                        | 18.9 ms                                          | 18.8 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), GS       | 8.82 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 16.3 ms                        | 22.3 ms                                          | 23.1 ms                        |
|                                |         |                                             | 8.79 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | $\geq 18.7 \text{ ms}^{(7)}$                     | $\geq 18.9 \text{ ms}^{(7)}$   |
|                                |         | Jacobi(0.67)                                | 8.84 ms             | 8.80 ms           | 8.79 ms                 | 17.5 ms                      | 17.5 ms                      | 17.5 ms                        | 18.7 ms                                          | 18.8 ms                        |
|                                |         | (GS)                                        | 8.81 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | 23.1 ms                                          | 23.5 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 8.82 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 16.3 ms                        | 18.9 ms                                          | 18.8 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), GS       | 8.82 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 16.3 ms                        | 22.3 ms                                          | 23.1 ms                        |
|                                |         |                                             | 8.79 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | $\geq 18.7 \text{ ms}^{(7)}$                     | $\geq 18.9 \text{ ms}^{(7)}$   |
|                                |         | Jacobi(0.67)                                | 8.84 ms             | 8.80 ms           | 8.79 ms                 | 17.5 ms                      | 17.5 ms                      | 17.5 ms                        | 18.7 ms                                          | 18.8 ms                        |
|                                |         | (GS)                                        | 8.81 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | 23.1 ms                                          | 23.5 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 8.82 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 16.3 ms                        | 18.9 ms                                          | 18.8 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), GS       | 8.82 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 16.3 ms                        | 22.3 ms                                          | 23.1 ms                        |
|                                |         |                                             | 8.79 ms             | 8.80 ms           | 8.79 ms                 | 17.4 ms                      | 17.5 ms                      | 17.5 ms                        | $\geq 18.7 \text{ ms}^{(7)}$                     | $\geq 18.9 \text{ ms}^{(7)}$   |
|                                |         | Jacobi(0.67)                                | 8.84 ms             | 8.80 ms           | 8.79 ms                 | 17.5 ms                      | 17.5 ms                      | 17.5 ms                        | 18.7 ms                                          | 18.8 ms                        |
| Variable-coefficient diffusion | $128^2$ | {Jacobi}                                    | 923 $\mu\text{s}$   | 805 $\mu\text{s}$ | 795 $\mu\text{s}$       | $\geq 2.54 \text{ ms}^{(1)}$ | $\geq 1.86 \text{ ms}^{(1)}$ | $\geq 1.64 \text{ ms}^{(1)}$   | $\geq 87.7 \text{ ms}^{(24)}$                    | $\geq 81.8 \text{ ms}^{(24)}$  |
|                                |         | Jacobi(0.67)                                | 939 $\mu\text{s}$   | 804 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.25 ms                      | 1.11 ms                      | 1.10 ms                        | 2.63 ms                                          | 2.31 ms                        |
|                                |         | (GS)                                        | 1.02 ms             | 877 $\mu\text{s}$ | 875 $\mu\text{s}$       | 1.96 ms                      | 1.90 ms                      | 1.88 ms                        | 3.86 ms                                          | 3.29 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 919 $\mu\text{s}$   | 806 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.45 ms                      | 1.13 ms                      | 1.10 ms                        | 2.94 ms                                          | 2.37 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), GS       | 915 $\mu\text{s}$   | 805 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.30 ms                      | 1.15 ms                      | 1.10 ms                        | 2.96 ms                                          | 2.54 ms                        |
|                                |         |                                             | 9.23 ms             | 8.05 ms           | 7.95 ms                 | $\geq 2.54 \text{ ms}^{(1)}$ | $\geq 1.86 \text{ ms}^{(1)}$ | $\geq 1.64 \text{ ms}^{(1)}$   | $\geq 87.7 \text{ ms}^{(24)}$                    | $\geq 81.8 \text{ ms}^{(24)}$  |
|                                |         | Jacobi(0.67)                                | 939 $\mu\text{s}$   | 804 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.25 ms                      | 1.11 ms                      | 1.10 ms                        | 2.63 ms                                          | 2.31 ms                        |
|                                |         | (GS)                                        | 1.02 ms             | 877 $\mu\text{s}$ | 875 $\mu\text{s}$       | 1.96 ms                      | 1.90 ms                      | 1.88 ms                        | 3.86 ms                                          | 3.29 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 919 $\mu\text{s}$   | 806 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.45 ms                      | 1.13 ms                      | 1.10 ms                        | 2.94 ms                                          | 2.37 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), GS       | 915 $\mu\text{s}$   | 805 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.30 ms                      | 1.15 ms                      | 1.10 ms                        | 2.96 ms                                          | 2.54 ms                        |
|                                |         |                                             | 9.23 ms             | 8.05 ms           | 7.95 ms                 | $\geq 2.54 \text{ ms}^{(1)}$ | $\geq 1.86 \text{ ms}^{(1)}$ | $\geq 1.64 \text{ ms}^{(1)}$   | $\geq 87.7 \text{ ms}^{(24)}$                    | $\geq 81.8 \text{ ms}^{(24)}$  |
|                                |         | Jacobi(0.67)                                | 939 $\mu\text{s}$   | 804 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.25 ms                      | 1.11 ms                      | 1.10 ms                        | 2.63 ms                                          | 2.31 ms                        |
|                                |         | (GS)                                        | 1.02 ms             | 877 $\mu\text{s}$ | 875 $\mu\text{s}$       | 1.96 ms                      | 1.90 ms                      | 1.88 ms                        | 3.86 ms                                          | 3.29 ms                        |
|                                |         | Jacobi, Jacobi(0.67)                        | 919 $\mu\text{s}$   | 806 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.45 ms                      | 1.13 ms                      | 1.10 ms                        | 2.94 ms                                          | 2.37 ms                        |
|                                |         | Jacobi, GS, Sym(GS), Jacobi(0.67), GS       | 915 $\mu\text{s}$   | 805 $\mu\text{s}$ | 794 $\mu\text{s}$       | 1.30 ms                      | 1.15 ms                      | 1.10 ms                        | 2.96 ms                                          | 2.54 ms                        |

## E.8 Overheads and amortisation

Table 31 lists the measured cost of every operation as a function of  $N$ , including one decision of the LSTM router of Section 7 (full-state input of size  $4N^2$ , hidden size 256, 4 layers): at  $128^2$  one LSTM decision costs **[pending]** ms, about **[pending]** Jacobi sweeps, which is why the wall-clock study uses the scalar-feature router ( $\approx 5 \mu\text{s}$ , independent of  $N$ ). In situ, inside the timed replays, the router’s decisions cost 7–21  $\mu\text{s}$  each (the first decision of a solve and every decision after a macro-action find cold caches; the 5  $\mu\text{s}$  figure is a warm-loop median), i.e. 0.2–3.3% of the router’s time to  $h^2$  (1–28 decisions) and 0.2–2.5% of its time to  $10^{-8}$ , as instrumented directly inside the replays. Separately, the paired difference between the router’s and the cost-aware oracle’s time at  $h^2$  is -176–1862  $\mu\text{s}$  per solve; the two run as separate replays with decisions that agree on 50–100% of the instances, so this gap is consistent with, but not a measurement of, the decision cost (an earlier version of this appendix put the decisions at “well under 1%” on the basis of the warm-loop figure). The corrector calls account for an estimated 57–100% of the router’s time to  $h^2$  (number of calls times the calibrated per-call cost; per-cell medians over the  $128^2$  pairwise cells). Two further accounting checks: the charged time of a router solve is 99–100% of the outer elapsed time of its replay, the remainder being the Python driver’s loop and timer overhead, which every method pays per iteration; and the router’s live time to  $h^2$  is 0.62–1.17 times its work-unit time, the per-operation costs being measured in the same form as the replay charges them. Table 32 accounts for training: the corrector (shared with HINTS) is fit in minutes, the router in seconds to a few minutes per pairing, and the break-even number of solves after which the router’s own training cost is recovered relative to HINTS ( $\tau=15$ ) with the same corrector is **[pending]**–**[pending]** solves for the GS pairing at each equation and grid (Table 32; **[pending]** of those cells have no positive saving, and **[pending]** have a positive saving but no recorded router-training time).

Table 31: Measured single-thread cost of one operation as a function of the grid size (Poisson; median of repeated calls). Multigrid uses a direct solve below  $64^2$  and is not listed there.

**[results pending: confirmatory run in progress]**

Table 32: Training-cost amortisation for the GS pairing: corrector training (data generation and least-squares fit, CPU), router training (oracle rollouts and DAgger rounds), median wall-clock saved per solve at  $\varepsilon = h^2$  relative to HINTS ( $\tau=15$ ) and to the solver alone, and the number of solves after which the router’s training cost is recovered relative to HINTS.

**[results pending: confirmatory run in progress]**

## 1428 E.9 Development versus confirmatory runs

1429 Several choices of this study were made after looking at test results: the corrector band and the  
 1430 anisotropic sensor grid, the length of the  $512^2$  router-training rollouts (set to cover the test horizon  
 1431 after the first Poisson Jacobi router, trained on 400-decision rollouts, kept calling the corrector beyond  
 1432 that horizon on 6 of 16 test instances, Section E.1), and the imitation budget of the ensemble routers  
 1433 (raised after the pairwise recipe imitated its oracle imperfectly on one development cell, Section E.7);  
 1434 the optimistic best-schedule baseline is tuned on the test set by design and labelled as such. All  
 1435 of these were made on the development set (seed 72), with the earlier numpy/scipy sweeps, the  
 1436 earlier forcing sampler and  $\tau = 25$  as the HINTS reference (Section E.1), which is why the table  
 1437 compares against  $\tau = 25$  in both runs. Every result in this appendix comes from a single evaluation  
 1438 of the frozen configurations on the confirmatory set (seed 73) with the compiled kernels and three  
 1439 timed replays. Table 33 lists, for every cell present in both, the development and the confirmatory  
 1440 speedups: the two differ both through the instances and through the kernels—the compiled sweeps  
 1441 are  $4\text{--}5\times$  faster, so the corrector is worth  $2\text{--}3\times$  more sweeps than in the development runs and the  
 1442 margins over fixed schedules at  $h^2$  shrink accordingly—and the table shows exactly how much of the  
 1443 development-phase margin survives.

Table 33: Development (seed 72, numpy/scipy sweeps, one timed replay) versus confirmatory (seed 73, compiled kernels, three timed replays) paired median speedups of the learned router, for every pairwise cell at  $\varepsilon = h^2$  against HINTS ( $\tau=25$ , the reference of the development runs) and against the best fixed schedule available in that run, and for every nested ensemble at  $10^{-8}$  in work units against the best single-member router and against its oracle; two-sided Wilcoxon  $p$ -values in parentheses, censoring classes as in the speed tables; bold: ratio  $\geq 1.1\times$  and significant after a Holm correction over the table.

| Equation              | $N$              | Pairing / ensemble                                                    | development (seed 72, numpy kernels)        |                                              | confirmatory (seed 73, compiled kernels)    |                                              |
|-----------------------|------------------|-----------------------------------------------------------------------|---------------------------------------------|----------------------------------------------|---------------------------------------------|----------------------------------------------|
|                       |                  |                                                                       | vs. HINTS-25 or best single                 | vs. best schedule or oracle                  | vs. HINTS-25 or best single                 | vs. best schedule or oracle                  |
| Poisson               | 128 <sup>2</sup> | [NO, Jacobi]                                                          | <b>2.68</b> ( $< 10^{-10}$ )                | <b>1.21</b> ( $4 \times 10^{-8}$ )           | <b>1.66</b> ( $< 10^{-10}$ )                | 0.96 $\times$ (slower, 0.100)                |
|                       | 128 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>2.90</b> ( $< 10^{-10}$ )                | <b>1.28</b> ( $4 \times 10^{-8}$ )           | <b>1.61</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $9 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, GS]                                                              | <b>6.75</b> ( $< 10^{-10}$ )                | <b>1.94</b> ( $< 10^{-10}$ )                 | <b>2.90</b> ( $< 10^{-10}$ )                | 0.96 $\times$ (slower, $7 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, SymGS]                                                           | <b>9.21</b> ( $< 10^{-10}$ )                | <b>2.34</b> ( $< 10^{-10}$ )                 | <b>3.72</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $3 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>10.3</b> ( $< 10^{-10}$ )                | <b>3.22</b> ( $< 10^{-10}$ )                 | <b>4.17</b> ( $< 10^{-10}$ )                | 0.96 $\times$ (slower, $9 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, Multigrid]                                                       | <b>1.96</b> ( $< 10^{-10}$ )                | <b>1.88</b> ( $< 10^{-10}$ )                 | 0.98 $\times$ (slower, $10^{-4}$ )          | 0.97 $\times$ (slower, $2 \times 10^{-5}$ )  |
|                       | 128 <sup>2</sup> | [Jacobi, GS] ( $10^{-8}$ , work units)                                | —                                           | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 1.00 $\times$ (slower, 0.072)               | 0.90 $\times$ (slower, $5 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS] ( $10^{-8}$ , work units)                         | —                                           | 0.99 $\times$ (slower, 0.012)                | 0.94 $\times$ (slower, $< 10^{-10}$ )       | 0.76 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 128 <sup>2</sup> | [Jacobi, Jacobi(0.67)] ( $10^{-8}$ , work units)                      | 1.00 $\times$ (0.194)                       | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 1.00 $\times$ (slower, $9 \times 10^{-7}$ ) | 0.99 $\times$ (slower, $2 \times 10^{-7}$ )  |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, GS] ( $10^{-8}$ , work units)                     | 0.92 $\times$ (slower, $2 \times 10^{-7}$ ) | 0.98 $\times$ (slower, $6 \times 10^{-8}$ )  | 0.99 $\times$ (slower, $10^{-7}$ )          | 0.99 $\times$ (slower, $10^{-7}$ )           |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, Jacobi(0.67)] ( $10^{-8}$ , work units)           | —                                           | 0.91 $\times$ (slower, $< 10^{-10}$ )        | 0.98 $\times$ (slower, 0.043)               | 0.96 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)] ( $10^{-8}$ , work units) | —                                           | 0.79 $\times$ (slower, $< 10^{-10}$ )        | 0.94 $\times$ (slower, $< 10^{-10}$ )       | 0.74 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 256 <sup>2</sup> | [NO, Jacobi]                                                          | <b>2.52</b> ( $10^{-9}$ )                   | <b>1.56</b> ( $3 \times 10^{-9}$ )           | <b>1.67</b> ( $7 \times 10^{-9}$ )          | 0.99 $\times$ (slower, 0.004)                |
|                       | 256 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>2.26</b> ( $8 \times 10^{-9}$ )          | <b>1.46</b> ( $10^{-9}$ )                    | <b>1.49</b> ( $3 \times 10^{-9}$ )          | 0.97 $\times$ (slower, $4 \times 10^{-9}$ )  |
|                       | 256 <sup>2</sup> | [NO, GS]                                                              | <b>5.77</b> ( $5 \times 10^{-10}$ )         | <b>1.70</b> ( $3 \times 10^{-9}$ )           | <b>3.29</b> ( $9 \times 10^{-10}$ )         | 0.95 $\times$ (slower, 0.005)                |
|                       | 256 <sup>2</sup> | [NO, SymGS]                                                           | <b>9.96</b> ( $5 \times 10^{-10}$ )         | <b>2.27</b> ( $6 \times 10^{-9}$ )           | <b>5.35</b> ( $5 \times 10^{-10}$ )         | 0.97 $\times$ (slower, 0.003)                |
|                       | 256 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>10.9</b> ( $5 \times 10^{-10}$ )         | <b>2.33</b> ( $5 \times 10^{-10}$ )          | <b>6.27</b> ( $5 \times 10^{-10}$ )         | 0.96 $\times$ (slower, $6 \times 10^{-9}$ )  |
|                       | 256 <sup>2</sup> | [NO, Multigrid]                                                       | <b>2.46</b> ( $5 \times 10^{-10}$ )         | <b>2.39</b> ( $5 \times 10^{-10}$ )          | <b>1.61</b> ( $9 \times 10^{-10}$ )         | 0.98 $\times$ (slower, 0.018)                |
|                       | 256 <sup>2</sup> | [Jacobi, Jacobi(0.67)] ( $10^{-8}$ , work units)                      | <b>1.26</b> ( $10^{-9}$ )                   | 0.99 $\times$ (slower, $8 \times 10^{-7}$ )  | <b>1.23</b> ( $8 \times 10^{-7}$ )          | 0.99 $\times$ (slower, $8 \times 10^{-7}$ )  |
|                       | 256 <sup>2</sup> | [Jacobi, Jacobi(0.67), GS] ( $10^{-8}$ , work units)                  | <b>1.26</b> ( $2 \times 10^{-9}$ )          | 0.99 $\times$ (slower, $8 \times 10^{-7}$ )  | <b>1.24</b> ( $8 \times 10^{-7}$ )          | 0.99 $\times$ (slower, $8 \times 10^{-7}$ )  |
|                       | 512 <sup>2</sup> | [NO, Jacobi]                                                          | <b>1.11</b> ( $10^{-12}$ , 0.121)           | <b>1.11</b> ( $10^{-12}$ , 0.352)            | <b>1.51</b> ( $3 \times 10^{-9}$ )          | 1.00 $\times$ (slower, 0.900)                |
|                       | 512 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>1.16</b> ( $3 \times 10^{-14}$ )         | <b>1.10</b> ( $4 \times 10^{-14}$ )          | <b>1.60</b> ( $9 \times 10^{-9}$ )          | 0.98 $\times$ (slower, 0.193)                |
|                       | 512 <sup>2</sup> | [NO, GS]                                                              | 1.11 $\times$ (0.005)                       | <b>1.11</b> ( $6 \times 10^{-9}$ )           | 1.10 $\times$ (0.003)                       | 0.99 $\times$ (slower, 0.495)                |
|                       | 512 <sup>2</sup> | [NO, SymGS]                                                           | 1.83 $\times$ (0.002)                       | 1.10 $\times$ (0.001)                        | 1.54 $\times$ (0.006)                       | 0.99 $\times$ (slower, 0.900)                |
|                       | 512 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>3.10</b> ( $3 \times 10^{-9}$ )          | <b>1.24</b> ( $3 \times 10^{-9}$ )           | <b>2.72</b> ( $3 \times 10^{-9}$ )          | 1.01 $\times$ (0.193)                        |
|                       | 512 <sup>2</sup> | [NO, Multigrid]                                                       | <b>2.08</b> ( $3 \times 10^{-9}$ )          | <b>2.08</b> ( $3 \times 10^{-9}$ )           | <b>1.63</b> ( $3 \times 10^{-9}$ )          | 0.99 $\times$ (slower, 1.000)                |
| Convection-diffusion  | 128 <sup>2</sup> | [NO, Jacobi]                                                          | <b>2.95</b> ( $< 10^{-10}$ )                | <b>1.34</b> ( $10^{-10}$ )                   | <b>1.67</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $10^{-9}$ )           |
|                       | 128 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>3.44</b> ( $< 10^{-10}$ )                | <b>1.32</b> ( $10^{-10}$ )                   | <b>1.51</b> ( $< 10^{-10}$ )                | 0.96 $\times$ (slower, $3 \times 10^{-10}$ ) |
|                       | 128 <sup>2</sup> | [NO, GS]                                                              | <b>6.76</b> ( $< 10^{-10}$ )                | <b>2.60</b> ( $< 10^{-10}$ )                 | <b>3.01</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $3 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, SymGS]                                                           | <b>8.70</b> ( $< 10^{-10}$ )                | <b>2.40</b> ( $< 10^{-10}$ )                 | <b>3.65</b> ( $< 10^{-10}$ )                | 0.94 $\times$ (slower, $3 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>8.89</b> ( $< 10^{-10}$ )                | <b>2.40</b> ( $< 10^{-10}$ )                 | <b>4.84</b> ( $< 10^{-10}$ )                | 0.96 $\times$ (slower, $5 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, Multigrid]                                                       | <b>2.07</b> ( $< 10^{-10}$ )                | <b>2.01</b> ( $< 10^{-10}$ )                 | 0.95 $\times$ (slower, 0.011)               | 0.95 $\times$ (slower, 0.012)                |
|                       | 128 <sup>2</sup> | [Jacobi, GS] ( $10^{-8}$ , work units)                                | —                                           | 0.78 $\times$ (slower, $< 10^{-10}$ )        | 1.01 $\times$ ( $9 \times 10^{-9}$ )        | 0.99 $\times$ (slower, $7 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS] ( $10^{-8}$ , work units)                         | —                                           | 0.96 $\times$ (slower, $6 \times 10^{-8}$ )  | 1.00 $\times$ (0.003)                       | 0.75 $\times$ (slower, $10^{-10}$ )          |
|                       | 128 <sup>2</sup> | [Jacobi, Jacobi(0.67)] ( $10^{-8}$ , work units)                      | 1.08 $\times$ ( $10^{-10}$ )                | 0.99 $\times$ (slower, $2 \times 10^{-10}$ ) | 1.00 $\times$ (slower, $8 \times 10^{-7}$ ) | 0.99 $\times$ (slower, $10^{-5}$ )           |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, GS] ( $10^{-8}$ , work units)                     | 0.99 $\times$ (slower, 0.309)               | 0.90 $\times$ (slower, $< 10^{-10}$ )        | 0.98 $\times$ (slower, $10^{-7}$ )          | 0.97 $\times$ (slower, $10^{-7}$ )           |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, Jacobi(0.67)] ( $10^{-8}$ , work units)           | —                                           | 0.82 $\times$ (slower, $< 10^{-10}$ )        | 0.95 $\times$ (slower, $< 10^{-10}$ )       | 0.94 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)] ( $10^{-8}$ , work units) | —                                           | 0.93 $\times$ (slower, $3 \times 10^{-10}$ ) | 0.96 $\times$ (slower, $< 10^{-10}$ )       | 0.93 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 256 <sup>2</sup> | [NO, Jacobi]                                                          | <b>1.92</b> ( $9 \times 10^{-9}$ )          | <b>1.30</b> ( $2 \times 10^{-9}$ )           | <b>1.59</b> ( $10^{-9}$ )                   | 0.98 $\times$ (slower, $2 \times 10^{-9}$ )  |
|                       | 256 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>1.80</b> ( $8 \times 10^{-9}$ )          | <b>1.25</b> ( $3 \times 10^{-9}$ )           | <b>1.23</b> ( $2 \times 10^{-9}$ )          | 0.98 $\times$ (slower, 0.011)                |
|                       | 256 <sup>2</sup> | [NO, GS]                                                              | <b>5.52</b> ( $5 \times 10^{-10}$ )         | <b>1.53</b> ( $9 \times 10^{-10}$ )          | <b>3.51</b> ( $5 \times 10^{-10}$ )         | 0.98 $\times$ (slower, 0.919)                |
|                       | 256 <sup>2</sup> | [NO, SymGS]                                                           | <b>6.26</b> ( $5 \times 10^{-10}$ )         | <b>1.69</b> ( $2 \times 10^{-9}$ )           | <b>4.23</b> ( $5 \times 10^{-10}$ )         | 0.97 $\times$ (slower, $4 \times 10^{-9}$ )  |
|                       | 256 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>7.99</b> ( $5 \times 10^{-10}$ )         | <b>1.96</b> ( $3 \times 10^{-9}$ )           | <b>5.29</b> ( $5 \times 10^{-10}$ )         | 0.98 $\times$ (slower, 0.009)                |
|                       | 256 <sup>2</sup> | [NO, Multigrid]                                                       | <b>2.19</b> ( $5 \times 10^{-10}$ )         | <b>2.19</b> ( $5 \times 10^{-10}$ )          | <b>1.54</b> ( $4 \times 10^{-9}$ )          | 0.97 $\times$ (slower, 0.011)                |
|                       | 256 <sup>2</sup> | [Jacobi, GS, SymGS] ( $10^{-8}$ , work units)                         | <b>1.42</b> ( $8 \times 10^{-7}$ )          | 0.99 $\times$ (slower, $8 \times 10^{-7}$ )  | <b>1.24</b> ( $8 \times 10^{-7}$ )          | 0.95 $\times$ (slower, $8 \times 10^{-7}$ )  |
|                       | 256 <sup>2</sup> | [Jacobi, GS, SymGS, GS] ( $10^{-8}$ , work units)                     | <b>1.42</b> ( $8 \times 10^{-7}$ )          | 0.99 $\times$ (slower, $8 \times 10^{-7}$ )  | <b>1.26</b> ( $8 \times 10^{-7}$ )          | 0.99 $\times$ (slower, $10^{-7}$ )           |
|                       | 512 <sup>2</sup> | [NO, Jacobi]                                                          | 1.17 $\times$ (0.005)                       | 1.14 $\times$ (0.018)                        | <b>1.36</b> ( $3 \times 10^{-9}$ )          | 1.00 $\times$ (slower, 0.900)                |
|                       | 512 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | 1.08 $\times$ (0.005)                       | 1.07 $\times$ ( $3 \times 10^{-9}$ )         | <b>1.40</b> ( $2 \times 10^{-9}$ )          | 0.94 $\times$ (slower, 0.105)                |
|                       | 512 <sup>2</sup> | [NO, GS]                                                              | 1.05 $\times$ (0.004)                       | 1.05 $\times$ (0.004)                        | 1.06 $\times$ ( $3 \times 10^{-9}$ )        | 1.01 $\times$ (0.074)                        |
|                       | 512 <sup>2</sup> | [NO, SymGS]                                                           | 1.55 $\times$ ( $2 \times 10^{-4}$ )        | 1.11 $\times$ (0.001)                        | <b>1.40</b> ( $3 \times 10^{-9}$ )          | 0.99 $\times$ (slower, 0.632)                |
|                       | 512 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>2.48</b> ( $3 \times 10^{-9}$ )          | 1.14 $\times$ (0.002)                        | <b>2.27</b> ( $3 \times 10^{-9}$ )          | 0.97 $\times$ (slower, 0.088)                |
|                       | 512 <sup>2</sup> | [NO, Multigrid]                                                       | <b>1.83</b> ( $3 \times 10^{-9}$ )          | <b>1.73</b> ( $3 \times 10^{-9}$ )           | <b>1.56</b> ( $3 \times 10^{-9}$ )          | 0.98 $\times$ (slower, 0.495)                |
| Anisotropic diffusion | 128 <sup>2</sup> | [NO, Jacobi]                                                          | <b>3.16</b> ( $< 10^{-10}$ )                | <b>1.25</b> ( $6 \times 10^{-10}$ )          | <b>1.63</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $3 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>3.31</b> ( $< 10^{-10}$ )                | <b>1.34</b> ( $< 10^{-10}$ )                 | <b>1.68</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $3 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, GS]                                                              | <b>7.61</b> ( $< 10^{-10}$ )                | <b>1.99</b> ( $< 10^{-10}$ )                 | <b>3.04</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 128 <sup>2</sup> | [NO, SymGS]                                                           | <b>11.8</b> ( $< 10^{-10}$ )                | <b>2.72</b> ( $< 10^{-10}$ )                 | <b>4.08</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $9 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>7.36</b> ( $< 10^{-10}$ )                | <b>2.04</b> ( $< 10^{-10}$ )                 | <b>3.06</b> ( $< 10^{-10}$ )                | 0.95 $\times$ (slower, $10^{-9}$ )           |
|                       | 128 <sup>2</sup> | [Jacobi, GS] ( $10^{-8}$ , work units)                                | —                                           | 0.99 $\times$ (slower, $2 \times 10^{-4}$ )  | 1.00 $\times$ (slower, $3 \times 10^{-9}$ ) | 0.88 $\times$ (slower, $< 10^{-10}$ )        |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS] ( $10^{-8}$ , work units)                         | —                                           | 0.99 $\times$ (slower, 0.003)                | 0.98 $\times$ (slower, $10^{-9}$ )          | 0.88 $\times$ (slower, $4 \times 10^{-9}$ )  |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, GS] ( $10^{-8}$ , work units)                     | 1.14 $\times$ ( $< 10^{-10}$ )              | 0.99 $\times$ (slower, $< 10^{-10}$ )        | 1.00 $\times$ (slower, 0.774)               | 0.99 $\times$ (slower, 0.053)                |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, GS] ( $10^{-8}$ , work units)                     | 1.15 $\times$ ( $< 10^{-10}$ )              | 0.99 $\times$ (slower, $2 \times 10^{-6}$ )  | 0.97 $\times$ (slower, $10^{-7}$ )          | 0.97 $\times$ (slower, $10^{-6}$ )           |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, Jacobi(0.67)] ( $10^{-8}$ , work units)           | —                                           | 0.95 $\times$ (slower, $4 \times 10^{-9}$ )  | 0.98 $\times$ (slower, $2 \times 10^{-7}$ ) | 0.96 $\times$ (slower, $5 \times 10^{-8}$ )  |
|                       | 128 <sup>2</sup> | [Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)] ( $10^{-8}$ , work units) | —                                           | 0.95 $\times$ (slower, $2 \times 10^{-9}$ )  | 0.85 $\times$ (slower, $< 10^{-10}$ )       | 0.89 $\times$ (slower, $4 \times 10^{-10}$ ) |
|                       | 256 <sup>2</sup> | [NO, Jacobi]                                                          | <b>1.58</b> ( $6 \times 10^{-9}$ )          | <b>1.11</b> ( $6 \times 10^{-9}$ )           | <b>1.18</b> ( $3 \times 10^{-9}$ )          | 1.00 $\times$ (slower, 0.144)                |
|                       | 256 <sup>2</sup> | [NO, Jacobi (0.67)]                                                   | <b>1.56</b> ( $6 \times 10^{-9}$ )          | <b>1.11</b> ( $4 \times 10^{-9}$ )           | <b>1.18</b> ( $3 \times 10^{-9}$ )          | 0.99 $\times$ (slower, 0.088)                |
|                       | 256 <sup>2</sup> | [NO, GS]                                                              | <b>3.18</b> ( $3 \times 10^{-9}$ )          | <b>1.37</b> ( $3 \times 10^{-9}$ )           | <b>1.71</b> ( $3 \times 10^{-9}$ )          | 0.99 $\times$ (slower, 0.117)                |
|                       | 256 <sup>2</sup> | [NO, SOR (1.5)]                                                       | <b>2.98</b> ( $3 \times 10^{-9}$ )          | <b>1.38</b> ( $3 \times 10^{-9}$ )           | <b>1.70</b> ( $3 \times 10^{-9}$ )          | 0.99 $\times$ (slower, 0.001)                |
|                       | 256 <sup>2</sup> | [Jacobi, GS, SymGS] ( $10^{-8}$ , work units)                         | 1.03 $\times$ (0.604)                       | 1.00 $\times$ (slower, 0.004)                | 1.00 $\times$ (slower, 0.005)               | 1.00 $\times$ (slower, 0.196)                |
|                       | 256 <sup>2</sup> | [Jacobi, GS, SymGS, GS] ( $10^{-8}$ , work units)                     | 0.97 $\times$ (slower, 0.234)               | 0.92 $\times$ (slower, $4 \times 10^{-4}$ )  | 0.99 $\times$ (slower, 0.006)               | 1.00 $\times$ (slower, 0.179)                |

## 1444 E.10 Verification of the theoretical assumptions

1445 Theorem 4.1 and Propositions 4.2–4.3 assume that every error propagation map  $G_j = I - C_j \circ \mathcal{L}_h^a$  is  
1446 Lipschitz with constant  $\rho_j \leq 1$  (with at least one strict contraction in the optimal sequence, respec-  
1447 tively a shared unitary eigenbasis), zero-preserving, and—for postfix monotonicity—simultaneously  
1448 diagonalisable by a unitary matrix; Theorem 5.1 assumes that at every state at least one solver leaves  
1449 a positive fraction of the error. Table 34 checks these conditions numerically for every operation  
1450 of the study—finite diagnostics (a power-iteration estimate of the Lipschitz constant, commutators  
1451 on random vectors, exhaustive search over short horizons), none of which is a certificate over the  
1452 whole input space— on the mean-free subspace (the constant mode is the null space of the pe-  
1453 riodic operator and is excluded from the data and from every iteration): the Euclidean Lipschitz  
1454 constant  $\|G_j\|_2$  (power iteration on  $G_j^T G_j$ ; exact from the Fourier symbol for Jacobi), the Lipschitz  
1455 constant  $\|G_j\|_A = \|A_s^{1/2} G_j A_s^{-1/2}\|_2$  in the energy norm of the symmetric part  $A_s$  of the discrete  
1456 operator, the same for the macro-action  $G_j^{m_j}$ , the spectral radius, the smallest observed contraction  
1457  $\sigma_{\min} = \min \|G_j x\| / \|x\|$ ,  $\|C_j(0)\|$ , and the commutator  $\|G_j G_{\text{NO}} - G_{\text{NO}} G_j\|$  on random vectors.

1458 Table 36 evaluates the quantities of Theorem 4.1 and Theorem 5.1 along the cost-aware oracle’s own  
 1459 path to  $\varepsilon = 10^{-8}$  (a horizon of tens of macro-actions; to  $h^2$  the path has one or two steps).

1460 **What holds, and in which norm.** (i) Every classical solver is zero-preserving exactly and, on the  
 1461 mean-free Nyquist-free subspace on which the forcing lives (undamped Jacobi has eigenvalue  $-1$   
 1462 on the checkerboard mode, which the forcing excludes and the Jacobi-type maps never generate;  
 1463 the table’s Jacobi values include the Nyquist modes as its caption states), contracts in the sense that  
 1464 matters for convergence: the spectral radius of  $G_j$  is below 1 in every setting (at most 1.000000), and  
 1465 so is the energy-norm Lipschitz constant of the solver and of its macro-action (at most 0.999760).  
 1466 The *Euclidean* Lipschitz constant, however, exceeds 1 for Gauss–Seidel, SOR and symmetric GS  
 1467 (up to 1.89 for SOR), whose iteration matrices are non-normal; the proof of Proposition 4.2 uses  
 1468 only the triangle inequality and the composition rule for Lipschitz constants, so it holds verbatim  
 1469 with  $\|\cdot\|_2$  replaced by the energy norm, in which the condition is satisfied by all classical solvers  
 1470 in all settings—but that statement concerns the greedy rule and objective defined with the energy  
 1471 norm, whereas the oracle and the router of this study minimise the Euclidean error. For the evaluated  
 1472 Euclidean rule, the strict-contraction premise of Proposition 4.2 therefore holds for the Jacobi-type  
 1473 solvers and fails for the non-normal sweeps, whose Euclidean expansion is confined to a single sweep  
 1474 (their spectral radii and macro-action contraction show the iteration is convergent). For (damped)  
 1475 Jacobi both constants agree (the maps are Fourier-diagonal); the values are close to 1 because the  
 1476 slowest modes (the smoothest ones, and the near-Nyquist mode for undamped Jacobi) are barely  
 1477 damped by a point smoother—which is exactly why the corrector is needed. (ii) The corrector’s  
 1478 map is zero-preserving, contracts every band mode to  $1.3 \times 10^{-1}$  or better (on average to  $10^{-4}$ –  
 1479  $10^{-6}$ ) and leaves the modes outside the band untouched, so its Lipschitz constant is exactly 1 (not  
 1480  $< 1$ ): Theorem 4.2 in its strict form does not apply to the corrector, but Theorem 4.3 does, and its  
 1481 second condition—a shared unitary eigenbasis—holds exactly for the exact band-limited inverse  
 1482 together with (damped) Jacobi, all of which are diagonal in the Fourier basis, and, for the learned  
 1483 corrector, the measured commutators are  $\approx 10^{-6}$ , at the corrector’s single-precision accuracy (a  
 1484 diagnostic; replacing the learned corrector by the exact inverse did not change the oracle’s decisions  
 1485 in the development run, Section E.1), so for the ensembles  $\{\text{NO}, \text{Jacobi}\}$ ,  $\{\text{NO}, \text{Jacobi} (0.67)\}$  and  
 1486  $\{\text{NO}, \text{Jacobi}, \text{Jacobi} (0.67)\}$  with the exact band-limited inverse the objective is supermodular and  
 1487 postfix monotone and Theorem 4.1 applies with  $\alpha = \mu = 1$ ; for the fitted corrector we measure  
 1488 the fit error and the commutators but do not propagate them through a perturbation bound (a small  
 1489 commutator is a diagnostic, not a certificate), so no guarantee is stated for it, and the exhaustive  
 1490 short-horizon check below is the evidence that the fitted maps behave like the exact ones. For GS,  
 1491 SOR, symmetric GS and multigrid the commutators are small but nonzero ( $10^{-3}$ – $10^{-1}$ ) and no  
 1492 guarantee is claimed at any horizon. (iii) Invertibility, which an earlier version of the theory used for  
 1493 postfix monotonicity, is neither needed for the corrected result nor sufficient in general (Section A.3);  
 1494 for the record, a Gauss–Seidel or symmetric-GS sweep annihilates an error concentrated at the  
 1495 first grid point ( $\sigma_{\min} = 0$ ), SOR leaves  $(1 - \omega)$  of it ( $\sigma_{\min} = 0.72$ – $0.83$  in Table 34), undamped  
 1496 Jacobi annihilates the mode  $(N/4, N/4)$ , and the corrector annihilates the band. Postfix monotonicity  
 1497 ( $\mu = 1$ ) is therefore guaranteed only for the Jacobi-type ensembles with the exact band-limited  
 1498 inverse, and is measured for the others. (iv) Table 35 checks Theorem 4.1 by exhaustive enumeration  
 1499 (in floating point, with errors clipped at a relative floor of  $10^{-11}$ ) for short horizons on the evaluated  
 1500 ensembles, in the Euclidean objective the oracle optimises and for the deployed cost-aware rule: for  
 1501 every pairing (including line GS on anisotropic diffusion), for  $\{\text{Jacobi}, \text{Jacobi} (0.67)\}$ , for the two  
 1502 non-commuting sweeps  $\{\text{GS}, \text{SOR}\}$ , for  $\{\text{GS}, \text{multigrid}\}$  and for all point smoothers together, the  
 1503 optimal sequence  $O$  of  $T$  macro-actions ( $T = 4$ , or 3 for ensembles with more than four actions) is  
 1504 found by exhaustive search on 8 test instances (4 for ensembles with more than four actions), and  
 1505  $\mu$  (over the greedy prefixes),  $\alpha(O)$  (over the greedy prefixes, and over every sequence of length  
 1506 below  $T$ ) and the bound itself are computed. Over all 296 instance–ensemble pairs the largest  $\mu$   
 1507 is 1.104, the largest  $\alpha(O)$  over the greedy prefixes is 1.003 and over all short prefixes 1.003; the  
 1508 numerical bound holds on 296/296 pairs, and the on-path premises that the recurrence proof of  
 1509 Theorem 4.1 uses—prefix monotonicity and the weak-supermodularity inequality at the reported  $\alpha$   
 1510 on the greedy prefixes (not the global definition), the deployed rule equal to Algorithm 1, and an  
 1511 exhaustive optimum resolved above the round-off floor—are verified, as a finite numerical check, on  
 1512 183/296 pairs (on 80 the optimum is at the floor and the instance is numerically unresolved rather  
 1513 than certified; the deployed rule differs from Algorithm 1 on 0, and the clipping would have changed  
 1514 the deployed path on 6); 38 greedy prefixes and 174 prefixes overall show an expansion of the error  
 1515 by a single action of  $O$ , i.e. a failure of prefix monotonicity in the Euclidean norm, and 0 greedy

1516 prefixes (0 prefixes overall) violate the inequality at the reported  $\alpha$  (the largest candidate  $\alpha$  over  
 1517 greedy prefixes, 1.003, includes the pairs that fail). The greedy sequence’s error is within  $8606.0\times$  of  
 1518 the exhaustive optimum on every one of the 216 pairs with a resolved optimum (on the others the  
 1519 ratio is one of clipped objectives; the table gives medians and the worst cases per ensemble). The  
 1520 check supports the optimality of the greedy rule over a few macro-actions; the horizons at which  
 1521 the oracle runs to  $10^{-8}$  are 10–400 macro-actions (Table 36), where no constant is established for  
 1522 the non-normal sweeps: Proposition 4.2 would need contraction in the Euclidean norm, which they  
 1523 lack (Table 34), and the worst-case constant in Table 36 is computed with energy-norm constants for  
 1524 orientation only. The worst-case  $\alpha(O)$  of Theorem 4.2, which only uses the Lipschitz constants, is  
 1525 loose in comparison: along the oracle’s path to  $10^{-8}$  (Table 36)  $\sum_i \rho_{O_i}^2$  is close to  $T$  because each  $\rho$   
 1526 is dominated by the slowest mode, so the bound ranges from 1 (the multigrid pairing, whose cycle  
 1527 contracts everything) to  $10^2$ – $10^4$  for the point smoothers, and is infinite on the few instances where  
 1528  $\sum_i \rho_{O_i}^2 \geq T$ . (The last two columns of Table 36 evaluate the ratio of Equation (7) with the oracle’s  
 1529 own suffix in place of the unknown long-horizon  $O$ ; this is only a diagnostic, and its maximum is  
 1530 attained trivially at the last prefix.) (v) Theorem 5.1 is stated for the per-state normalised weights  
 1531  $\tilde{c}_j / \sum_k \tilde{c}_k$  that the router is trained with; its one hypothesis is that at every visited state at least  
 1532 one solver leaves a positive fraction of the current error, i.e.  $\max_j \tilde{c}_j / \|e^{(t)}\|^2 > 0$  (last column of  
 1533 Table 36; the column before it shows that the per-step errors never exceed the current error), so the  
 1534 surrogate’s Bayes consistency applies at every visited state.

Table 34: Numerical check of the assumptions of Propositions 4.2–4.3 for every operation of the  
 study (mean-free subspace). **[Provisional: development run; the confirmatory run regenerates  
 this table with the compiled kernels’ macro-action sizes.]**  $\|\cdot\|_2$ : Euclidean Lipschitz constant  
 of the error propagation map (power-iteration estimate; exact from the symbol for Jacobi);  $\|\cdot\|_A$ :  
 Lipschitz constant in the energy norm of the symmetric part of  $\mathcal{L}_h$ , for one iteration and for the  
 macro-action of  $m_j$  iterations;  $\rho$ : spectral radius;  $\sigma_{\min}$ : smallest observed contraction (a unit error  
 at the first grid point for the triangular sweeps, the minimum over Fourier modes otherwise; for the  
 corrector, the largest remaining fraction of a band mode);  $\|C_j(0)\|$ : zero preservation; last column:  
 commutator with the corrector’s map on random unit vectors (Proposition 4.3). For (damped) Jacobi  
 the constants are exact from the Fourier symbol (the value with the Nyquist modes included; without  
 them undamped Jacobi has  $\|G\|_2 = \rho = 0.9994$  at  $128^2$ ).

| Equation                                | Operation          | $m_j$ | $\ I - C_j \mathcal{L}_h\ _2$ | $\ I - C_j \mathcal{L}_h\ _A$ | $\ (I - C_j \mathcal{L}_h)^{m_j}\ _A$ | $\rho(I - C_j \mathcal{L}_h)$ | $\sigma_{\min}$ | $\ C_j(0)\ $ | $\ [G_j, G_{\text{SO}}]\ $ |
|-----------------------------------------|--------------------|-------|-------------------------------|-------------------------------|---------------------------------------|-------------------------------|-----------------|--------------|----------------------------|
| Poisson, $128^2$                        | DeepONet corrector | 1     | 1.000000                      | 1.000000                      | 1.000000                              | 1.000000                      | 4.4e-05 (band)  | 0            | –                          |
|                                         | Jacobi             | 32    | 0.999398                      | 0.999398                      | 0.980906                              | 0.999398                      | 0               | 0            | 5.7e-06                    |
|                                         | Jacobi (0.67)      | 32    | 0.999596                      | 0.999596                      | 0.987168                              | 0.999596                      | 1.4e-04         | 0            | 4.1e-06                    |
|                                         | GS                 | 11    | 1.021515                      | 0.998595                      | 0.986837                              | 0.998797                      | 0               | 0            | 3.1e-02                    |
|                                         | SymGS              | 7     | 1.006607                      | 0.997396                      | 0.983422                              | 0.997623                      | 0               | 0            | 2.2e-02                    |
|                                         | SOR (1.5)          | 11    | 1.449417                      | 0.996467                      | 0.961304                              | 0.996411                      | 7.2e-01         | 0            | 8.5e-02                    |
|                                         | Multigrid          | 2     | 0.098203                      | 0.097860                      | 0.009575                              | 0.096877                      | 1.9e-03         | 0            | 8.8e-03                    |
| Convection–diffusion, $128^2$           | DeepONet corrector | 1     | 1.000000                      | 1.000000                      | 1.000000                              | 1.000000                      | 1.5e-05 (band)  | 0            | –                          |
|                                         | Jacobi             | 30    | 0.999398                      | 0.999398                      | 0.982089                              | 0.999398                      | 0               | 0            | 3.1e-06                    |
|                                         | Jacobi (0.67)      | 34    | 0.999596                      | 0.999596                      | 0.986371                              | 0.999596                      | 1.4e-04         | 0            | 2.2e-06                    |
|                                         | GS                 | 12    | 1.025726                      | 0.998460                      | 0.983255                              | 0.998589                      | 0               | 0            | 3.1e-02                    |
|                                         | SymGS              | 8     | 1.010112                      | 0.997353                      | 0.980965                              | 0.997571                      | 0               | 0            | 2.3e-02                    |
|                                         | SOR (1.5)          | 12    | 1.554544                      | 0.996981                      | 0.931772                              | 0.993973                      | 7.5e-01         | 0            | 9.1e-02                    |
|                                         | Multigrid          | 2     | 0.076967                      | 0.074905                      | 0.005613                              | 0.074592                      | 1.5e-03         | 0            | 6.9e-03                    |
| Anisotropic diffusion, $128^2$          | DeepONet corrector | 1     | 1.000000                      | 1.000000                      | 1.000000                              | 1.000000                      | 4.4e-04 (band)  | 0            | –                          |
|                                         | Jacobi             | 30    | 0.999988                      | 0.999988                      | 0.999642                              | 0.999988                      | 1.1e-16         | 0            | 2.4e-05                    |
|                                         | Jacobi (0.67)      | 30    | 0.999992                      | 0.999992                      | 0.999760                              | 0.999992                      | 4.4e-03         | 0            | 1.6e-05                    |
|                                         | GS                 | 12    | 1.131668                      | 0.999528                      | 0.999396                              | 0.999976                      | 0               | 0            | 5.7e-02                    |
|                                         | SymGS              | 8     | 1.102789                      | 0.999489                      | 0.998994                              | 0.999953                      | 0               | 0            | 5.5e-02                    |
|                                         | SOR (1.5)          | 12    | 1.889774                      | 0.999393                      | 0.999108                              | 0.999928                      | 8.3e-01         | 0            | 1.5e-01                    |
|                                         | Line GS            | 7     | 1.125366                      | 0.997628                      | 0.983361                              | 0.997599                      | 2.5e-03         | 0            | 2.4e-03                    |
| Variable-coefficient diffusion, $128^2$ | DeepONet corrector | 1     | 1.118533                      | –                             | –                                     | 1.000000                      | 1.3e-01 (band)  | 0            | –                          |
|                                         | Jacobi             | 20    | 0.998841                      | –                             | –                                     | 1.000000                      | 6.6e-04         | 0            | 1.7e-03                    |
|                                         | Jacobi (0.67)      | 21    | 0.999226                      | –                             | –                                     | 0.999656                      | 4.4e-03         | 0            | 1.1e-03                    |
|                                         | GS                 | 6     | 1.022959                      | –                             | –                                     | 0.998972                      | 1.3e-16         | 0            | 3.1e-02                    |
|                                         | SymGS              | 4     | 1.008153                      | –                             | –                                     | 0.997969                      | 2.3e-16         | 0            | 2.2e-02                    |
|                                         | SOR (1.5)          | 6     | 1.458326                      | –                             | –                                     | 0.996862                      | 7.3e-01         | 0            | 8.5e-02                    |
|                                         | Multigrid          | 1     | 0.151899                      | –                             | –                                     | 0.111987                      | 1.5e-03         | 0            | 8.9e-03                    |

Table 35: Exhaustive short-horizon check of Theorem 4.1 ( $128 \times 128$ ,  $n$  test instances per row,  $T$  macro-actions,  $K$  actions including the corrector, deployed cost-aware rule): the optimal sequence  $O$  is found by exhaustive search;  $\mu = \max_{t < T} g(S^t \oplus O)/g(O)$  over the greedy prefixes;  $\alpha(O)$  over the greedy prefixes and over all sequences of length below  $T$ ; whether  $g(S^T) \leq \mu(1 - \phi_T)g(O) + \phi_T g(\emptyset)$  holds; the ratio  $g(S^T)/g(O)$  of the greedy to the optimal error (median and maximum over instances); the number of instances on which the greedy sequence is the optimum; and the numbers of instances on which the exhaustive optimum is at the round-off floor (numerically unresolved), on which the deployed rule coincides with Algorithm 1, of greedy prefixes and of all prefixes at which the weak-supermodularity inequality fails at the reported  $\alpha$ , and of instances on which the on-path premises of the proof were verified. Errors below  $10^{-11}$  relative are clipped to that floor.

| Equation                                | Ensemble                                            | $K$ | $T$ | $n$ | $\mu$ | $\alpha(O)$ greedy prefixes | $\alpha(O)$ all $ S  < T$ | bound holds | $g(S^T)/g(O)$ med (max) | greedy $= O$ | $g(O)$ at floor $f$ rule – Alg. 1 <sup>†</sup> violations (greedy prefixes, all prefixes) / on-path premises verified |
|-----------------------------------------|-----------------------------------------------------|-----|-----|-----|-------|-----------------------------|---------------------------|-------------|-------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------|
| Poisson, $128^2$                        | {Jacobi}                                            | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 6/8          | 0/8/0/0/8                                                                                                             |
|                                         | {Jacobi(0.67)}                                      | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 8/8/0/0/0                                                                                                             |
|                                         | {GS}                                                | 2   | 4   | 8   | 1.003 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SymGS}                                             | 2   | 4   | 8   | 1.004 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SOR(1.5)}                                          | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.001 (1.02)            | 4/8          | 0/8/0/0/8                                                                                                             |
|                                         | {Multigrid}                                         | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 7.545 (21.54)           | 1/8          | 5/8/0/0/3                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67)}                              | 3   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 8/8/0/0/0                                                                                                             |
|                                         | {GS, SOR(1.5)}                                      | 3   | 4   | 8   | 1.003 | 1.000                       | 1.000                     | 8/8         | 1.000 (2.29)            | 6/8          | 0/8/0/0/8                                                                                                             |
|                                         | {GS, Multigrid}                                     | 3   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 7.545 (21.54)           | 1/8          | 5/8/0/0/3                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5)}         | 6   | 3   | 4   | 1.000 | 1.000                       | 1.000                     | 4/4         | 1.000 (58860.85)        | 3/4          | 1/4/0/0/3                                                                                                             |
| Convection-diffusion, $128^2$           | {Jacobi}                                            | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 5/8          | 0/8/0/0/8                                                                                                             |
|                                         | {Jacobi(0.67)}                                      | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 8/8/0/0/0                                                                                                             |
|                                         | {GS}                                                | 2   | 4   | 8   | 1.046 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SymGS}                                             | 2   | 4   | 8   | 1.034 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SOR(1.5)}                                          | 2   | 4   | 8   | 1.020 | 1.000                       | 1.000                     | 8/8         | 1.165 (1.56)            | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {Multigrid}                                         | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 6.936 (11.45)           | 0/8          | 4/8/0/0/4                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67)}                              | 3   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 8/8/0/0/0                                                                                                             |
|                                         | {GS, SOR(1.5)}                                      | 3   | 4   | 8   | 1.046 | 1.000                       | 1.000                     | 8/8         | 1.000 (2.34)            | 7/8          | 0/8/0/0/8                                                                                                             |
|                                         | {GS, Multigrid}                                     | 3   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 6.936 (11.45)           | 0/8          | 4/8/0/0/4                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5)}         | 6   | 3   | 4   | 1.000 | 1.000                       | 1.000                     | 4/4         | 1.000 (413405.00)       | 3/4          | 1/4/0/0/3                                                                                                             |
| Anisotropic diffusion, $128^2$          | {Jacobi}                                            | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 7/8          | 0/8/0/0/8                                                                                                             |
|                                         | {Jacobi(0.67)}                                      | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (63.46)           | 6/8          | 1/8/0/0/7                                                                                                             |
|                                         | {GS}                                                | 2   | 4   | 8   | 1.007 | 1.000                       | 1.000                     | 8/8         | 1.447 (39.70)           | 2/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SymGS}                                             | 2   | 4   | 8   | 1.002 | 1.000                       | 1.000                     | 8/8         | 1.154 (1.58)            | 2/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SOR(1.5)}                                          | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.377 (144.56)          | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {SymGS}                                             | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 1/8          | 8/8/0/0/0                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67)}                              | 3   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.250 (50.26)           | 0/8          | 6/8/0/0/2                                                                                                             |
|                                         | {GS, SOR(1.5)}                                      | 3   | 4   | 8   | 1.007 | 1.000                       | 1.000                     | 8/8         | 1.752 (19.70)           | 0/8          | 0/8/0/0/8                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5), LineGS} | 7   | 3   | 4   | 1.000 | 1.000                       | 1.000                     | 4/4         | 1.268 (3.66)            | 2/4          | 1/4/0/0/3                                                                                                             |
|                                         | {Jacobi}                                            | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 6/8          | 0/8/0/0/4                                                                                                             |
| Variable-coefficient diffusion, $128^2$ | {Jacobi(0.67)}                                      | 2   | 4   | 8   | 1.002 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.07)            | 3/8          | 4/8/0/0/1                                                                                                             |
|                                         | {GS}                                                | 2   | 4   | 8   | 1.002 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/4                                                                                                             |
|                                         | {SymGS}                                             | 2   | 4   | 8   | 1.010 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/4                                                                                                             |
|                                         | {SOR(1.5)}                                          | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 0/8/0/0/4                                                                                                             |
|                                         | {Multigrid}                                         | 2   | 4   | 8   | 1.012 | 1.000                       | 1.000                     | 8/8         | 1.167 (1.00)            | 4/8          | 0/8/0/0/4                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67)}                              | 3   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 0/8          | 8/8/0/0/0                                                                                                             |
|                                         | {GS, SOR(1.5)}                                      | 3   | 4   | 8   | 1.002 | 1.000                       | 1.000                     | 8/8         | 1.000 (6.20)            | 7/8          | 0/8/0/0/4                                                                                                             |
|                                         | {GS, Multigrid}                                     | 3   | 4   | 8   | 1.012 | 1.000                       | 1.000                     | 8/8         | 6.154 (19.94)           | 3/8          | 0/8/0/0/4                                                                                                             |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5)}         | 6   | 3   | 4   | 1.005 | 1.003                       | 1.003                     | 4/4         | 1.000 (8605.92)         | 3/4          | 0/4/0/0/2                                                                                                             |
|                                         | {Jacobi}                                            | 2   | 4   | 8   | 1.000 | 1.000                       | 1.000                     | 8/8         | 1.000 (1.00)            | 6/8          | 0/8/0/0/4                                                                                                             |

Table 36: Quantities of Theorem 4.1, Proposition 4.2 and Theorem 5.1 along the cost-aware oracle’s path to  $\varepsilon = 10^{-8}$  (8 test instances; medians unless stated): number of macro-actions  $T$ ,  $\sum_i \rho_{O_i}^2$  with the energy-norm constants of the chosen macro-actions, the per-instance margin  $T - \sum_i \rho_{O_i}^2$  (median) and the resulting per-instance  $\alpha(O)$  of Proposition 4.2 (median and range over the instances on which the margin is positive;  $\infty$  otherwise), the empirical supermodularity ratio  $\hat{\alpha}$  of Equation (7) evaluated with  $S$  the oracle’s prefix and  $S'$  its suffix (maximum and median over prefixes and instances), and the normalised bounds of Theorem 5.1: the largest ratio  $\max_j \tilde{c}_j / \|e^{(t)}\|^2$  over visited states ( $\bar{E}$ ) and the smallest value of  $\max_j \tilde{c}_j / \|e^{(t)}\|^2$  ( $E_{\min}$ ).

| Equation                                | Ensemble                                            | $T$ | $T - \sum_i \rho_{O_i}^2$ | $\alpha(O)$ (Prop. 4.2): median [min, max] | $\hat{\alpha}$ max | $\hat{\alpha}$ median | $\max_i \max_j \tilde{c}_j / \ e^{(t)}\ ^2$ | $\min_i \max_j \tilde{c}_j / \ e^{(t)}\ ^2$ |
|-----------------------------------------|-----------------------------------------------------|-----|---------------------------|--------------------------------------------|--------------------|-----------------------|---------------------------------------------|---------------------------------------------|
| Poisson, $128^2$                        | {Jacobi}                                            | 180 | 6.733                     | 1.00 [1.00, 26.44]                         | 1.000              | 0.151                 | 1.000                                       | 4.6e-01                                     |
|                                         | {Jacobi(0.67)}                                      | 3   | 0.025                     | 39.22 [39.22, 39.22]                       | 1.000              | 1.000                 | 1.000                                       | 4.6e-01                                     |
|                                         | {GS}                                                | 5   | 0.052                     | 19.12 [19.12, 38.24]                       | 1.014              | 0.500                 | 1.000                                       | 3.8e-01                                     |
|                                         | {SymGS}                                             | 4   | 0.066                     | 15.21 [15.21, 30.41]                       | 1.008              | 0.540                 | 1.000                                       | 3.4e-01                                     |
|                                         | {SOR(1.5)}                                          | 6   | 0.266                     | 3.84 [3.29, 13.18]                         | 1.000              | 0.392                 | 1.000                                       | 2.8e-01                                     |
|                                         | {Multigrid}                                         | 3   | 2.500                     | 1.00 [1.00, 1.00]                          | 1.000              | 0.500                 | 0.988                                       | 6.1e-10                                     |
|                                         | {Jacobi, Jacobi(0.67)}                              | 3   | 0.025                     | 39.22 [15.79, 39.22]                       | 1.000              | 1.000                 | 1.000                                       | 4.6e-01                                     |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5)}         | 3   | 0.025                     | 39.22 [13.18, 39.22]                       | 1.000              | 1.000                 | 1.000                                       | 4.6e-01                                     |
|                                         | {Jacobi}                                            | 223 | 7.881                     | 1.00 [1.00, 14.08]                         | 1.000              | 0.129                 | 1.000                                       | 4.9e-01                                     |
|                                         | {Jacobi(0.67)}                                      | 3   | 0.027                     | 36.94 [36.94, 36.94]                       | 1.000              | 1.000                 | 1.000                                       | 4.6e-01                                     |
| Convection-diffusion, $128^2$           | {GS}                                                | 5   | 0.066                     | 15.06 [15.06, 30.11]                       | 1.017              | 0.500                 | 1.000                                       | 3.8e-01                                     |
|                                         | {SymGS}                                             | 5   | 0.075                     | 13.26 [13.26, 26.52]                       | 1.009              | 0.500                 | 1.000                                       | 3.7e-01                                     |
|                                         | {SOR(1.5)}                                          | 6   | 0.395                     | 2.53 [2.53, 7.59]                          | 1.000              | 0.421                 | 1.000                                       | 3.3e-01                                     |
|                                         | {Multigrid}                                         | 3   | 3.000                     | 1.00 [1.00, 1.00]                          | 1.000              | 0.500                 | 0.998                                       | 2.5e-09                                     |
|                                         | {Jacobi, Jacobi(0.67)}                              | 3   | 0.027                     | 36.94 [15.98, 36.94]                       | 1.000              | 1.000                 | 1.000                                       | 4.6e-01                                     |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5)}         | 3   | 0.027                     | 36.94 [7.59, 36.94]                        | 1.000              | 1.000                 | 1.000                                       | 4.6e-01                                     |
|                                         | {Jacobi}                                            | 116 | 0.082                     | 12.70 [8.52, 1398]                         | 1.000              | 0.225                 | 1.000                                       | 1.9e-01                                     |
|                                         | {Jacobi(0.67)}                                      | 4   | 0.001                     | 1043 [1043, 2086]                          | 1.000              | 0.615                 | 1.000                                       | 2.5e-01                                     |
|                                         | {GS}                                                | 4   | 0.002                     | 414 [276, 828]                             | 1.001              | 0.539                 | 1.000                                       | 2.3e-01                                     |
|                                         | {SymGS}                                             | 4   | 0.004                     | 249 [249, 497]                             | 1.014              | 0.533                 | 1.000                                       | 1.9e-01                                     |
| Anisotropic diffusion, $128^2$          | {SOR(1.5)}                                          | 6   | 0.005                     | 187 [187, 561]                             | 1.000              | 0.434                 | 1.000                                       | 1.1e-01                                     |
|                                         | {LineGS}                                            | 3   | 0.033                     | 30.30 [30.30, 30.30]                       | 1.000              | 1.000                 | 1.000                                       | 4.1e-02                                     |
|                                         | {Jacobi, Jacobi(0.67)}                              | 4   | 0.001                     | 837 [837, 2086]                            | 1.000              | 0.591                 | 1.000                                       | 2.5e-01                                     |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5), LineGS} | 3   | 0.033                     | 30.30 [30.30, 30.30]                       | 1.000              | 1.000                 | 1.000                                       | 2.5e-01                                     |
|                                         | {Jacobi}                                            | 352 | -1.260                    | $\infty$                                   | 1.000              | 0.155                 | 1.002                                       | 4.3e-01                                     |
|                                         | {Jacobi(0.67)}                                      | 4   | 0.026                     | 39.04 [39.04, 78.09]                       | 1.000              | 0.623                 | 1.003                                       | 3.5e-01                                     |
|                                         | {GS}                                                | 6   | -0.032                    | $\infty$                                   | 1.000              | 0.422                 | 1.002                                       | 2.8e-01                                     |
|                                         | {SymGS}                                             | 6   | 0.012                     | 85.74 [68.59, 206]                         | 1.000              | 0.459                 | 1.002                                       | 2.6e-01                                     |
|                                         | {SOR(1.5)}                                          | 9   | -1.789                    | $\infty$                                   | 1.000              | 0.299                 | 1.002                                       | 2.9e-01                                     |
|                                         | {Multigrid}                                         | 5   | 3.908                     | 1.00 [1.00, 1.00]                          | 1.000              | 0.334                 | 1.002                                       | 3.3e-04                                     |
| Variable-coefficient diffusion, $128^2$ | {Jacobi, Jacobi(0.67)}                              | 4   | 0.009                     | 108 [78.09, 108] ( $\infty$ on 1/8)        | 1.000              | 0.605                 | 1.003                                       | 3.5e-01                                     |
|                                         | {Jacobi, Jacobi(0.67), GS, SymGS, SOR(1.5)}         | 4   | 0.003                     | 108 [78.09, 108] ( $\infty$ on 4/8)        | 1.000              | 0.598                 | 1.003                                       | 3.5e-01                                     |

## 1535 F Limitations

1536 A primary limitation of our approach is the higher on-time computational cost requires to learn the  
1537 greedy routing strategy. In contrast, the simpler baselines such as HINTS incur no training overhead.  
1538 While this cost is amortized during inference time and leads to improved solution quality, it may be  
1539 prohibitive in settings with limited computational resources or when rapid deployment is required.

On the deployment side, Section E shows what survives cost accounting: with a corrector that covers the smooth half of the spectrum and a classical smoother that covers the rest, the router makes one corrector call on 80–100% of the instances at truncation level and executes the same decision sequence as the best static schedule on 0–100% of them, without a search; below truncation level the comparison with the best fixed schedule is pairing-dependent, and the paired ratio of multigrid alone to the router is  $0.03\times$ – $20.5\times$  at  $128^2$ . The corrector we use is specific to the constant-coefficient periodic setting (a fixed Fourier trunk basis on a coarse sensor grid), its cost grows with the fourth power of the sensor-grid size, and undamped Jacobi remains limited far below truncation level by the near-Nyquist modes that neither component damps. Our wall-clock study is restricted to single-thread CPU execution; with a corrector band that does not grow with the grid, the router’s gains over fixed schedules persist at  $512^2$  in the Jacobi pairings ( $1.12$ – $1.24\times$  at  $10^{-8}$  against the strongest fixed schedule on the two equations run at that grid) but shrink to parity for the faster point smoothers, whose long tails of sweeps then dominate the time, and multigrid alone becomes the best classical method there; pairing the corrector with geometric multigrid keeps a  $1.3$ – $1.6\times$  advantage over multigrid alone (Section E). Ensembles of classical solvers can pay off only where the error passes through regimes that favour different members; in the development screening a single well-chosen smoother sufficed at truncation level (Section E.7). For the constant-coefficient periodic test problems the discrete operator is diagonalised by the FFT, and the resulting direct solve, reported in every table, took  $1/3.25\times$ – $1/3.08\times$  of the best router’s time at  $128^2$ ; the hybrid is meant for problems on which no such direct diagonalisation exists—where the band-limited inverse has to be assembled or fitted rather than read off a symbol, and the corrector is then inexact—which is why Section E also reports a variable-coefficient diffusion problem.

Additionally, the effectiveness of the learned router depends on the neural operator exhibiting heterogeneous performance across samples. When the neural operators produces predictions of relatively uniform quality, the benefit of adaptive routing diminishes and it is more practical to set a fixed schedule based on the quality of the neural operator. In practice, achieving such heterogeneity can be sensitive to training choices, and neural operators themselves can be finicky to train. In the wall-clock study the number of corrector calls to truncation level is one on 80–100% of the instances (Section 7); the instance dependence of the router’s decisions appears below truncation level, in when the corrector is worth calling again (Figure 1).

## 1570 G LLM Usage

LLMs were used substantially in preparing this paper, and we disclose this precisely. The text of the original submission was drafted by the authors, with ChatGPT and Gemini used for feedback on grammar and clarity. The revision that responds to the reviews was produced with Claude Code (Anthropic), operated and directed by the authors: the wall-clock benchmark pipeline (the compiled stencil kernels, the cost-aware macro-action environment, the scalar-feature router and its training loop, the corrector fitting, the baseline, table and figure generators, and the assumption and theorem checkers) was written largely by the assistant under the authors’ direction and was reviewed, tested and run by the authors; the corrections to the theory in this revision (the  $\mu$ -slack in Theorem 4.1, the corrected constant and hypotheses of Proposition 4.2, the counterexample to postfix monotonicity, the unitary-eigenbasis condition, and the restatement of Theorem 5.1 for normalised weights) were drafted with the assistant’s help in response to reviewer findings and were verified by the authors; and much of the new text of Section 7 and Section E was drafted by the assistant from the generated tables and edited by the authors. The theorems and proof strategies of the original submission predate the use of the assistant. The research questions, the framing of the method, and the decisions about what to run and what to claim are the authors’. Independent reviews of the manuscript and code were also obtained from LLM-based reviewers; their findings are addressed in this revision.

## 1587 NeurIPS Paper Checklist

### 1588 1. Claims

1589 Question: Do the main claims made in the abstract and introduction accurately reflect the  
1590 paper’s contributions and scope?

1591 Answer: [Yes]

1592 Justification: The abstract and introduction state the two-regime empirical outcome (parity  
1593 with the best static schedule at truncation level; mixed results against the best fixed schedule  
1594 below it; multigrid alone faster below truncation level) with counts and ranges computed  
1595 from the tables, and the theoretical claims with their hypotheses (unitary eigenbasis for  
1596 postfix monotonicity, measured slack otherwise; per-decision consistency for the full-  
1597 information router). At the time of this draft several confirmatory cells are still being  
1598 regenerated and are marked as pending in the text.

1599 Guidelines:

- 1600 • The answer [N/A] means that the abstract and introduction do not include the claims  
1601 made in the paper.
- 1602 • The abstract and/or introduction should clearly state the claims made, including the  
1603 contributions made in the paper and important assumptions and limitations. A [No] or  
1604 [N/A] answer to this question will not be perceived well by the reviewers.
- 1605 • The claims made should match theoretical and experimental results, and reflect how  
1606 much the results can be expected to generalize to other settings.
- 1607 • It is fine to include aspirational goals as motivation as long as it is clear that these goals  
1608 are not attained by the paper.

### 1609 2. Limitations

1610 Question: Does the paper discuss the limitations of the work performed by the authors?

1611 Answer: [Yes]

1612 Justification: The paper discusses its limitations in the Limitations section (Appendix F),  
1613 including the constant-coefficient periodic scope of the corrector, the single-thread CPU  
1614 setting, the test-set-selected optimistic baselines and the pairing-dependent results below  
1615 truncation level.

1616 Guidelines:

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1621 violations of these assumptions (e.g., independence assumptions, noiseless settings,  
1622 model well-specification, asymptotic approximations only holding locally). The authors  
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1624 implications would be.
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1626 only tested on a few datasets or with a few runs. In general, empirical results often  
1627 depend on implicit assumptions, which should be articulated.
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1630 is low or images are taken in low lighting. Or a speech-to-text system might not be  
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1632 technical jargon.
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1634 and how they scale with dataset size.
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